How Reliable Is Inference in Staggered Difference-in-Differences?

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Abstract

We study the reliability of inference in staggered difference-in-differences designs in empirical finance. Our census of articles published in the top three finance journals from 2010 to 2025 shows increasing use of heterogeneity-robust estimators in recent years, while the use of finite-sample refinements to inference remains limited. Many articles rely solely on cluster-robust standard errors for inference and provide little information for assessing their reliability. We further show that the reliability of inference in staggered difference-in-differences designs depends on both the characteristics of the sample and researchers' choice of estimators. These features characterize an interval for the effective number of clusters, which helps us assess when conventional clustered inference may be unreliable for a given estimator and when alternative inference methods should be considered. Evidence from simulations and reanalyses of published studies shows that adapting inference according to the effective number of clusters can affect conclusions about treatment effects and lower false rejection rates. The reanalyses also show that a published finding can be sensitive to the choice of inference even when the point estimate remains unchanged.

Keywords: staggered difference-in-differences; cluster-robust inference; effective number of clusters; heterogeneity-robust estimators; cluster imbalance; randomization inference.
JEL: C12, C21, C23, G21, G28.

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1 Introduction

Difference-in-differences (DiD) is a central research design in empirical finance, where laws and regulations often take effect in different states or countries at different times and researchers use the resulting variation to study firms, households and financial markets. The credibility of these studies depends on both the comparisons used to estimate a treatment effect and the inference used to assess its statistical significance. Early work established that serial correlation (Bertrand et al., 2004) and common group errors can lead conventional standard errors to overstate the precision of difference-in-differences estimates (Moulton, 1990; Donald and Lang, 2007), while recent work shows that staggered adoption and treatment effect heterogeneity can complicate the interpretation of estimates in workhorse two-way fixed effects regressions (Sun and Abraham, 2021), because already-treated units enter as controls (Goodman-Bacon, 2021; Baker et al., 2022). Therefore, the development of estimators that restrict comparisons to appropriate control groups has changed empirical practice. The estimator of Callaway and Sant'Anna (2021, CS) estimates treatment effects for each cohort and period before aggregating them, while Sun and Abraham (2021, SA) estimate cohort-specific effects at each event time and aggregate across cohorts. The imputation estimator of Borusyak et al. (2024) uses untreated observations to predict treated units' outcomes without treatment and compares those predictions with their observed outcomes. These approaches use different comparisons and weights, so the choice of inference remains consequential for the conclusions drawn from their estimates (Baker et al., 2026).

We ask when and which inference is reliable in the staggered difference-in-differences designs used in finance. We show that the answer can differ for average treatment effects, for each horizon of an event study, and for the joint pre-trend test, because each estimate can draw on a different set of treatment and comparison clusters. We take the estimator and its identifying assumptions as given and examine how the adoption schedule, cluster sizes, fixed effects and estimator weights affect the inference and interpretation of the resulting estimate.

The motivation for this question follows from the way staggered designs combine policy variation with clustered data. Policies adopted by states can affect many firms or counties within a state, so standard errors clustered at the level of adoption are a natural response to shared disturbances, yet the effective number of clusters can be much smaller than the number of observations (Carter et al., 2017) or the number of states in the sample (Hu and Spamann, 2020). Restricting comparisons to units that have not yet received treatment can also leave a small number of clusters in a cohort or comparison group, making inference sensitive to features of the empirical design that differ across estimators (Weiss, 2026; Karim et al., 2026). The recent staggered difference-in-differences literature has devoted considerable

attention to estimator choice, while finite-sample inference for the resulting estimates remains an active question (Baker et al., 2026; Karim et al., 2026). These features matter for whether a policy effect can be distinguished from zero and for how researchers interpret evidence about the timing of an effect and the plausibility of their research design.

We extend the effective cluster framework of Carter et al. (2017) to staggered difference-in-differences designs and examine how sample characteristics and estimator choice determine the information available for inference. For each estimator, we use the adoption schedule, cluster sizes, fixed effects and estimator weights to characterize an interval for the effective number of clusters over a family of disturbances. We then calibrate estimator-specific thresholds for the effective number of clusters for given samples. By comparing the thresholds with the effective cluster interval, we examine the accuracy and power of multiple inference methods. In the calibration designs, conventional clustered Wald tests perform well when the lower end of the effective cluster interval exceeds the corresponding threshold. However, the relationship between the sample and inference can change with the estimator chosen by researchers. Even though the lower end of the effective cluster interval exceeds the corresponding threshold and the conventional clustered Wald tests work well for the two-way fixed effects (TWFE) estimator for a sample, they can be less reliable in some designs when researchers switch to heterogeneity-robust estimators such as Callaway and Sant'Anna (2021) (CS) or Sun and Abraham (2021) (SA). This is because the information supporting an estimate can become concentrated in fewer clusters in the process of constructing cleaner comparisons with untreated or not-yet-treated units. These results show how the adoption schedule, the distribution of clusters, the observation level of the panel and the choice of estimator jointly determine the information available for inference. The choice of estimator and the choice of inference method are therefore linked: under the limits of a given sample, selecting a particular estimator can make conventional cluster-robust inference unreliable. Evidence from Monte Carlo experiments and the applications to published research demonstrate that consequence.

Our census of published articles in the *Journal of Finance*, the *Journal of Financial Economics* and the *Review of Financial Studies* illustrates how developments in estimation and inference have been reflected in empirical finance. Consistent with the evidence of the earlier census in Baker et al. (2022), among 198 articles from 2010 to 2025 whose main analysis uses a staggered DiD design, 181 use a TWFE estimator and 184 use cluster-robust standard errors, corresponding to 91.4 and 92.9 percent of the sample. Of the 198 articles, 136 were published through 2021 and 62 from 2022 to 2025. Heterogeneity-robust estimators appear in the main results of twenty-two later articles. However, only six articles report the number of clusters and none reports the effective number of clusters, leaving readers with

limited information about how the sample and estimator affect the reliability of inference. The growing use of alternative estimators therefore increases the importance of reporting the design features that support inference for each estimate.

To understand which features of the design matter for inference, we begin with the contribution of each cluster to the variance of an estimate. Following Carter et al. (2017), the effective number of clusters summarizes the concentration of these contributions as an equivalent number of equally contributing clusters. It can be much smaller than the number of clusters directly observed in the sample when a few clusters account for a large share of the variance, making conventional clustered Wald tests sensitive to the disturbances in those clusters. Each cluster's contribution depends on the estimator's weights and on the dependence of observations within the cluster. For policies adopted by states, firms or counties in the same state and period may share a common shock, and disturbances can also be correlated over time. Since the strength of this dependence is not determined by the observed sample structure, we characterize the effective number of clusters over a family of disturbances, yielding an interval. For example, in the elder financial protection design of Carlin et al. (2023), the TWFE regression uses 51 state clusters, while its effective cluster interval ranges from 10.25 to 28.46. The blind share captures a separate feature of the variance calculation by measuring the modeled contribution associated with single-cluster cohorts or comparison groups whose group-specific scores cannot reflect their own shocks. The minimum detectable effect complements these measures by describing the size of an effect that a method detects with a specified probability across repeated samples under stated variance and inference assumptions. Together, the effective cluster interval and the blind share help assess the reliability of conventional inference against the calibration evidence, while the minimum detectable effect informs the interpretation of estimates that do not reject a null hypothesis.

The effective number of clusters can also vary across disturbances when the observation level coincides with the clustering level. In a staggered design, the timing of adoption and the estimator determine how observations within each cluster contribute across periods, so serial correlation can change each cluster's contribution to the variance even when there is only one observation per cluster and period. The bank deregulation design of Beck et al. (2010) illustrates this case, since the panel contains one observation per state and year and the TWFE regression is estimated on 49 state clusters. Across the serial correlation values considered, its effective cluster interval ranges from 35.01 to 46.52. The reanalysis of Baker et al. (2022) uses a restricted sample of 36 states to construct comparisons with untreated or not-yet-treated states, and the effective cluster intervals for the four heterogeneity-robust estimators on this sample range from approximately 1 to 12. These differences reflect both

the sample restrictions and the weights assigned to treatment and comparison groups, with the last state to deregulate serving as the sole comparison group for the SA estimator. 10 of the 17 treated cohorts in the restricted sample contain only one state, so the conventional cluster-robust variance estimator can also miss some of the variance associated with these states for some cohort-based estimates. In simulations based on these samples and adoption schedules, the conventional Wald test rejects a true zero 29.0 percent of the time for CS and 35.0 percent for SA, compared with 4.0 percent for TWFE. These results illustrate how the sample and the comparisons used by an estimator can affect the reliability of conventional clustered inference.

The conditions supporting reliable conventional inference can also differ across estimators, which makes the choice of inference specific to the estimate being tested. In a Compustat design, we assume a two-component disturbance with a common state-by-year shock and an idiosyncratic firm-year error under a true zero effect. The conventional clustered Wald test uses variation in cluster contributions to measure uncertainty while allowing observations within a cluster to be dependent. To assess its accuracy, we examine how often it rejects a true zero effect, which is intended to occur in no more than 5% of repeated samples at the nominal 5% significance level. We calculate this rejection probability under the assumed disturbances, and the largest value across the common-shock shares considered gives the exact size for each design. The stacked regression pools cohort-specific comparison samples, the switching estimator compares changes for newly treated and untreated units, and the two-stage estimator first fits unit and period effects using untreated observations. In these calibration designs, the exact size differs from the nominal five percent by at most one percentage point once the lower end of the effective cluster interval reaches about 14 effective clusters for two-way fixed effects, about 27 for the stacked regression, and about 55 to 60 for Callaway and Sant'Anna, Sun and Abraham, and the imputation estimator. The corresponding thresholds are about 64 for the switching estimator of de Chaisemartin and D'Haultfœuille (2020) and about 9.5 for the two-stage estimator of Gardner (2022).

The relationship between the effective cluster interval and these calibrated thresholds provides information about the conditions under which different inference methods may be considered. When the whole interval lies above the threshold and the blind share is zero, the calibration evidence indicates more favorable conditions for conventional clustered inference in the designs examined. The cluster jackknife can provide a sensitivity analysis by re-estimating the same treatment effect after leaving out one cluster at a time, provided that the original effect remains identifiable in each reduced sample. In contrast, when the whole interval lies below the threshold, the accuracy of the conventional test requires closer attention, and randomization inference can be considered when adoption dates follow a specified assignment

mechanism conditional on the design information used to select the method. When the interval spans the threshold or lies above it with a positive blind share, reporting alternative inference methods can help researchers assess the sensitivity of their conclusions to the disturbance and to the construction of the variance estimator. These comparisons require attention to the hypothesis tested by each method, since the clustered Wald test and the cluster jackknife assess an average treatment effect while the randomization inference considered here tests a sharp null under the specified conditional assignment mechanism. A sharp null of no effect specifies that treatment has no effect on any unit in any period, allowing the researcher to determine the outcomes associated with alternative adoption dates under the null. A zero average effect can also arise when positive and negative effects offset one another, so the finite-sample exactness of the randomization test for the sharp null does not automatically extend to the average-effect null. Reporting the three methods together can therefore reveal how conclusions depend on their assumptions and the hypotheses being tested, while their joint rejection can be interpreted as evidence against the sharp null under the stated assignment mechanism. The thresholds provide a reference from the designs and disturbances used in their calibration, and the validity of each inference method depends on its own assumptions.

To examine the finite-sample performance of design-adapted inference and alternative inference methods, we use Monte Carlo experiments based on the bank deregulation design of Beck et al. (2010) (BLL) and the elder financial protection design of Carlin et al. (2023) (CUY). Our main comparison focuses on the two-way fixed effects estimator (TWFE), the stacked estimator (Gormley and Matsa, 2011; Cengiz et al., 2019), Callaway and Sant'Anna (CS) (Callaway and Sant'Anna, 2021), Sun and Abraham (SA) (Sun and Abraham, 2021), and the imputation estimator of Borusyak et al. (2024). We also examine the switching estimator of de Chaisemartin and D'Haultfœuille (2020) and the two-stage DiD estimator (2S-DiD) of Gardner (2022) separately in the appendix. For the main comparison, we report the conventional cluster-robust Wald test based on CV_1, the cluster jackknife based on CV_3, randomization inference (RI) and the classic restricted wild cluster bootstrap (WCR-C) where the corresponding calculations are available (Young, 2019; MacKinnon et al., 2023b). The wild cluster bootstrap generates alternative outcomes by applying random weights to cluster-level residual contributions under the null, and WCR-C uses conventional restricted scores. We also report the jackknife-based bootstrap variants WCR-S and WCR-B for the TWFE and stacked regressions in both designs, where both use transformed scores and differ in how their test statistics are standardized. The CV_3 results are not reported for SA in either design because deleting certain states removes identifying support for the original treatment effect. The design-adapted calculations use CV_1, CV_3 and RI in light of the calibration evidence,

the sensitivity of estimates to individual clusters and the assumed assignment of treatment, while the bootstrap calculations provide additional comparisons for the specifications on which they are available.

For each estimator and design, we hold the sample and adoption schedule fixed and examine a true zero effect and a homogeneous reference effect under repeatedly drawn disturbances. The reference effect is the same for every available inference method applied to a given estimator and disturbance setting, allowing us to compare their rejection frequencies under the same alternative. The corresponding zero-effect frequencies show how often each method rejects when there is no treatment effect. In several specifications the conventional Wald test rejects a true zero substantially more often than its nominal level, while the design-adapted calculations and alternative methods have different rejection frequencies under both scenarios. We also calculate the minimum detectable effect of each available method in the same calibrated model. The reference effect and these method-specific values answer complementary questions, since the former holds the effect fixed across methods and the latter shows how large an effect each method requires to reach 95% power. The switching and two-stage calculations use separate specifications and are presented in the appendix.

To examine how these findings inform the interpretation of empirical evidence, we apply our design-adapted inference approach and alternative inference methods to the bank deregulation study of Beck et al. (2010), who examine the effect of state bank deregulation on income inequality using a panel of states and years. In their influential reanalysis, Baker et al. (2022) show how staggered adoption and treatment effect heterogeneity can affect the comparisons underlying TWFE and use alternative estimators to construct comparisons with untreated or not-yet-treated states. Building on their analysis, we examine how the information supporting these alternative estimates affects inference. The original TWFE estimate remains statistically significant under the published tests, while inference for the alternative estimates is sensitive to the choice of inference. For SA and the imputation estimator, CV_1 rejects a zero average effect, while RI does not reject the sharp null of no effect. For three of the four alternative estimators, RI rejects the hypothesis that the effect of deregulation was the same in every state and period after adoption and equal to the TWFE estimate. The average effect may nevertheless equal the TWFE estimate if treatment effects differ across states or periods. Together, these results complement the findings of Baker et al. (2022) by showing that the interpretation of the alternative estimates depends on both the comparisons used to construct them and the assumptions underlying inference. Differences in samples and in the treatment effects aggregated by each estimator also matter when comparing them with the original TWFE estimate.

In our second reanalysis, the elder financial protection study of Carlin et al. (2023)

examines the effects of state laws on reported elder financial exploitation using a county-month panel in which multiple counties share the same state policy. This sample structure makes the information available for inference sensitive to dependence among counties within the same state. For the TWFE estimate, CV_1 rejects a zero average effect, while CV_3 does not reject and RI rejects the sharp null of no effect. The three tests therefore do not all reject, and the evidence for a reduction in reported exploitation depends on the inference method and the hypothesis being tested. The event studies also yield different pointwise results across horizons and specifications. For the pre-treatment periods, RI tests a hypothesis that the policy has no effect in any period, including after adoption, so a failure to reject does not establish that treated and comparison states would have followed parallel trends without the policy. These results illustrate how the effective cluster interval and the blind share can inform the assessment of empirical findings, while the interpretation of each test depends on the treatment effect of interest and its assumptions about disturbances and treatment assignment.

We contribute to three strands of literature.

First, we contribute to the literature on staggered DiD by examining how the sample and the choice of estimator affect the reliability of inference. de Chaisemartin and D'Haultfœuille (2020) and Goodman-Bacon (2021) characterize the comparisons underlying TWFE estimates, and subsequent estimators use untreated or not-yet-treated units to recover stated treatment effects under their identifying assumptions (Callaway and Sant'Anna, 2021; Sun and Abraham, 2021; Borusyak et al., 2024). Baker et al. (2022) bring these developments to finance and demonstrate their importance in published applications, while recent work examines the precision and finite-sample inference of alternative estimators (Hull, 2026; Weiss, 2026; Karim et al., 2026). Baker et al. (2026) emphasize that the target effect, comparison groups and assumptions about uncertainty are integral to the design of a DiD study. We contribute to this literature by examining how the sample and comparisons used by each estimator, together with dependence within clusters, affect the information available for inference. Our analysis shows why the choice of estimator can change the reliability of conventional clustered inference and why this assessment can differ across average effects and event-study horizons.

Second, we contribute to the literature on cluster-robust inference by extending the effective cluster framework to the estimators used in staggered designs. Carter et al. (2017) show that differences in cluster size, covariates and within-cluster dependence affect the reliability of clustered tests and propose the effective number of clusters as a diagnostic, and MacKinnon et al. (2023a) recommend disclosing effective cluster counts and leverage when assessing inference. Hu and Spamann (2020) document the practical importance of cluster imbalance for corporate finance designs based on state laws. We extend the effective cluster

framework to the estimators considered in staggered designs and characterize the range of effective cluster counts over a family of disturbances. We also examine a separate source of uncertainty that arises when conventional cluster-robust variance estimators miss part of a cluster's contribution to the variance, including when a treatment cohort or comparison group contains a single cluster. The effective cluster interval and the blind share together connect the distribution of adoption dates and estimator weights to the assessment of inference.

Third, we contribute to the literature on design-based inference by examining how randomization inference can complement cluster-robust methods in staggered designs. Abadie et al. (2020) and Abadie et al. (2023) relate uncertainty and clustering to sampling and assignment mechanisms, and Young (2019) studies randomization inference under an explicit assignment mechanism. In staggered designs, Athey and Imbens (2022) examine estimation and inference under assumptions about the assignment of adoption dates. For corporate finance designs with unequal state sizes, Hu and Spamann (2020) propose a permutation test that holds the dominant state's treatment status fixed. We draw on these approaches to examine randomization inference alongside clustered inference and the cluster jackknife in the published staggered designs. By relating their performance to the effective cluster interval and the blind share, we show how these diagnostics can inform the assessment of alternative methods and the sensitivity of empirical conclusions. The interpretation of these comparisons depends on the assumptions supporting each method and on the distinction between a sharp null and a null concerning an average treatment effect.

The rest of the paper proceeds as follows. Section 2 presents the census of staggered designs in finance, and Section 3 develops the diagnostics and examines the conditions of alternative inference procedures. Section 4 reports exact calculations and simulation evidence, and Section 5 applies the framework to the two published studies. Section 6 concludes.

2 Motivation: Staggered DiD and Inference in Finance

2.1 Estimation and inference in staggered designs

The expansion of difference-in-differences in empirical finance has been accompanied by two closely related developments in causal inference. The first concerns the comparisons used to estimate treatment effects. When policies are adopted at different times, researchers can follow changes in treated units and compare them with changes in units that have not yet been treated. The conventional two-way fixed-effects regression combines these comparisons with others in which already-treated units serve as controls, so treatment effects that vary across cohorts or over time can affect the interpretation of both average effects and event studies (Goodman-Bacon, 2021; Sun and Abraham, 2021). Heterogeneity-robust estimators

address this concern by restricting the comparisons and defining the effects they aggregate (Callaway and Sant'Anna, 2021; Borusyak et al., 2024). The choice of comparison group and the assumptions needed to interpret its outcomes remain central to the resulting causal claim.

The second development concerns inference in samples where observations share common influences. In finance, a state law can affect many firms or counties in the same jurisdiction, and outcomes within that jurisdiction can also respond to common economic conditions. Standard errors that account for within-cluster dependence address an important source of overstated precision, as shown by Bertrand et al. (2004) for difference-in-differences and by Petersen (2009) for finance panels. Subsequent work shows that the accuracy of clustered inference also depends on the number of clusters and on differences in their size and influence (Cameron and Miller, 2015; Carter et al., 2017; MacKinnon et al., 2023a). Hu and Spamann (2020) illustrate the practical consequences in corporate finance, where firms are unevenly distributed across states of incorporation.

The two developments meet in staggered designs because the estimator determines which clusters contribute to each comparison. Restricting a comparison to units that have not yet been treated can leave a small number of clusters in a treatment cohort or comparison group, even when the full sample contains many clusters. The resulting estimate may have a clearer causal interpretation while its standard error becomes sensitive to the distribution of information across those clusters. For empirical finance, this sensitivity matters for the published significance of an average effect, for the timing of effects in an event study, and for the tests of pre-treatment differences that accompany the interpretation of a design.

2.2 The use of staggered designs and estimators

Baker et al. (2022) document the widespread use of staggered difference-in-differences in finance and accounting and explain the consequences of treatment effect heterogeneity for conventional estimates. Their survey covers articles published or accepted from 2000 to 2019 in five finance and five accounting journals, and their reading of published applications illustrates why the choice of estimator matters for substantive conclusions. We examine a later period and a narrower set of journals, in order to pay particular attention to the inference exercises presented alongside the estimates in the more recent articles.

Our corpus contains 4,953 research articles with recoverable text published in the *Journal of Finance*, the *Journal of Financial Economics* and the *Review of Financial Studies* from 2010 to 2025, of which 1,093 describe a difference-in-differences design. We read those articles in full and classify a main analysis as staggered when a binary treatment begins at two or more distinct times and those differences in timing provide the identifying variation. We identify 198 articles whose main analysis uses a staggered DiD design. The classification

follows the research design described in each article, including cases in which the authors do not call the design staggered.

Table 1 reports the estimator and inference choices in the 198 staggered articles. Consistent with the prevalence documented by Baker et al. (2022), two-way fixed effects remain the dominant specification, appearing in 181 articles, or 91.4 percent of the sample. Twenty-two articles use at least one of the heterogeneity-robust estimators considered in the table, with the stacked regression used most often. These counts describe the use of the estimators in the articles and allow an article to use more than one method.

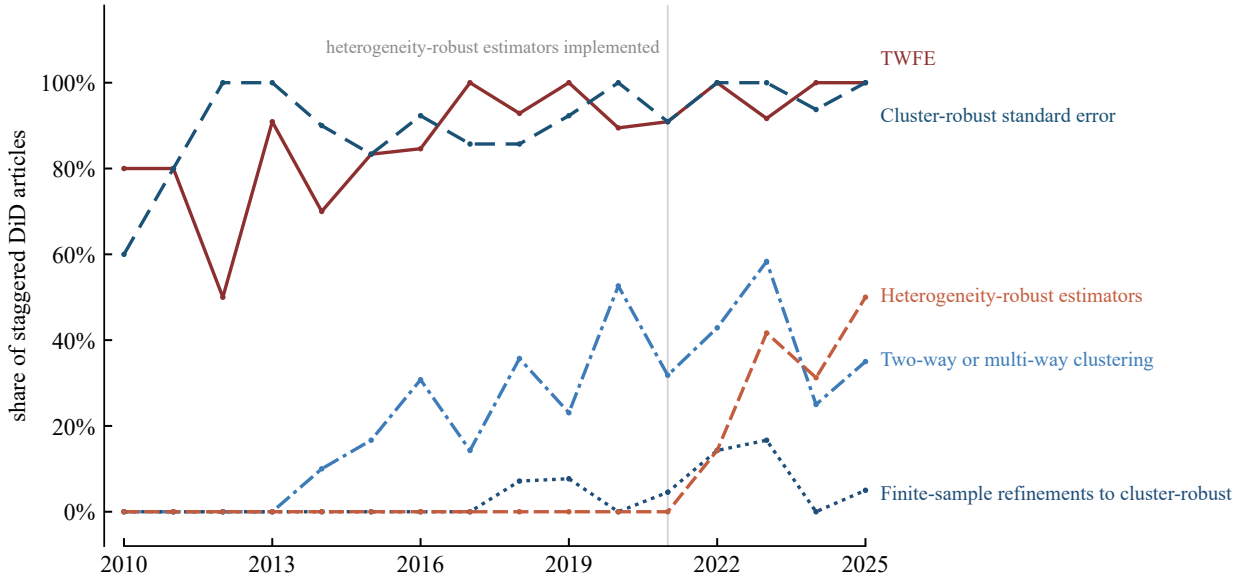
Table 1: A census of staggered DiD articles in the top three finance journals, 2010–2025.

	JF ($n = 38$)	JFE ($n = 88$)	RFS ($n = 72$)	All ($n = 198$)
Panel A. Estimators				
TWFE	35 (92.1%)	83 (94.3%)	63 (87.5%)	181 (91.4%)
Heterogeneity-robust estimators	3 (7.9%)	8 (9.1%)	11 (15.3%)	22 (11.1%)
Panel B. Inference				
Cluster-robust standard error	34 (89.5%)	81 (92.0%)	69 (95.8%)	184 (92.9%)
Two-way or multi-way clustering	9 (23.7%)	21 (23.9%)	27 (37.5%)	57 (28.8%)
Finite-sample refinements to cluster-robust	3 (7.9%)	2 (2.3%)	3 (4.2%)	8 (4.0%)
Reports the number of clusters	0 (0.0%)	2 (2.3%)	4 (5.6%)	6 (3.0%)

Note 1: The denominator is the 198 articles in the *Journal of Finance* (JF), the *Journal of Financial Economics* (JFE), and the *Review of Financial Studies* (RFS) from 2010 to 2025 whose main result is a staggered difference-in-differences.
Note 2: TWFE refers to a specification that includes unit and time fixed effects, as reported in the specification or the table notes. Heterogeneity-robust estimators include the group-time estimator of Callaway and Sant'Anna (2021) (CS), the interaction-weighted estimator of Sun and Abraham (2021) (SA), stacked regressions (Gormley and Matsa, 2011; Cengiz et al., 2019) (stacked), and the imputation estimator of Borusyak et al. (2024) (imputation). An article is counted once if it applies at least one of these estimators in the text or the table notes. The Callaway and Sant'Anna estimator is used in 5 articles, the Sun and Abraham estimator in 4, stacked regressions in 15, and the imputation estimator in 2. An article may use more than one estimator.
Note 3: Finite-sample refinements and related inference procedures include the bias-reduced cluster variance estimator of Bell and McCaffrey (2002) (CV_2), the cluster jackknife (MacKinnon et al., 2023b) (CV_3), the wild cluster bootstrap (Cameron et al., 2008) (WCR), and randomization inference (Young, 2019) (RI). An article is counted once if it uses at least one of these procedures. Neither the bias-reduced cluster variance estimator nor the cluster jackknife appears in the sample. The wild cluster bootstrap is used in 2 articles and randomization inference in 7. One article uses both, so the total number using at least one procedure is 8.
Note 4: Cluster-robust standard errors and two-way or multi-way clustering are identified from the text and the table notes. Multi-way clustering includes clustering in three dimensions. The last row counts articles that state the number of clusters alongside the reported inference.

Figure 1 places those choices in their publication years. None of the 136 articles published through 2021 uses one of the four heterogeneity-robust estimators in the table, while 22 of the 62 articles published from 2022 to 2025 use at least one, including ten of the twenty articles published in 2025. The pattern is consistent with an increasing response to the concerns about staggered estimation discussed by Baker et al. (2022), although the journal coverage and publication years of the two surveys differ. The continued prevalence of two-way fixed effects also means that inference for conventional estimates remains relevant alongside inference for the newer estimators.

Figure 1: Estimators and inference in staggered DiD finance articles



Note 1: The horizontal axis is the publication year and the vertical axis is the share of that year's staggered articles. A year contains between four and twenty-two such articles. Estimators are red and inference is blue, and each series is labeled at its right end. TWFE is the solid red line. Heterogeneity-robust estimators are the dashed red line. The cluster-robust standard error is the long-dashed blue line, two-way or multi-way clustering is the dash-dot blue line, and finite-sample refinements to cluster-robust inference are the dotted blue line.
Note 2: The gray vertical line marks 2021 and is labeled as the implementation of the heterogeneity-robust estimators. It marks the publication year of Callaway and Sant'Anna (2021) and Sun and Abraham (2021) and of the TWFE decomposition in Goodman-Bacon (2021). Some estimators shown in the figure were introduced earlier.

2.3 Inference in published research

The same articles show a much more stable reliance on clustered inference. Panel B of Table 1 shows that 184 of the 198 articles use cluster-robust standard errors, and 57 cluster in two or more dimensions. Figure 1 shows that cluster-robust standard errors are common throughout the sample period, while the use of two-way or multi-way clustering rises over time. These choices recognize that observations can share disturbances within a cluster and, in the case of multi-way clustering, across more than one dimension.

The use of clustering in more than one dimension raises a further question about the information supporting inference. For example, clustering by state and period allows researchers to account for dependence across states within a period as well as dependence within states over time. These sources of dependence can coexist with the concentration of an estimate's variance in a small number of states. The use of more than one clustering dimension therefore calls attention both to the sources of common shocks and to the information available for estimating their contribution to uncertainty.

The information needed to assess the accuracy of a clustered test is reported less often. Six articles report the number of clusters used for inference, and none reports the effective number of clusters. Eight articles use at least one of the finite-sample refinements listed in

the table, with two using a wild cluster bootstrap and seven using randomization inference. These counts describe what the articles report and do not establish that their individual tests are invalid. They do show that a reader usually cannot assess the contribution of individual clusters or the sensitivity of a published result to a test designed for a limited number of informative clusters.

The distinction matters because the number of clusters alone does not describe how much each contributes to the estimate. Carter et al. (2017) recommend measuring the effective number of clusters and explain that its exact value depends on features of the disturbance that are not observed directly, while MacKinnon et al. (2023a) recommend examining cluster leverage and effective cluster counts when choosing inference. In a staggered design, differences in adoption dates and in the sizes of clusters can concentrate an estimator's information, and a common shock affecting units within a cluster and period can change the extent of that concentration. Heterogeneity-robust estimators can add another source of sensitivity when a cohort or comparison group contains only one cluster and some of its contribution to the estimate is not reflected in the conventional standard error.

The census therefore points to a question that applies to both conventional and heterogeneity-robust estimates. Researchers need to know when the standard error attached to their chosen estimator provides a reliable basis for judging an effect, and how to proceed when the design supplies limited information about the disturbance. Examining the range of effective cluster counts supported by the design and the variance that may be hidden from the standard error connects the literature on clustered inference to the comparisons used in staggered designs. This connection also helps readers distinguish an imprecise estimate from evidence of no effect and interpret event-study and pre-trend results in light of the information available in the sample.

3 Inference in Staggered DiD Designs

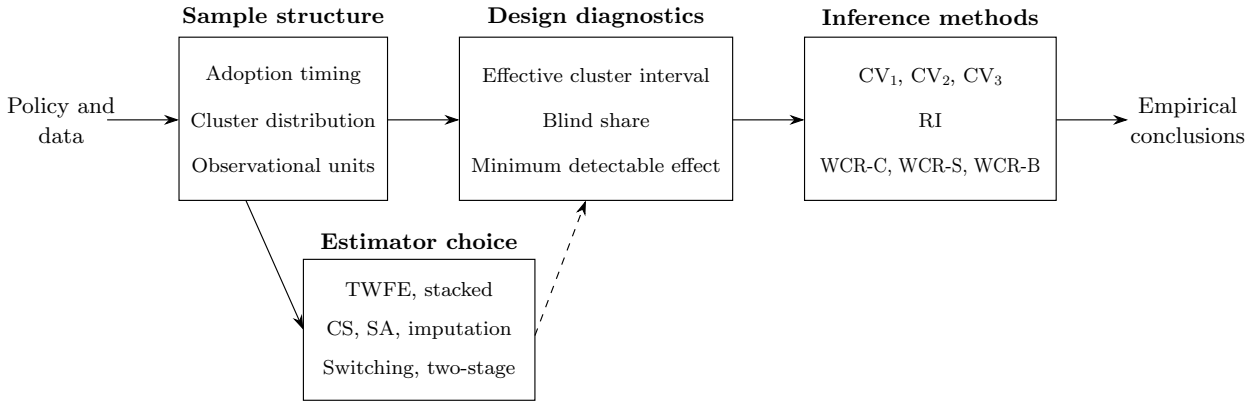
Financial policies are often adopted at different times across the jurisdictions in which researchers observe firms, households or other economic units. For example, state anti-takeover statutes are enacted in different years, elder financial exploitation rules take effect in different states in different months, and bank branch deregulation rolled out across states on their own schedules. These policies give rise to staggered DiD designs in which the timing of adoption, the distribution of observations across clusters and the available comparison groups shape the sample used to estimate a treatment effect.

Several features of these designs affect the information available for inference. The timing of adoption and dependence within clusters over time influence how each cluster contributes

to the variance of an estimate, while shocks shared by units within the same cluster and period provide an additional source of dependence when the observation level is finer than the clustering level. Heterogeneity-robust estimators construct comparisons using untreated or not-yet-treated units, and the weights associated with these comparisons can concentrate information in a smaller number of clusters. For some estimates, conventional cluster-robust variance estimators can also miss part of the variance when a treatment cohort or comparison group contains a single cluster.

Figure 2 summarizes how the sample and the choice of estimator enter the assessment of inference. Researchers choose an estimator in light of the policy, the structure of the data and the comparisons needed for the treatment effect of interest. Different estimators can assign different weights to treatment and comparison groups, so the information supporting an estimate can be distributed differently across clusters even when the underlying sample is the same. In this section, we develop a framework that relates these features of the sample and the choice of estimator to the effective cluster interval and the blind share, and we examine how these diagnostics can inform the assessment of inference. The interpretation of alternative inference methods also depends on their assumptions about disturbances and treatment assignment and on the hypothesis being tested.

Figure 2: Sample structure, estimator choice, and inference



Note 1: The figure illustrates how sample structure and estimator choice enter the assessment of inference. The sample constrains the available comparisons and estimators. TWFE is the two-way fixed effects estimator, CS is the estimator of Callaway and Sant'Anna (2021), SA is the estimator of Sun and Abraham (2021), the imputation estimator is from Borusyak et al. (2024), the switching estimator is from de Chaisemartin and D'Haultfœuille (2020), and the two-stage estimator is from Gardner (2022).

Note 2: The effective cluster interval (discussed in Section 3.1.1) summarizes the range over the evaluated disturbance settings. The blind share (discussed in Section 3.2) concerns variance contributions that a particular conventional cluster-score construction can miss. The minimum detectable effect (discussed in Section 3.1.2) depends on the inference method and the specified variance, significance level and power.

Note 3: CV_1 is the conventional cluster-robust variance estimator (Cameron and Miller, 2015), CV_2 is the bias-reduced cluster-robust variance estimator (Bell and McCaffrey, 2002), and CV_3 is the cluster jackknife variance estimator (MacKinnon et al., 2023b). RI is randomization inference (Young, 2019). WCR-C is the classic restricted wild cluster bootstrap, while WCR-S and WCR-B are restricted wild cluster bootstrap variants that use jackknife-transformed scores, with WCR-B also using jackknife-based studentization (MacKinnon et al., 2023b).

Note 4: The diagnostics do not by themselves establish the validity of an inference method. Methods can concern different null hypotheses and require additional assumptions, including assumptions about treatment assignment and the availability of the original target after cluster deletion.

3.1 A Framework for Clustered Inference

We begin with a workhorse two-way regression with unit and period effects[1] and a cluster-level treatment indicator,

$$y_{it} = \alpha_i + \delta_t + \theta D_{g(i)t} + \nu_{g(i)t} + \mu_{it}, \tag{1}$$

where unit i belongs to cluster $g(i)$ and is observed in period t. The indicator $D_{g(i)t}$ turns on when cluster $g(i)$ adopts the policy. Adoption is staggered, so clusters adopt in different periods, some are treated in every period the panel covers, and some never adopt. The policy is assigned at the cluster level, so $D_{g(i)t}$ takes the same value for every unit in a cluster in a given period. The unit effect α_i absorbs characteristics that are permanent for a unit, including permanent characteristics of its cluster, and the period effect δ_t absorbs shocks shared by every cluster in a period. Neither generally absorbs $\nu_{g(i)t}$, the shock common to the units of cluster $g(i)$ in period t, and a fixed effect for that cluster and period would absorb $D_{g(i)t}$ with it. The remaining term, μ_{it}, is idiosyncratic to the unit.

Researchers estimate the coefficient θ to study the treatment effect of the policy and often cluster the standard error at the level where the policy is assigned to assess whether the estimate is distinguishable from zero. The reliability of this inference depends on the assumptions about disturbances across clusters (Cameron and Miller, 2015) and on how the clusters contribute to the variance of the estimate. When a few clusters carry a large share of the variance, the effective number of clusters can be much smaller than the number observed in the sample, making conventional clustered inference less reliable (Carter et al., 2017).

To examine how these considerations extend across estimators, we first consider the choice of clustering level and then characterize the statistical structure underlying their estimates and standard errors.

The choice of clustering level depends on the sampling and assignment processes and on the sources of dependence in the data (Abadie et al., 2023; Cameron and Miller, 2015). In the state policy designs studied here, clustering by state allows observations exposed to the same policy and common state disturbances to be dependent. Abadie et al. (2020) and Rambachan and Roth (2026) study conditions under which conventional cluster-robust variance estimators can be conservative in design-based settings. Those results depend on their sampling and assignment assumptions and do not by themselves establish the finite-sample accuracy of a clustered test in the staggered designs considered here. The cross-sectional refinements of Abadie et al. (2023) also require treatment variation within clusters, a condition that does not follow from state-level assignment. For the resulting clustered test, the distribution of variance contributions across clusters provides additional information about the plausibility

[1] Unit and period fixed effects are also referred to as individual and time fixed effects.

of a large-sample approximation (Carter et al., 2017; MacKinnon et al., 2023a).

When outcomes in different states respond to the same economic conditions, disturbances can also be correlated across states within a period. Period fixed effects absorb a shock that affects all observations equally, but need not remove the dependence created when states have different exposures to that shock. Clustering by both state and period allows for dependence within each state over time and across states within a period (Cameron et al., 2011; Petersen, 2009). These sources of dependence can coexist with shocks shared by observations within a state and period, and together with the weights of an estimator they determine the variance of its estimate. The additional cross-state covariance can change the precision of an estimate and affect the accuracy of a test that clusters only by state. At the same time, allowing for an additional clustering dimension does not by itself increase the number of states that support a comparison or ensure that the scores capture the contribution of a single-state group. Appendix A.4 describes how these considerations enter the variance calculation. The effective cluster interval developed below characterizes information across states under the disturbance family considered here, while the choice of clustering dimensions also reflects the sources of dependence relevant to the application.

The next issue is how estimators construct their comparisons and standard errors. In the standard staggered-adoption setting, the TWFE coefficient combines two-by-two comparisons, including comparisons that use already-treated units as controls (Goodman-Bacon, 2021). CS compares changes for each treated cohort with changes for an untreated or not-yet-treated group and aggregates the resulting cohort-period effects, while SA estimates cohort-specific event-time effects relative to a comparison cohort before aggregating them (Callaway and Sant'Anna, 2021; Sun and Abraham, 2021). A stacked regression combines cohort-specific samples of treated and comparison units in a pooled regression (Gormley and Matsa, 2011; Cengiz et al., 2019; Deshpande and Li, 2019). The imputation estimator fits an untreated-outcome model using untreated observations and averages the differences between treated outcomes and their predicted untreated outcomes (Borusyak et al., 2024). Each method defines a treatment-effect target and constructs comparisons intended to identify it under its respective assumptions. These comparisons determine the weights assigned to observations, and the analysis below considers how those weights affect inference.

To study inference, we use a linear representation for the specifications considered in the calculations. Consider a panel of N observations indexed by row $r \in \{1, \ldots, N\}$, partitioned into G mutually exclusive policy clusters (e.g., states) indexed by $g \in \{1, \ldots, G\}$. Let $y = (y_1, \ldots, y_N)'$ denote the $N \times 1$ vector of stacked outcomes, and let m index the estimator of interest, such as TWFE, Callaway–Sant'Anna (CS), Sun–Abraham (SA), stacked regression, or imputation.

Conditional on the panel structure, the policy adoption schedule and the other design inputs used to construct the weights, the fixed-weight specifications considered here can be written as a linear combination of the observed outcomes,

$$\hat{\theta}_m = c_m' y = \sum_{r=1}^{N} c_{m,r} y_r, \tag{2}$$

where $c_m = (c_{m,1}, \ldots, c_{m,N})'$ is an $N \times 1$ vector of estimator-specific weights, with scalar $c_{m,r}$ denoting the weight assigned to row r. The weights c_m are fixed conditional on the timing of adoption, unit and time fixed effects, cluster affiliations and the other inputs used in the calculation. This fixed-weight representation connects each specification to the distribution of information across clusters (see Borusyak et al., 2024).

The expectation of estimator m defines its statistical target,

$$\theta_m = c_m' \mathbb{E}[y] = \sum_{r=1}^{N} c_{m,r} \mathbb{E}[y_r], \tag{3}$$

which represents a specific weighted average of treatment effects when the identifying and support conditions for that estimator hold (Roth and Sant'Anna, 2023b; Rambachan and Roth, 2023). Let $u_r = y_r - \mathbb{E}[y_r]$ be the unobserved disturbance for row r. The estimation error $\hat{\theta}_m - \theta_m = c_m' u$ decomposes into a sum of G cluster-level scores:

$$\hat{\theta}_m - \theta_m = \sum_{g=1}^{G} S_g, \qquad S_g = \sum_{r \in g} c_{m,r} u_r, \tag{4}$$

where $r \in g$ denotes the subset of rows belonging to cluster g, and S_g is cluster g's unobserved contribution to the estimation error. If disturbances are independent across clusters, the scores are uncorrelated ($\mathbb{E}[S_g S_h] = 0$ for $g \neq h$), and the true sampling variance of $\hat{\theta}_m$ is simply the sum of the G cluster score variances:

$$\mathrm{Var}(\hat{\theta}_m) = \sum_{g=1}^{G} \mathbb{E}[S_g^2]. \tag{5}$$

For the fixed-weight score constructions considered here, the conventional cluster-robust variance estimator takes the following form,

$$\mathrm{CV}_1^m = \mathrm{ssc}_m \sum_{g=1}^{G} \hat{S}_g^2, \qquad \hat{S}_g = \sum_{r \in g} c_{m,r} \hat{e}_{m,r}, \tag{6}$$

where $\hat{e}_{m,r}$ denotes the residual or centered contribution used in the specified score construction,

$\hat{S}_g$ is the empirical score for cluster g, and ssc_m is the small-sample degrees-of-freedom adjustment factor (e.g., $\frac{G}{G-1}\frac{N-1}{N-K}$). This expression adapts the classic one-way cluster variance CV_1 of MacKinnon et al. (2023a) to estimator m.

The representation in (2) separates three considerations for inference. The variance is a sum of cluster-level terms under independence across clusters, and a large-sample approximation requires conditions on the distribution of those contributions as well as their number. The weights differ across estimators, so the same sample can provide different amounts of information for different estimates. Randomization inference instead uses a distribution of adoption dates under a specified assignment mechanism and a sharp null (Young, 2019). Estimators whose weights depend directly on realized outcomes, including synthetic-control weights fitted to pre-treatment outcomes, fall outside the fixed-weight representation. A least-squares adjustment for pre-treatment trends can be included when its regressors are fixed conditional on the design.

3.1.1 Effective Cluster Number Interval

Cluster-robust tests can over-reject when clusters are few, and that what matters is less their nominal count than their heterogeneity in size and leverage (e.g. Bertrand et al., 2004; Cameron et al., 2008; Conley and Taber, 2011; Cameron and Miller, 2015; MacKinnon and Webb, 2017; Djogbenou et al., 2019). Carter et al. (2017) formalize this insight by defining the "effective number of clusters," denoted G^*. Let $\gamma_g = \mathbb{E}[S_g^2]$ denote cluster g's contribution to the total sampling variance in (5). The effective count G^* is the inverse Herfindahl index of these variance shares:

$$G^* = \frac{\left(\sum_{g=1}^{G} \gamma_g\right)^2}{\sum_{g=1}^{G} \gamma_g^2}. \tag{7}$$

This quantity summarizes a feature relevant to the normal approximation for the t-statistic (see also Kojevnikov and Song, 2023; Chiang et al., 2026; MacKinnon et al., 2023a). In their simulations, size rises sharply "only when the feasible effective number of clusters falls below 10" (Carter et al., 2017).

In practice, calculating G^* is challenging because each cluster's variance contribution γ_g depends linearly on the unknown intra-cluster disturbance covariance matrix Ω_g. To circumvent this, Carter et al. (2017) propose a feasible measure, G^{*A}, which replaces Ω_g "with a matrix of 1's," the case of perfect within-cluster correlation ($\Omega_g = \iota\iota'$), and which they note "may be conservative for G^*". At the other extreme, MacKinnon et al. (2023c) interpolate to independence, $\Omega_g(\rho) = (1-\rho)I + \rho\,\iota\iota'$, and the count MacKinnon et al. (2023a) recommend reporting is the independence end ($\rho = 0$), $G^*(0)$, which corresponds to the inverse Herfindahl index of clusters' shares of partial leverage (a cluster-level analogue of the

effective sample size in Aronow and Samii, 2016).

Two features of the panel and the disturbance are particularly relevant when applying these measures to staggered designs. The first complication is a known consequence of fixed effects (MacKinnon et al., 2023c). When unit fixed effects are nested within clusters (e.g., firm fixed effects within states), the estimator's weights sum to zero within every cluster: $\sum_{r\in g} c_{m,r} = 0$. Consequently, a shock that is constant within each cluster over time has no contribution to the estimate. The equicorrelated component of γ_g vanishes, rendering Carter et al.'s feasible measure G^{*A} mathematically indeterminate (0/0). Under nesting, the limit of this feasible count as the equicorrelation approaches one is $G^*(0)$. If units relocate across clusters, G^{*A} is technically defined, but its value is driven entirely by the movers' rows (Appendix A.2). The feasible counts of 1.3 to 3.2 reported by Hu and Spamann (2020) for state-level corporate law panels use a perfect within-cluster correlation assumption and illustrate how the assessment depends on the assumed covariance structure.

A second feature is the cluster-by-period shock $\nu_{g(i)t}$ in (1). Unit and period effects remove what is constant for a unit and what is common to every cluster in a period. They leave $\nu_{g(i)t}$, and a fixed effect for the cluster and period would remove the treatment indicator with it. The shock is the group-level disturbance studied by Donald and Lang (2007) and Angrist and Pischke (2009), with the group a cluster and period. We examine how this shock affects the effective number of clusters for the estimators and designs studied here. Because γ_g is linear in the disturbance covariance, substituting the covariance implied by (1) into $\gamma_g = \mathbb{E}[S_g^2]$ yields (Appendix A.2):

$$\gamma_g(\lambda) \propto (1-\lambda)\sum_{r\in g} c_{m,r}^2 + \lambda \sum_t \left(\sum_{r\in(g,t)} c_{m,r}\right)^2, \qquad G^*(\lambda) = \frac{\left(\sum_g \gamma_g(\lambda)\right)^2}{\sum_g \gamma_g(\lambda)^2}, \tag{8}$$

where $\nu_{g(i)t}$ and μ_{it} have mean zero and variances σ_ν^2 and σ_μ^2, $r \in (g,t)$ indexes the rows of cluster g in period t, and $\lambda \equiv \sigma_\nu^2/(\sigma_\nu^2+\sigma_\mu^2) \in [0,1]$ denotes the share of total disturbance variance attributable to the cluster-by-period shock.

Under serially uncorrelated components, equation (8) separates each cluster's variance contribution into two channels. In the first, idiosyncratic channel, each observation's weight is squared individually before summing. At the boundary $\lambda = 0$, only idiosyncratic shocks operate, and $G^*(0)$ coincides with the partial-leverage count of MacKinnon et al. (2023a). In the second, common-shock channel, open whenever $\lambda > 0$, weights belonging to the same cluster and period are summed before being squared, because common shocks move units inside that cluster and period in tandem.

This distinction formalizes the aggregate regressor problem of Moulton (1990) at the level of the cluster and period. Because policy treatment is fixed within each cluster and period,

a common shock in a cluster and period of n observations with similar weights scales the variance contribution by approximately n^2, compared to a factor of n in the idiosyncratic channel (Hu and Spamann, 2020). Unlike time-invariant state effects, which unit fixed effects completely eliminate, shocks common to a cluster and period survive and propagate through this n^2 multiplier.

Which panel the regression is fit to decides whether λ matters, and the two applications in Section 5 are the two corresponding cases. Where the panel is already one observation per cluster and period, the two channels coincide and λ cannot move G^*: a common shock and an idiosyncratic error are the same object, so the variance share of $\nu_{g(i)t}$ is not an unknown. That is the state-year panel of Beck et al. (2010). Serial correlation can still widen its interval. Where the panel is finer than the cluster, many rows share a cluster and period, and the shift from the idiosyncratic channel to the common-shock channel happens at very small values of λ. That is the county-month panel of Carlin et al. (2023), clustered by state: the counties of a state-month share $\nu_{g(i)t}$, the design does not reveal its variance share, and G^* depends on the share. The transition is governed by the index $\psi \equiv \varphi\, n_{\text{eff}}$, where $\varphi = \lambda/(1-\lambda)$ and

$$n_{\text{eff}} = \frac{\sum_g \sum_t \left(\sum_{r\in(g,t)} c_{m,r}\right)^2}{\sum_g \sum_{r\in g} c_{m,r}^2} \tag{9}$$

denotes the effective number of observations inside the cluster and period. When the variance share of the common shock λ is of the order of $1/n_{\text{eff}}$, ψ approaches one and the aggregate common-shock and idiosyncratic components are of comparable magnitude.

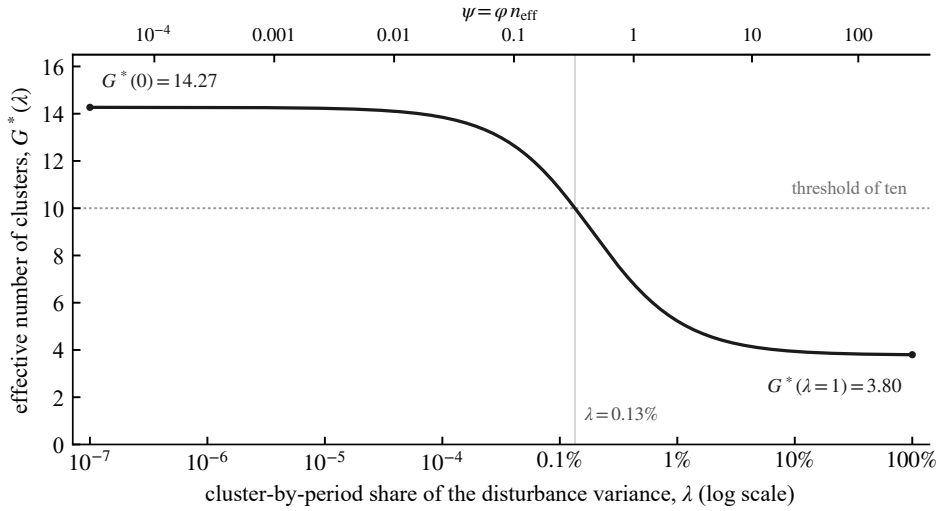
Conditional on the specified disturbance family, the design determines the trajectory $G^*(\lambda)$, while the unknown disturbance mix determines where the empirical panel lies along that path. The sample structure alone does not determine λ, so we summarize the resulting range with an effective cluster interval,

$$[G^*_{\text{lo}}, G^*_{\text{hi}}]\,, \tag{10}$$

computed over a grid of λ values spanning $[0, 1]$ and a grid of first-order serial correlation coefficients $\rho_e \in \{0, 0.25, 0.5, 0.75, 0.9, 0.99\}$ (or finer granularity) common to both error components. Because the grid contains $\rho_e = 0$, allowing serial correlation can only widen the interval (Appendix A.2). On a panel already one observation per cluster and period, serial correlation is the only source of the interval's width. The interval is computed from treatment timing, cluster sizes, the allocation of observations to cluster-periods, and the specified regressors and estimator weights.

Figure 3 illustrates how a small common shock can substantially reduce the effective

Figure 3: The effective number of clusters as the share of the common shock varies, on a synthetic design



Notes. $G^*(\lambda)$ of equation (8) for the two-way coefficient with unit and period effects, against the share of the common shock λ of the disturbance variance (bottom axis) and $\psi = \varphi\, n_{\text{eff}}$ (top axis); both components serially uncorrelated. 50 clusters, 20 periods, 3,999 units; the largest cluster holds 25% of them and the others' sizes decay geometrically from 361 to 1; adoption dates lie between periods 6 and 15, and 12 clusters never adopt. The left end, $\lambda = 10^{-7}$, equals $G^*(0)$ to two decimals. Dotted line: the threshold of ten that Carter et al. (2017) report for the feasible count G^{*A}, used here as a reference.

number of clusters in a synthetic design. Two counts that can be computed from the sample provide useful reference points: the mass-based count G_{eff}, which ignores the adoption schedule, is 10.15, and the leverage-based count $G^*(0)$, the independence end of the path, is 14.27, both clearing Carter et al.'s threshold of 10. Yet as λ increases from zero, $G^*(\lambda)$ plummets along the curve, crossing below 10 when the common shock accounts for a mere 0.13% of the total disturbance variance, with the fall centered where $\psi \approx 1$ ($n_{\text{eff}} = 292$, so $\lambda \approx 0.34\%$), and ultimately leveling off at 3.80. Against the threshold that Section 4.1 calibrates for the two-way regression on a Compustat panel, $\tau_{\text{TWFE}} = 13.92$, the mass-based count no longer clears, and the curve falls below 13.92 at a still smaller share of the common shock. The count $G^*(0)$ recommended by MacKinnon et al. (2023a) describes the independence case, which is the upper end of this particular curve. The curve shows how the count can change under other disturbances in the family.

Finally, one might consider eliminating λ by collapsing the micro data to one observation per cluster and period, the classic remedy for Moulton-type bias (Moulton, 1986; Donald and Lang, 2007; Angrist and Pischke, 2009). In staggered designs, this strategy does not resolve the interval. When the regressors are constant within each cluster and period, a collapsed regression weighted by the number of observations in each cell reproduces the micro-level point estimate. In this setting, weighting transforms the common shock ν_{gt} into

heteroskedasticity across cluster-periods of different sizes, so that both channels and the resulting interval carry over exactly (Appendix A.3). An unweighted collapse generally has different weights and, under heterogeneous effects, may target a different estimand; equal cell sizes and regressors constant within cells are sufficient for equivalence (MacKinnon and Webb, 2017, online Appendix A.9). The observation level and the variation of regressors within cluster-periods determine how aggregation relates to the original estimator.

3.1.2 Thresholds and Minimum Detectable Effects

A natural question is whether one could simply estimate the variance share λ from sample residuals and evaluate G^* at that single point estimate $\hat{\lambda}$. Its usefulness depends on the precision with which the relevant variance components can be estimated and on the disturbance model. In micro-level panels, the effective number of observations inside each cluster and period, n_{eff}, can run to the hundreds (292 in the design of Figure 3). As a result, the effective cluster count $G^*(\lambda)$ is hyper-sensitive near the origin: as shown above, a common shock accounting for about $1/n_{\text{eff}}$ of the total variance—a fraction of one percent—is enough to move the count from well above ten to a handful. To determine whether a panel clears its estimator's threshold τ_m (Section 4.1) or falls below it, an empirical estimate $\hat{\lambda}$ would need to achieve sampling precision on the order of $1/n_{\text{eff}}$.

Variance-component estimators cannot be counted on for such precision. At shares this close to zero, the sampling error of ANOVA-type or residual-based estimates can be of the same order as the share itself, moment estimates can turn negative, and every such estimate presumes that the two components of (1) describe the disturbance. Because the uncertainty around $\hat{\lambda}$ can span the transition region of $G^*(\lambda)$, a plug-in estimate can be sensitive to the variance model and sampling error. Using a residual-based estimate of λ would also make the classification depend on the estimated disturbance model.[2] We therefore do not attempt to pinpoint λ directly, treating G^* as an interval $[G^*_{\text{lo}}, G^*_{\text{hi}}]$ and asking instead whether any empirical conclusion actually depends on where the panel lies within it.

If an empirical decision is invariant across the entire interval, locating where the panel sits along the curve is unnecessary. We formalize this through two distinct invariance checks: Routing invariance asks whether the design routes to the same inference procedure at both endpoints of the interval, evaluated against the estimator's own threshold τ_m (Section 4.1).

[2]The degrees-of-freedom corrections for the bias-reduced estimator CV_2 of Bell and McCaffrey (2002), which Imbens and Kolesár (2016) recommend computing routinely, either fix the disturbance or estimate it. Bell and McCaffrey (2002) assume a disturbance covariance proportional to the identity (MacKinnon et al., 2023a), the single point $\lambda = \rho_e = 0$ of (1); Imbens and Kolesár (2016) estimate a random-effects correlation from the residuals, an approach that MacKinnon et al. (2023a) call inapplicable once cluster fixed effects are partialled out, which unit fixed effects nested in clusters do implicitly.

Verdict invariance asks whether the 5% statistical significance of the cluster-robust t-statistic changes when evaluated against a Student's t critical value with $G^* - 1$ degrees of freedom across the interval. Because t critical values decrease monotonically as degrees of freedom rise, a rejection decision can flip at most once—at the unique threshold G^*_{flip} where $t_{G^*_{\text{flip}}-1,\,0.975} = |t|$. The two endpoint decisions therefore determine whether the verdict is invariant across the interval, with a boundary value treated according to the stated rejection rule. Such a point exists only when $|t|$ exceeds the normal critical value 1.96, the limit of $t_{\text{df},0.975}$ as the degrees of freedom grow; otherwise the test rejects at no degrees of freedom and the verdict is invariant. While referencing test statistics to t_{G^*-1} has precedents (e.g., Kott, 1994; Carter et al., 2017), we do not propose t_{G^*-1} as a valid inference procedure (Hu and Spamann, 2020, a practice rejected for the feasible count by); rather, we use it solely as a diagnostic to assess whether an empirical finding hinges on unobserved disturbance properties.

The minimum detectable effect (MDE) connects the precision of an estimate to the magnitude of the policy effect a study can detect. At a five percent significance level, an MDE for 95 percent power is an effect for which the test rejects a zero effect in 95 percent of repeated samples under the specified model. The effect needed to achieve this probability depends on the sample, the estimator and the inference method. A standard approximation expresses these features at significance level α and target power π as

$$\text{MDE} = c\Big(G^*_{\text{used}} - 1\Big) \cdot \sigma\sqrt{\kappa} \cdot \mathcal{A}^{-1/2} \cdot \rho_{\mathcal{A}}^{-1/2}, \qquad c(\text{df}) = t_{\text{df},1-\alpha/2} + t_{\text{df},\pi}, \tag{11}$$

where the sum of critical values gives an approximate power calculation, $c(\text{df})$ is the critical value factor for degrees of freedom df, σ is the disturbance standard deviation, and κ accounts for variance inflation due to intra-cluster dependence. The simulation comparison uses a common reference effect based on this approximation with the observed number of clusters in the degrees of freedom. The application tables also report a separate MDE for each inference method, calculated from its simulated rejection frequencies under the same calibrated disturbances.

An MDE describes detection across repeated samples, while the outcome of a particular test depends on the estimate and uncertainty observed in that sample. A realized estimate smaller in magnitude than the MDE can therefore be statistically significant, and an estimate larger than the MDE can be insignificant. Comparing an estimate with the MDE places its magnitude alongside the precision of the design under the specified model, while confidence intervals describe the range of effects compatible with the observed data under the assumptions of the corresponding inference method. Economically meaningful effects specified independently of the estimate provide a further reference for assessing power (Hu and Spamann, 2020; Abadie,

2020; Weiss, 2026).

The remaining factors in (11) isolate the adoption structure, the distribution of cluster sizes, the number of observations inside each cluster and period of the panel and the estimator-specific weights. Specifically, $\mathcal{A} = \|\ddot{x}\|^2$ is the identifying variation remaining after partialling out fixed effects and controls (where $\ddot{x}$ is the residualized treatment), $\rho_{\mathcal{A}} \in (0, 1]$ is the share of $\mathcal{A}$ generated by observations that directly target the causal estimand, and G^*_{used} is the effective cluster count of those informative rows. For a general linear estimator m, the effective identifying variation $\mathcal{A}_m \equiv \|c_m\|^{-2}$ replaces the product $\rho_{\mathcal{A}}\mathcal{A}$.

In this decomposition, $\rho_{\mathcal{A}}^{-1/2}$ summarizes the effect of restricting the measure of identifying variation to the rows used for the stated target. In a balanced panel with unit and period fixed effects, units treated in every observed period have residualized treatment $\ddot{x}_{it} = \bar{D} - \bar{D}_t$, which is non-zero unless the aggregate adoption path $\bar{D}_t$ is completely flat. In standard two-way fixed effects regressions, these "always-treated" units serve as effective comparison groups (Goodman-Bacon, 2021); they generate mechanical identifying variation in $\mathcal{A}$ while diluting the share $\rho_{\mathcal{A}}$ that actually isolates newly treated units. The diagnostic parameter $\rho_{\mathcal{A}}$ summarizes this composition of the residualized treatment under the specified design.

3.2 Heterogeneity-Robust Estimators: Leverage Concentration and Variance Blindness

Heterogeneity-robust estimators use comparisons designed to identify interpretable effects when treatment effects vary across cohorts or time. Their comparisons can also change the distribution of information across clusters, which makes inference depend on the structure of the sample as well as the estimator. Baker et al. (2026) show how the choice between never-treated and not-yet-treated comparison groups relates to the identifying assumptions and the information used for estimation. In a reanalysis of published economics studies, Hull (2026) reports differences in the precision of conventional and heterogeneity-robust estimates. Weiss (2026) finds that coverage and power of modern DiD estimators vary with sample size and treatment timing, while Karim et al. (2026) document substantial over-rejection for CS when treated clusters are few. These findings motivate an assessment of the weights and cluster scores associated with each estimate, since the effect of changing the comparison group depends on the design.

The linear representation $\hat{\theta}_m = c_m' y$ from (2) shows how the choice of estimator changes the weights used in the diagnostics. Because each estimator m constructs its own weight

vector c_m, each estimator induces its own cluster-level partial leverage:

$$L_g^m = \frac{\|c_{m,g}\|^2}{\|c_m\|^2}, \qquad \sum_{g=1}^{G} L_g^m = 1, \tag{12}$$

where $c_{m,g}$ denotes the subvector of weights assigned to rows in cluster g. Substituting these estimator-specific weights into (8) yields an estimator-specific effective cluster interval, $[G^*_{m,\text{lo}}, G^*_{m,\text{hi}}]$. At the independence baseline ($\lambda = 0$), the endpoint $G^*_m(0) = 1/\sum_g (L_g^m)^2$ is precisely the leverage diagnostic proposed by Hull (2026); Borusyak et al. (2024, Assumption 6, pp. 3269–70) build a related effective sample size from the imputation estimator's own weights, summed by cluster under the worst case of correlation within it, and require only that it grow.

The estimator-specific weights show how the choice of comparisons can concentrate information. When the last-treated cohort supplies the comparison group in a design without never-treated units, the clusters in that cohort can enter many contrasts (Sun and Abraham, 2021; Weiss, 2026). Their contributions can consequently account for a large share of the estimator's variance, and the effective cluster interval can differ substantially from that of a two-way regression. The extent of this concentration depends on the adoption schedule, the sample, and the target of the estimate.

A second feature concerns variance contributions that the effective cluster interval alone does not describe. Estimators such as Callaway and Sant'Anna (2021), Sun and Abraham (2021), and Borusyak et al. (2024) can aggregate cohort-specific effects with fixed weights, comparing a treated cohort with a comparison group in each period. For the score constructions studied here, conventional cluster-robust variance estimation uses within-group deviations to measure each cluster's contribution.

The construction is easiest to see for a treatment cohort. When the cohort contains several states, the group-specific score uses the deviation of each state's contribution from the cohort average. When only one state belongs to the cohort, its contribution is also the cohort average, and its within-group deviation is zero. The same logic applies to a comparison group containing a single cluster. A shock to that cluster-period can therefore affect $\hat{\theta}_m$ while the corresponding group-specific score contribution is zero (for a single-cluster treated cohort in the saturated specification of Sun and Abraham (2021) without controls, $\hat{S}_g = 0$ exactly; Appendix A.4). The corresponding group-specific score does not capture that contribution. A cluster may also enter other comparisons, so the aggregate variance estimate depends on the full set of scores. This phenomenon manifests the classic "single treated cluster" problem of Conley and Taber (2011) and MacKinnon and Webb (2017), but recurring cohort by cohort.[3]

[3] Borusyak et al. (2024) discuss the treated-side case in which the leave-out cluster variance is undefined

For the fixed-weight group score considered here under independent cluster-period disturbances, the relative contribution of a cluster-period shock to the expected variance estimate is described by $f_{gt} \geq 0$, which reflects group-level leverage,

$$f_{gt} = 1 - 2w_{gt} + \sum_h w_{ht}^2, \qquad b_m(\lambda) = \frac{\sum_{(g,t):\, f_{gt}=0} v_{gt}(\lambda)}{\sum_{g,t} v_{gt}(\lambda)}, \qquad \bar{b}_m(\lambda) = \frac{\sum_{g,t}(1 - f_{gt})\, v_{gt}(\lambda)}{\sum_{g,t} v_{gt}(\lambda)}, \tag{13}$$

where w_{ht} is cluster h's share of its group's weight in period t, and $v_{gt}(\lambda) = (1-\lambda)\sum_{r\in(g,t)} c_{m,r}^2 + \lambda\left(\sum_{r\in(g,t)} c_{m,r}\right)^2$ is the variance term for that cluster and period from (8). In a group of k equally weighted clusters, $f = 1-1/k$, which is precisely the factor by which the cluster-robust variance estimator understates a treated cluster's contribution with i.i.d. errors (MacKinnon and Webb, 2017). When $k = 1$, $w_{gt} = 1$ and $f_{gt} = 0$; when another cluster dominates the group, f_{gt} can exceed one.

The metric $b_m(\lambda)$ is the blind share. It measures the fraction of modeled variance associated with single-cluster group contributions that are absent from the corresponding group-specific scores. A larger value indicates that the estimate relies more heavily on contributions exposed to this problem. Because a cluster can enter several comparisons, its contribution can also appear in other group scores, so the blind share need not equal the fraction missing from the aggregate variance estimate. The measure $\bar{b}_m(\lambda)$ extends the calculation to the attenuation associated with groups of different sizes. The effective cluster interval describes the concentration of variance across clusters, whereas these measures describe how the score construction represents their contributions (Appendix A.4).

In the saturated regression construction studied for Sun and Abraham (2021), a leverage correction does not necessarily recover the missing variance contribution. The bias-reduced estimator CV_2 of Bell and McCaffrey (2002) rescales each cluster's residuals by $M_{gg}^{-1/2}$, the inverse symmetric square root of that cluster's diagonal block of the residual-maker matrix (MacKinnon et al., 2023a); in a group of $k \geq 2$ equally weighted clusters with i.i.d. errors, this removes the factor $1 - 1/k$ exactly. In the saturated specifications analyzed in Appendix A.4, when a cohort or reference group is a single cluster, the relevant cluster block M_{gg} is singular even after the unit fixed effects are partialled out, because the singularity arises from parameters that this cluster alone identifies (Appendix A.4). The original CV_2 then cannot be computed, as MacKinnon et al. (2023b) note for a single treated cluster. The generalization of Pustejovsky and Tipton (2018), which replaces the inverse with a Moore–Penrose inverse, can be computed, but its unbiasedness result assumes that the coefficients remain identified when any one cluster is dropped (their Theorem 1). That condition fails

because one cluster carries the weight of a cohort-by-period group.

here, and the generalized estimator is then as blind to these shocks as CV_1^m.

A single-cluster cohort need not be invisible to every inferential procedure, since an influence function can include the contribution from estimated aggregation weights and a cluster jackknife changes the data used to construct the variance. The leave-one-cluster calculation can also change the composition and weights of the aggregate, as discussed for CS by Karim et al. (2026). Its interpretation therefore depends on how the reduced samples relate to the original target. For Sun–Abraham in both applications, the original aggregate loses identifying support after some state deletions, so a reaggregated jackknife cannot be interpreted as a fixed-target variance for that aggregate. Appendix A.7 examines these cases.

The diagnostics also extend to the switching estimator of de Chaisemartin and D'Haultfœuille (2020) and the two-stage estimator of Gardner (2022). The switching calculation in Appendix B.5 averages effects in the first treated period using changes for units that remain untreated, while the two-stage calculation estimates unit and time effects from untreated observations and can average adjusted outcomes over different treated periods. Other estimators considered here also permit treatment effects to vary across horizons, and comparisons across methods therefore require attention to the target periods as well as the comparison groups and score construction.

3.3 Small-Sample Procedures

When the effective cluster interval and the calibrated threshold indicate that conventional clustered inference may be sensitive to the design, researchers can consider procedures developed for settings with limited or unequal cluster information (MacKinnon et al., 2023a). These methods have different requirements concerning the distribution of disturbances, the treatment assignment and the target effect. Hu and Spamann (2020) discuss such requirements in state-of-incorporation designs. Table A.1 in Appendix A.1 summarizes the assumptions and interpretation of the procedures considered here.

Karim et al. (2026) propose cluster jackknife inference for CS and find that it improves rejection frequencies relative to the influence-function and multiplier-bootstrap calculations in their simulations. The calculation re-estimates the effect after excluding each cluster, linking the resulting variance estimate to the sensitivity of the estimate to individual clusters. Weiss (2026) examines a nonparametric block bootstrap that resamples entire cluster histories and finds improved coverage for several modern DiD estimators in the simulated designs. These approaches use different resampling constructions and provide complementary perspectives on inference in staggered panels.

Consider first the score bootstrap family. An unrestricted multiplier bootstrap based on fixed-weight cluster scores cannot restore a variance component that is absent from every

score, although it can change the reference distribution of the statistic. The classic restricted wild cluster bootstrap uses a different set of scores, while the WCR-S and WCR-B procedures of MacKinnon et al. (2023b) use jackknife-transformed restricted scores and differ in their studentization. The restricted leave-one-cluster calculations required for these procedures must be checked for the target and design under study, and the simulated comparisons in Table 2 describe the regression specifications for which they are available.

Design-based procedures instead use assumptions about the treatment assignment mechanism. Under a sharp null and a correctly specified conditional assignment distribution, randomization inference (RI) provides a finite-sample test without relying on a cluster-level central limit theorem (Young, 2019). Its validity for a heterogeneous average-effect null requires additional conditions. Design-based standard errors also rely on assumptions about treatment assignment (Aronow and Samii, 2017; Athey and Imbens, 2022; Roth and Sant'Anna, 2023a).

The role of treatment assignment also distinguishes the setting here from the cross-sectional procedures of Abadie et al. (2023). Their causal cluster variance and two-stage cluster bootstrap use within-cluster treatment variation to estimate cluster-level effects and improve inference when a substantial fraction of clusters is observed. At a given date all units in a policy state share the same treatment status in the designs studied here, while staggered adoption supplies variation over time that calls for a separate panel analysis. The conditions for those cross-sectional procedures therefore do not follow solely from the presence of many observations within a state.

A second class of remedies aggregates clusters into q coarse groups: the approximate randomization test (ART) of Canay et al. (2017), which changes the signs of group-level estimates, and the t-test of Ibragimov and Müller (2010) (IM) applied to the same group estimates. While both methods permit arbitrary error dependence within groups, they encounter two severe limitations in staggered designs: First, they can change the target of inference. With cluster-level assignment and period fixed effects, the treatment effect cannot be identified within a single cluster (Cai et al., 2023). Their proposed Remedy A refits fixed effects within each multi-cluster group, and the resulting ART and IM tests evaluate $\bar{\theta}_j$, the unweighted average of the within-group two-way fixed effects coefficients, which generally differs from the pooled two-way coefficient.

Second, coarsening clusters partitions the effective cluster information available in the design across groups. Let L_g denote cluster g's share of partial leverage, $W_j = \sum_{g \in J_j} L_g$ be group j's leverage share, and $G_j^* = W_j^2 / \sum_{g \in J_j} L_g^2$ denote group j's internal effective cluster

count. A direct variance accounting identity reveals:

$$\sum_{j=1}^{q} \frac{W_j^2}{G_j^*} = \sum_{g=1}^{G} L_g^2 = \frac{1}{G^*} \qquad \Rightarrow \qquad \min_j G_j^* \leq G^* \sum_{j=1}^{q} W_j^2 \leq G^* \max_j W_j. \tag{14}$$

Under equal group leverage shares ($W_j = 1/q$), equation (14) simplifies to $\min_j G_j^* \leq G^*/q$. Because the smallest p-value ART can attain with q groups is 2^{1-q}, a test at the 5% level can reject at all only if $2^{1-q} \leq 0.05$, that is, $q \geq 6$ (Cai et al., 2023); requiring each group to reach the illustrative reference of ten effective clusters would then require, with equal shares, an aggregate effective cluster count of at least 60.

In the two applications, the upper end of every group's effective cluster interval is below ten when six groups are formed, with the largest values equal to 8.69 and 6.80 (Appendix A.5). We report the grouped calculations as supplementary evidence in the applications, where they assess the average of the group-specific estimates.

Finally, small-sample procedures can fail simply because the comparison group runs out. When a design lacks never-treated clusters, the two-sample cluster-level t-test of Ibragimov and Müller (2016)—which requires at least two treated and two never-treated clusters—cannot be computed. Furthermore, once all clusters have adopted the policy, no untreated comparisons exist, forcing modern robust estimators to truncate their point-identified region at the final adoption date (Callaway and Sant'Anna, 2021; Sun and Abraham, 2021; Borusyak et al., 2024). When an aggregate assigns positive weight to periods after all clusters have adopted, parallel trends alone leaves the common untreated evolution in those periods unrestricted under the conditions in Appendix A.6, which prevents point identification of that aggregate. Bounding effects past the last adoption date requires imposing explicit restrictions on the post-treatment evolution of the counterfactual path, in the spirit of Rambachan and Roth (2023).

3.4 Using the Diagnostics to Assess Inference

The effective cluster interval and the blind-share diagnostic provide complementary information for assessing the inference attached to an estimate. We illustrate their use by comparing each estimator's interval with a threshold calibrated for the corresponding design and disturbance family, and by recording whether the score construction has a blind share. The resulting classes organize the calculations reported in the simulations and applications, while the validity of each procedure continues to depend on its assumptions and on the target and null under study.

For class W, the effective cluster interval lies at or above the estimator's calibrated

threshold at both endpoints and the stated score construction has no blind share. We report the conventional cluster-robust Wald test as the main calculation and use the cluster jackknife as a sensitivity analysis, subject to identification and preservation of the original target after each state deletion. The classification describes the calibration designs and their disturbances, and it does not by itself guarantee the validity of either procedure for another sample.

For class F, the effective cluster interval lies below the calibrated threshold at both endpoints, and we report studentized randomization inference using adoption dates drawn under an explicit assignment mechanism with the dominant cluster's date held fixed. The calculation tests a sharp null that specifies effects for all relevant clusters and periods, and inversion of constant-effect sharp nulls describes a common effect under that model. If the class varies across admissible assignments, the reference distribution must also condition on the selected class as described in Appendix A.10.

For class U, the interval straddles the threshold or clears it with a positive blind share, and we report the conventional Wald test, the cluster jackknife when the fixed target is identified after deletion, and randomization inference. The illustrative combined calculation records a rejection when all three reject, and its interpretation depends on a common null and the assumptions supporting at least one component under the same conditional probability model. Appendix A.10 gives the conditional argument and distinguishes the sharp null from a heterogeneous average-effect null.

Each event-study horizon has its own estimate and effective cluster interval, which may differ from those of the pooled coefficient. In the reported event studies, the horizon classifications use the estimator-specific interval and threshold. The cohort-based horizons lie below their thresholds, while the pooled regressions are classified using their intervals under the convention for the single-group diagnostic. We also report maximum statistics over the pre-period horizons using Gaussian and randomization reference distributions. The randomization reference distribution is constructed under a global sharp null of no treatment effect, so it does not by itself provide a test of a pre-trend null that permits unrestricted post-treatment effects. Appendix A.11 describes the calculation and its interpretation.

3.5 Discussions

A causal claim from a staggered design requires an interpretable treatment effect and an assessment of uncertainty around its estimate. The estimator literature studies the comparisons and assumptions needed for the first (Roth et al., 2023), while the analysis here examines how the sample and the estimator shape the second. When parallel trends is in doubt, sensitivity analysis addresses a separate aspect of the research design (Rambachan

and Roth, 2023; Roth and Sant'Anna, 2023b).

The randomization test and the clustered Wald tests address related hypotheses that need not coincide. A sharp null specifies the treatment effect for each cluster and period, while a null about an average effect allows effects to differ and offset one another. Randomization inference is exact for the sharp null under the stated assignment mechanism, and a studentized statistic does not by itself extend that finite-sample result to a null about an average. A rejection of a constant effect equal to a published estimate therefore challenges that constant-effect schedule without establishing that the average effect differs from the published estimate. The two interpretations coincide under a common-effect model, while inference on heterogeneous average effects requires a separate justification. Appendix A.8 gives the formal argument.

The validity of the design-adapted calculation also depends on the assumptions supporting its component tests and on the selection of the inference class. For class F, the randomization test controls rejection of the sharp null when its reference distribution matches the assignment mechanism conditional on the information used to select that class. For class U, requiring all three procedures to reject controls rejection of a common null whenever at least one component is valid for that null under the same conditional probability model. The threshold and the requirement of agreement do not by themselves establish validity for a heterogeneous average effect, and the finite-sample simulations describe performance on the designs and disturbances examined. Appendix A.10 derives these results.

The interval records how the effective cluster count varies over the disturbance family considered, and the classification relates that interval to the calibrated threshold and the blind share. Widening that family can change the class assigned to an estimate, while the calibrated thresholds summarize the designs used to construct them. The classification therefore organizes the comparison of inference methods in relation to the sample, the estimator and the disturbance family.

The wild cluster bootstrap is reported alongside the classified calculations because its performance depends on the score construction and the allocation of information across clusters. The classic restricted bootstrap and its transformed-score variants can behave differently in the same design, while neither the calibrated threshold nor the bootstrap comparison establishes validity for all disturbance processes. The randomization calculation rests on a separate assumption about the assignment of adoption dates.

4 Estimator-Specific Thresholds and Monte Carlo Evidence

The preceding analysis identifies several ways in which the distribution of information and the construction of an inference procedure can affect empirical conclusions. We first use exact

calculations on Compustat-based designs to examine the relationship between the effective cluster interval and the size of the conventional Wald test. We then study rejection rates for conventional clustering, the cluster jackknife, randomization inference, and available wild cluster bootstrap variants on the sample structures and adoption schedules of two published finance studies. These exercises describe performance under the designs and disturbances specified in the calculations.

4.1 Estimator-Specific Thresholds

To locate the lower end beyond which the cluster-robust Wald test stays within one percentage point of its nominal size, we examine a balanced benchmark panel of 3,290 nonfinancial U.S. Compustat firms observed over the ten-year period from 1986 to 1995 (Appendix B.1). Across these 49 states of incorporation, Delaware accounts for roughly half of all firms, mirroring the extreme cluster imbalance commonly encountered in corporate finance research.

Within this empirical framework, we examine policy adoption across three structural design families that introduce varying degrees of leverage concentration. The first family divides the firms into an expanding number of equal-sized clusters, varying the group count from 25 to 250, with half of the clusters adopting in four successive cohorts and the rest never treated. The second family fixes the total number of clusters at 50, but intensifies identifying leverage by expanding the share of firms in a single treated state from 5 percent to 50 percent. The third family keeps the 49 incorporation states, once at their empirical sizes and otherwise with one state holding between 3 and 90 percent of the firms and the other 48 sharing the rest equally; that state adopts in the first cohort or is never treated.

Throughout these designs, the outcome follows (1) with a true effect of zero: a common state-by-year shock alongside an idiosyncratic firm-year error, both Gaussian and serially uncorrelated, with the common shock's share of total disturbance variance between zero and 75 percent (Appendix B.1 lists the shares computed for each design). Because the error components are Gaussian and all five estimators are linear in outcomes, the cluster-robust Wald statistic can be represented as a ratio of quadratic forms in normal variables. We compute the exact finite-sample rejection probability of CV_1^m under the true null by numerical inversion (Imhof, 1961). This yields exact tail probabilities across 66 distinct designs for each estimator, free of Monte Carlo sampling noise.

We define each estimator's threshold as the lower end of the effective cluster interval beyond which the exact size stays within one percentage point of the nominal level. For each design, we report the least favorable of the common-shock shares computed for it, the share at which the common shock produces the highest false rejection rate. A design is classified as failing size control when this worst-case rejection probability exceeds six percent,

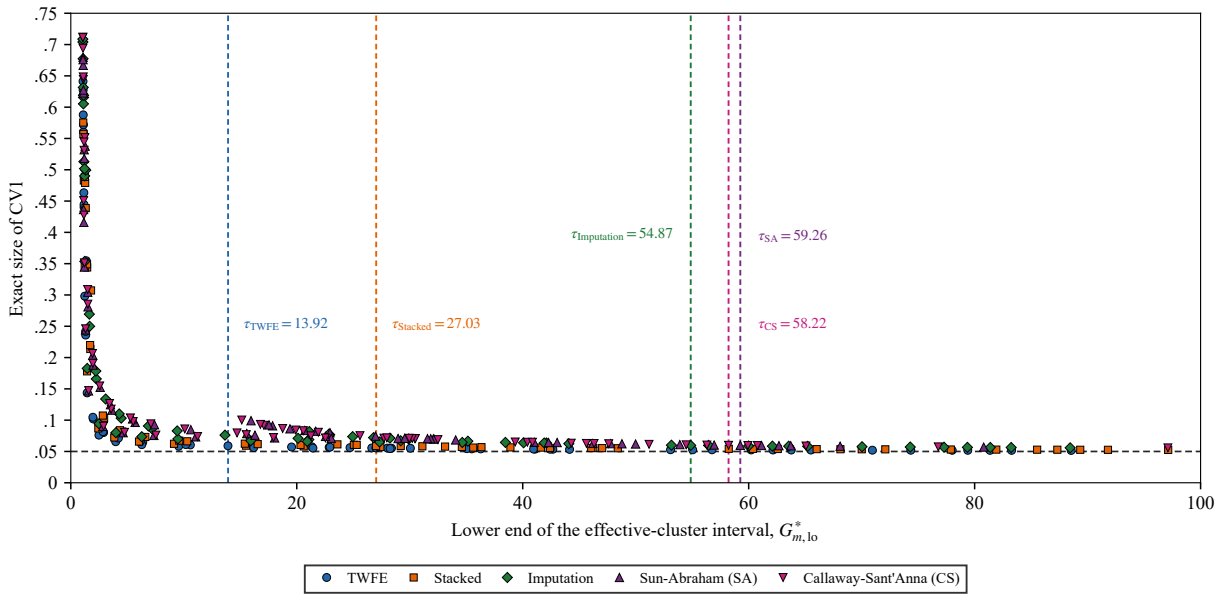
one percentage point above the nominal level. The calibrated threshold is then the smallest lower end among passing designs that strictly exceeds the lower ends of all failing designs, rounded up to the next hundredth. The lower end on the horizontal axis is the interval of Section 3.1, computed over the disturbance-mix and serial-correlation grids, a wider family than the shares at which the size is computed; the size calculations concern only the shares and serial dependence specified in the calibration. The lower end summarizes the smallest effective cluster count in the specified family. It need not correspond to the disturbance with the largest rejection probability, since size also depends on features not captured by that count.

Figure 4 plots these worst-case exact null rejection probabilities against the lower end of the effective cluster interval for all five estimators. Across all methods, exact test size reaches between 57 and 71 percent when effective clusters are few, and settles toward the nominal five percent benchmark as the lower end rises.

Crucially, the lower end at which the worst exact size comes within one percentage point of nominal differs sharply across estimators. The traditional two-way fixed effects estimator gets there first, once the lower end of its interval reaches about 14 effective clusters. The stacked regression needs about 27 effective clusters, while the three cohort-based robust estimators—Callaway and Sant'Anna, Sun and Abraham, and the imputation estimator—need about 55 to 60.

The designs that set the thresholds differ in kind, in the way Section 3.2 leads one to expect. The last failures of the two-way regression are designs in which one treated state holds 8 to 10 percent of the firms; its threshold is set by how concentrated the leverage is. For Callaway–Sant'Anna, Sun–Abraham and the imputation estimator the last failures are equal-group designs with 88, 85 and 60 groups. Every cohort there has at least two clusters, so $b_m = 0$, and each treated cohort has 11, 10 to 11, and 7 to 8 clusters. In a cohort of k equally weighted clusters, the group-mean deviation through which a cluster enters CV_1^m removes a share $1/k$ of each treated cluster's contribution ($f = 1 - 1/k$ in equation (13)). In these designs, the number of clusters in each cohort affects the conventional scores in addition to the overall effective cluster count, and the corresponding calibrated thresholds are higher. The importance of treated-cluster and cohort size is also evident in the simulations of Karim et al. (2026) for CS and Weiss (2026) for modern DiD estimators, including SA in her Appendix F.6. Our thresholds describe the relationship between these features and the effective cluster interval in the calibration designs. The stacked regression's threshold lies between the two. Its coefficient is pooled across stacks, which do not demean residuals within cohort-by-period groups (Section 3.2), so this account does not cover it, and its threshold is best read as an empirical regularity of this panel.

Figure 4: Exact size of CV_1^m and the lower end of the effective-cluster interval



Note 1: Each marker is one design in the Compustat calibration designs of Section 4.1. The horizontal axis is the lower end of the effective-cluster interval and the vertical axis is that design's largest exact rejection probability of the cluster-robust Wald variance CV_1^m across the state-year shock shares computed for it. Five series of markers are drawn, one per estimator. The dashed horizontal line is 0.05.
Note 2: A design fails when that probability exceeds 0.06. The vertical lines are the thresholds τ_m: for each estimator, the smallest lower end above the lower end of every failing design, rounded up to the next hundredth. They are 13.92 for TWFE, 27.03 for the stacked regression, 54.87 for the imputation estimator, 58.22 for Callaway–Sant'Anna and 59.26 for Sun–Abraham. Appendix B.1 describes the designs.

The differences across estimators show the limits of applying a common reference value to these calibration designs. The value of ten reported by Carter et al. (2017) refers to their feasible count and to a different setting, and several designs in our calculations have rejection probabilities above six percent at larger values of the lower end. A higher value suitable for one estimator can also classify designs using another estimator below the reference even when their calculated rejection probabilities are within one percentage point of nominal. The estimator-specific thresholds summarize the calibration designs and their disturbance shares, and their interpretation outside these designs requires separate assessment.

4.2 Performance in Published Finance Applications: Simulation Evidence

We next examine inference on the sample structures and adoption schedules of Beck et al. (2010) and Carlin et al. (2023). For each estimator, the classification compares the effective cluster interval with a threshold calibrated on designs based on the corresponding application (Appendix B.2). We hold the sample, adoption schedule and estimator specification fixed and draw disturbances over 1,000 replications. The bank deregulation panel contains one observation per state and year, while the elder protection panel contains multiple counties per

state and month. For the latter we consider a common state-month shock that accounts for 75 percent of the disturbance variance, so that both the method comparisons and the application power calculations reflect dependence within states. The scale of the disturbances is calibrated to the conventional variance estimated for each application and estimator. Randomization inference uses 199 assignments with the dominant cluster's adoption date held fixed, and the available restricted wild cluster bootstrap calculations use 999 state-level sign draws.

We examine inference under a true zero effect and under a homogeneous reference effect. Within each application and estimator, the reference effect is the same for all available methods and is constructed using a t approximation based on the number of clusters and a 95 percent power target. Its value therefore provides a common alternative for comparing rejection frequencies, though a particular method need not attain 95 percent power at that value. We also calculate the 95 percent minimum detectable effect separately for each method under the same calibrated disturbance setting. The reference effect and the method-specific minimum detectable effects are reported with the application design features, while the corresponding null and alternative rejection frequencies appear in Table 2.

The comparison of rejection frequencies under the alternative is informative alongside the rejection frequencies under the null. The conventional Wald test rejects a true zero between 4.0 and 39.8 percent of the time across the ten estimator and application combinations, while the design-adapted calculation ranges from 3.6 to 5.8 percent. The methods do not generally have the same actual null rejection rate, so a higher rejection frequency under the alternative cannot by itself establish greater power at a common size. The original Sun–Abraham aggregate loses identifying support after some state deletions, so its fixed-target cluster jackknife is not reported. Randomization inference tests a sharp null under its stated assignment mechanism, and the assignment is held fixed in the disturbance simulations.

The bootstrap results distinguish the classic restricted procedure from the transformed-score variants where the latter can be calculated for the original regression. For the bank deregulation stacked regression, the null rejection frequencies for WCR-C, WCR-S and WCR-B are 0.065, 0.055 and 0.059, respectively. For the elder protection stacked regression, the corresponding frequencies are 0.046, 0.046 and 0.048. For Sun–Abraham, WCR-C rejects a true zero with frequencies of 0.000 and 0.173 in the two designs, while the transformed-score variants are not reported for the original aggregate after it loses support in some state deletions. The minimum detectable effects describe each method's rejection behavior in the same calibrated model and should be read together with the null frequencies.

The choice of estimator also matters for inference in the switching estimator of de Chaisemartin and D'Haultfœuille (2020) and the two-stage DiD estimator of Gardner (2022). Table B.1 reports calculations for both application designs using balanced samples without

covariates. The switching estimate averages effects in the first treated period, while the two-stage estimator is calculated for both the first treated period and all treated periods. In the bank deregulation design, the conventional Wald test rejects a true zero with probabilities of 13.3 percent for switching and 4.7 and 26.2 percent for the two-stage estimates targeting the first treated period and all treated periods, respectively. In the elder protection design, the corresponding probabilities are 49.1, 6.7 and 6.8 percent, where each is the largest across the state-month common-shock shares considered. These differences show how the treatment periods and comparisons used to construct an estimate can affect the information available for inference. The calculations use the specifications and disturbance settings described in Appendix B.5, and the corresponding rejection frequencies under a homogeneous alternative are reported in Table B.2.

The randomization calculations also depend on the assignment mechanism used to form their reference distribution. In a supplementary bank-deregulation experiment under a sharp zero-effect null, random assignment of adoption dates gives a rejection frequency of 0.054, while assigning earlier dates to states with larger untreated trends gives 0.978. The second design departs from both random assignment and parallel trends, so the comparison illustrates sensitivity to a joint departure from these two conditions. Appendix A.9 also explains how undefined statistics are included in the randomization calculation, and Appendix A.8 distinguishes the sharp null from a zero-average null under heterogeneous effects.

Separate exact calculations over the disturbance shares also show that the two designs do not sit equally close to the Compustat relation between the lower end and size. On the bank-deregulation design no row's exact size exceeds the largest among the Compustat designs of the same estimator whose lower ends lie within a factor of 1.5 of its own, and the class-W row's 0.050 compares with 0.053 to 0.056 there. On the elder-protection design every row's does. For the three cohort-based estimators, at 0.218 to 0.390 against at most 0.104, the gap is consistent with a contribution from their blind shares, which the lower end does not describe and which are absent from the nearby calibration designs. The classification considers the blind share separately. The single-group blind-share diagnostic is not defined for the pooled two-way and stacked coefficients, whose sizes of 0.071 and 0.077 exceed the largest of those Compustat designs, 0.061 and 0.067. The two-way excess arises only under the common shock. With no common shock the design's exact size is 0.061, no more than the worst size of the Compustat designs whose lower ends lie between 9.55 and 13.91; its worst size, 0.071, is at the 75 percent share, where the one of these designs computed at that share stays below 0.061. The excess therefore comes from features of the design that the lower end does not summarize and the calibration does not vary, of which its 84 months, its county and state trends and its ten controls are candidates. The baseline classifications

are largely stable across the threshold calibrations examined. In the alternative family with single-cluster cohorts, the stacked estimator in the elder-protection application moves from class U to class F, while the two-way specification remains in class U. The cohort-based estimators have no calibrated threshold in that family because every design has a positive blind share. On the bank-deregulation design, the published classifications remain unchanged across the families examined. The comparison illustrates that a classification can change when a threshold moves across an interval endpoint, and that its interpretation remains tied to the calibration designs.

Taken together, the exact calculations and simulations show that the effective cluster interval, the blind-share diagnostic, and the construction of each procedure provide complementary information about inference. In the Compustat calibration designs, the relationship between the lower end of the interval and null rejection differs across estimators, while the application-based simulations reveal additional sensitivity to sample structure and score construction. The classifications summarize these findings for the designs examined, and the next section considers how the same features affect the interpretation of the published estimates.

Table 2: Rejection rates on two published finance designs

Design	Class	Design-adapted	CV_1^m	CV_3^m	RI	WCR-C	WCR-S	WCR-B
Panel A. True zero								
BLL, TWFE	W	0.040	0.040	0.040	0.038	0.040	0.040	0.040
BLL, Callaway–Sant'Anna	F	0.051	0.290	0.043	0.051	—	—	—
BLL, Sun–Abraham	F	0.048	0.350	—	0.048	0.000	—	—
BLL, stacked	F	0.058	0.088	0.050	0.058	0.065	0.055	0.059
BLL, imputation	F	0.039	0.291	0.049	0.039	—	—	—
CUY, TWFE	U	0.040	0.067	0.046	0.047	0.046	0.047	0.047
CUY, Callaway–Sant'Anna	F	0.036	0.398	0.061	0.036	—	—	—
CUY, Sun–Abraham	F	0.037	0.296	—	0.037	0.173	—	—
CUY, stacked	U	0.039	0.073	0.042	0.050	0.046	0.046	0.048
CUY, imputation	F	0.049	0.208	0.058	0.049	—	—	—
Panel B. Common reference effect								
BLL, TWFE	W	0.951	0.951	0.947	0.952	0.948	0.948	0.948
BLL, Callaway–Sant'Anna	F	0.926	0.992	0.870	0.926	—	—	—
BLL, Sun–Abraham	F	0.897	0.997	—	0.897	0.254	—	—
BLL, stacked	F	0.932	0.967	0.929	0.932	0.937	0.933	0.925
BLL, imputation	F	0.909	0.993	0.872	0.909	—	—	—
CUY, TWFE	U	0.931	0.967	0.941	0.937	0.942	0.941	0.937
CUY, Callaway–Sant'Anna	F	0.889	0.999	0.913	0.889	—	—	—
CUY, Sun–Abraham	F	0.941	0.995	—	0.941	0.980	—	—
CUY, stacked	U	0.934	0.958	0.942	0.940	0.942	0.939	0.939
CUY, imputation	F	0.942	0.990	0.929	0.942	—	—	—

Note 1: BLL is Beck et al. (2010) and CUY is Carlin et al. (2023); Appendix B.3 describes the simulation design. The class is the estimator's own interval read against the threshold calibrated for that estimator's point estimate on that panel's research design and dataset (Appendix B.2). For the bank-deregulation rows the thresholds are 21.83 for the TWFE, 22.45 for the stacked estimator, 29.85 for SA, 59.52 for CS and 59.60 for the imputation estimator (Figure B.1). For the elder-protection rows they are 26.89, 25.41, 56.42, 56.12 and 59.84, in that order (Figure B.2).

Note 2: Class W uses CV_1^m. Class F uses studentized randomization inference with the dominant cluster's adoption date held fixed. Class U records a rejection only when CV_1^m, CV_3^m and randomization inference all reject. Randomization inference concerns a sharp null under the specified assignment mechanism. The original Sun–Abraham aggregate is not identified after deletion of some states, so its fixed-target jackknife is not reported. For the other jackknife calculations, deletion can change the aggregation weights or target (Appendix A.7).

Note 3: WCR-C is the classic restricted wild cluster bootstrap of MacKinnon et al. (2023b, §5). WCR-S uses restricted jackknife-transformed scores and CV_1^m studentization, and WCR-B uses those scores and CV_3^m studentization. Each uses 999 state-level sign draws. The WCR-S and WCR-B entries are available for the two-way and stacked regressions on both designs. The stacked regression holds its stacks fixed when forming jackknife-transformed scores, while the jackknife column can rebuild stacks after a state is deleted. A dash denotes a comparison not computed for the original specification. The comparison uses only rows for which the procedure is reported (Appendix B.6).

Note 4: Panel A reports rejection frequencies under a true zero effect, and Panel B reports rejection frequencies under the same homogeneous reference effect across the available methods for each estimator. The disturbance has one observation per state-year in BLL and a state-month common-shock share of 0.75 in CUY. The reference effect is based on a t approximation using one degree of freedom fewer than the number of clusters and a 95% power target, and it is reported in the corresponding application table. It need not give exactly 95% power for any particular method. The method-specific minimum detectable effects use the same disturbance setting. All rejection frequencies use 1,000 replications, and bootstrap results use 999 sign draws. Randomization inference uses 199 assignments and counts undefined statistics as at least as extreme as the observed statistic. Different rejection rates under the null qualify comparisons of rejection under the alternative. On the elder-protection two-way regression, each permutation rebuilds the county and state trends from the published outcome and the permuted dates.

5 Applications

We apply the diagnostics to the bank branch deregulation study of Beck et al. (2010) and the elder financial protection study of Carlin et al. (2023). Both studies use staggered state policies and report state-clustered inference, while their sample structures differ in ways that affect the distribution of information. The bank panel contains one observation per state and year, and the elder protection panel contains multiple counties per state and month, so a common state-month disturbance can have a different role in the latter design.

We reproduce the headline estimates and examine the samples, comparisons, and treatment-effect targets used by each specification. The simulations based on these designs provide evidence on rejection rates under the disturbances considered, and the reported empirical calculations illustrate how conclusions vary across procedures with different assumptions. The replicated coefficients differ from the published ones by -0.06% and -0.45%, and the replicated standard errors by -5.93% and $+0.01\%$.

Beside each article's TWFE estimate we compute the four heterogeneity-robust estimators of Section 4: Callaway and Sant'Anna (2021) (CS), Sun and Abraham (2021) (SA), the stacked regression as Baker et al. (2022) implement it after Cengiz et al. (2019), and the imputation estimator of Borusyak et al. (2024). We add the robust analyses that the literature itself reports on each design: for BLL the reanalysis of Baker et al. (2022), and for CUY the article's own CS event study.

5.1 Beck et al. (2010) (BLL)

5.1.1 Research design

From the 1970s through the 1990s most U.S. states removed their restrictions on intrastate bank branching, which intensified bank competition and improved bank performance. BLL ask whether this deregulation changed the distribution of income, a question that runs through two centuries of debate over whether restrictions on bank expansion protect or harm the poor. Exploiting the cross-state, cross-time variation in the timing of deregulation, they find that it tightened the distribution of income, mainly by raising incomes in the lower part of the distribution (Beck et al., 2010). Their headline estimate[4] comes from the static regression

$$\log(Gini)_{st} = \alpha_s + \delta_t + \theta D_{st} + \varepsilon_{st}, \tag{15}$$

where $Gini_{st}$ is the Gini coefficient of total personal income in state s and year t, computed from the March Current Population Survey for civilians aged 25 to 54 with nonnegative

[4]Column 2 of Panel A of the Table II of Beck et al. (2010).

income, excluding incomes below the 1st or above the 99th percentile; α_s and δ_t are state and year fixed effects; and D_{st} equals one in the years after state s permits branching through mergers and acquisitions and zero otherwise. The sample is the District of Columbia and the states other than Delaware and South Dakota, 49 clusters over 1976–2006, 1,519 state-years, and the standard errors are clustered by state. Their baseline estimate is -0.022, with a standard error of 0.008: deregulation lowers the Gini coefficient by about 2%. BLL's Figure 3 replaces D_{st} with event-time indicators,

$$\log(Gini)_{st} = \alpha_s + \delta_t + \sum_{k=-10,\, k\neq 0}^{15} \theta_k D_{st}^k + \varepsilon_{st}, \tag{16}$$

where D_{st}^k equals one when year t is k years after state s deregulates. [5]

In an influential methodological reanalysis, Baker et al. (2022) (BLW) use this design to show how staggered TWFE estimates can mislead without clean comparison. Their decomposition of the TWFE coefficient finds that its negative estimate comes from comparisons of later- with earlier-deregulating states, which carry 0.86 of its weight, while the clean comparisons of earlier- with later-deregulating states average close to zero. Two features of the panel make many of its state-years uninformative about the effect. Twelve states deregulated before the panel opens in 1976 and are treated throughout, and every state has deregulated by 1999, so from 2000 on no state is untreated; together these hold 631 of the 1,101 treated state-years. BLW therefore drop the states that deregulated before 1977 and the years after 1998, which leaves 36 states over 1976–1998 with the 1999 deregulator as the last-treated comparison.[6] Their CS event studies show marginal evidence of an increase in inequality several years after deregulation, and they conclude that the alternative estimators raise doubts about whether banking deregulation impacted income inequality. Our four robust estimators run on BLW's 36 states: CS and the stacked regression as in their Table 2, Sun–Abraham with the 1999 deregulator as the comparison cohort, and the imputation estimator averaged over event years 0 to 10. Panel (1) of Figure 5 reproduces BLL's Figure 3, which BLW draw as their Fig. 9(A), and panel (4) reproduces BLW's Fig. 10(ii)(B).

Table 3 shows what this design implies for inference. Treatment is assigned by state and each state is one unit, so the 49 clusters carry equal mass, no state is dominant, and

[5] The indicators at -10 and $+15$ also cover every earlier and every later year, and the year of deregulation is omitted. The figure draws each coefficient less the mean of the pre-period ones, with 95% intervals, and Beck et al. (2010) read it as showing no trend before deregulation, an effect that appears at once and a decline that grows for about eight years and then levels off.

[6] The sample used here ends in 1998. On that sample Table 2, Panel B of Baker et al. (2022) estimates the average effect over event years 0 to 10 by CS and by a stacked regression on windows from five years before to ten after each deregulation, both against not-yet-treated states: 0.001 (0.007) and 0.000 (0.005).

Table 3: Beck et al. (2010) (BLL): the design features that govern inference.

	TWFE	CS	SA	Stacked	Imputation
Clusters	49	36	36	36	36
Observations	1,519	828	828	1,116	828
Leverage-based count	46.52	3.63	1.86	11.78	2.62
Common-shock count	46.52	3.63	1.86	11.78	2.62
Effective cluster interval	[35.01, 46.52]	[1.48, 3.63]	[1.12, 1.86]	[4.17, 11.78]	[1.56, 2.62]
Blind share, idiosyncratic	—	0.365	0.845	—	—
Blind share, common shock	—	0.365	0.845	—	—
Design class	W	F	F	F	F
Largest partial-leverage share	0.037	0.516	0.731	0.208	0.602
Common reference effect	0.0278	0.0373	0.0355	0.0304	0.0381
Minimum detectable effect at 95% power					
Design-adapted	0.0277	0.0401	0.0399	0.0316	0.0430
CV_1	0.0277	0.0274	0.0257	0.0291	0.0284
CV_3	0.0279	0.0455†	—	0.0325†	0.0473†
RI	0.0276	0.0401	0.0399	0.0316	0.0430
WCR-C	0.0279	—	0.0744	0.0314	—
WCR-S	0.0279	—	—	0.0322	—
WCR-B	0.0279	—	—	0.0329	—

Note 1: The leverage-based count is the effective number of clusters when the disturbance is idiosyncratic, $G^*(0)$, and the common-shock count is the count when the whole disturbance is common to a state-year, $G^*(1)$. Each state-year is one row of this panel, so the two coincide, and only serial correlation separates the ends of the interval. The interval, the class and the largest partial-leverage share use that estimator's own weights. The class reads the interval against the threshold calibrated for that estimator's point estimate on this research design and dataset (Figure B.1): 21.83 for TWFE, 59.52 for Callaway–Sant'Anna, 29.85 for Sun–Abraham, 22.45 for stacked and 59.60 for imputation.

Note 2: Twelve states are already treated when the panel opens and none is never treated. The last state deregulates in 1999, so its indicator first equals one in 2000, and from then on no state is untreated.

Note 3: No cluster is dominant. Each state is one unit, so the 49 clusters hold equal mass and the mass-based count is 49.

Note 4: Only 470 of the 1,101 treated state-years (0.427) lie where the effect is identified; the others belong to the twelve states treated throughout or fall in 2000 or later. Columns (2) to (5) use the sample of Baker et al. (2022): the 36 states that deregulate in 1977 or later, over 1976–1998.

Note 5: The blind-share diagnostic uses fixed weights for CS and SA and reports the fraction of modeled variance associated with single-cluster treatment cohorts or comparison groups under idiosyncratic and common cluster-period disturbances, both without serial correlation. The diagnostic identifies variance exposed to the single-cluster score problem. Contributions can also enter other scores when comparisons overlap, so these entries need not equal the exact variance omitted by the aggregate score. The diagnostic is not defined for the pooled TWFE and stacked coefficients, and its imputation value is not reported for these applications. A dash therefore does not establish that a variance estimator is free of finite-sample bias. Appendix A.4 describes the score construction.

Note 6: The common reference effect is used for every available inference method in the corresponding row of Table 2. The minimum detectable effects (MDEs) use the same calibrated disturbance setting and report the homogeneous effect at which each method reaches 95% rejection at a nominal 5% level. The design-adapted MDE for class U uses joint rejection by all three tests. For each estimator, the disturbance scale matches the expected conventional variance to its estimate in the application. The entries describe repeated-sample rejection, and a smaller MDE can be accompanied by a higher rejection rate under a true zero. The cluster jackknife entries marked † reaggregate the surviving observations after each state deletion. The homogeneous simulated effect remains the same, while aggregation weights may change under heterogeneous effects. The original Sun–Abraham target loses support after some state deletions, so its fixed-target jackknife is not reported. WCR-S and WCR-B hold the stacked regression fixed when constructing the transformed scores. RI tests a sharp null under the stated assignment mechanism. A dash indicates that the corresponding calculation is not reported. Appendix B.4 describes the calculations.

Table 4: Beck et al. (2010) (BLL): replication, alternative inference, and routing.

	(1) TWFE	(2) CS	(3) SA	(4) Stacked	(5) Imputation
Panel A: Replication with conventional inference					
Bank deregulation	−0.0220	0.0009	0.0177	−0.0004	0.0135
Standard error	0.0075	0.0073	0.0044	0.0050	0.0058
p value	0.0053	0.9046	0.0003	0.9341	0.0258
95% interval	[−0.0371, −0.0069]	[−0.0134, 0.0152]	[0.0087, 0.0267]	[−0.0102, 0.0094]	[0.0017, 0.0252]
Joint pre-trend test p	0.3646	0.0389	0.0134	0.3855	0.2228
Pre-trend rejection at 5%	NO	YES	YES	NO	NO
Observations	1,519	828	828	1,116	828
Panel B: Comparison among alternative inference methods					
Cluster-robust Wald (CV_1^m)					
Standard error	0.0075	0.0055	0.0044	0.0075	0.0058
p value	0.0053	0.8740	0.0003	0.9562	0.0258
95% interval	[−0.0371, −0.0069]	[−0.0103, 0.0120]	[0.0087, 0.0267]	[−0.0155, 0.0147]	[0.0017, 0.0252]
Cluster jackknife (CV_3^m)					
Standard error	0.0076	0.0122	—	0.0086	0.0173
p value	0.0058	0.9431	—	0.9620	0.4426
95% interval	[−0.0373, −0.0067]	[−0.0239, 0.0256]	—	[−0.0179, 0.0170]	[−0.0217, 0.0487]
Randomization inference (RI)					
Standard error	0.0079	0.0157	0.0211	0.0092	0.0151
p value	0.004 (0.0020)	0.958 (0.0063)	0.363 (0.0152)	0.965 (0.0058)	0.383 (0.0154)
95% interval	[−0.0371, −0.0067]	[−0.0261, 0.0334]	[−0.0192, 0.0550]	[−0.0191, 0.0175]	[−0.0155, 0.0459]
Restricted wild cluster bootstrap (WCR-C)					
Standard error	0.0080	—	0.0133	0.0071	—
p value	0.003	—	0.342	0.956	—
95% interval	[−0.0374, −0.0066]	—	[−0.0407, 0.0716]	[−0.0193, 0.0153]	—
Panel C: Routing by the effective cluster interval					
Design class	W	F	F	F	F
Reject at 5%	YES	NO	NO	NO	NO
Pre-period calculation p	0.382	0.494	0.390	0.561	0.140
Calculation rejects at 5%	NO	NO	NO	NO	NO

Note 1: Columns (2) and (4) are the regressions in Baker et al. (2022)'s Table 2, Panel B, which prints 0.001 (0.007) and 0.000 (0.005).

Note 2: Panel A reports conventional inference: for TWFE the article's cluster-robust test by state, on Student's t with 48 degrees of freedom; for Callaway–Sant'Anna the clustered multiplier bootstrap and for stacked clustering by state × stack (373 clusters), both as Baker et al. (2022) report them. Neither Beck et al. (2010) nor Baker et al. (2022) reports Sun–Abraham or imputation estimates on this panel, so these two columns repeat Panel B's cluster-robust Wald test.

Note 3: Panel B clusters by state on the estimator's own weights. The cluster-robust Wald test, CV_1^m, uses Student's t with one degree of freedom fewer than the number of clusters and, for Callaway–Sant'Anna, holds the aggregation weights fixed, which is mainly why its standard error (0.0055) is below Panel A's (0.0073). The cluster jackknife, CV_3^m, leaves out one state at a time and reaggregates surviving cells. The Sun–Abraham original aggregate is not identified after some state deletions, so its fixed-target jackknife is not reported (Appendix A.7).

Note 4: RI is studentized randomization inference on 999 draws of the adoption dates, holding the dominant cluster's date fixed; no state dominates here, so the first state in the panel's order is held. Its standard error is the standard deviation of the estimate across the draws, the parentheses give the Monte Carlo standard error of the p value, and its interval inverts the sharp null of a constant effect. Randomization tests of constant effects concern sharp nulls. Their inversion gives a confidence set for a common effect under the constant-effect model, and rejection of a constant effect does not by itself reject an average effect of the same magnitude under heterogeneity.

Note 5: WCR-C, the restricted wild cluster bootstrap, is reported as an additional check alongside the design-adapted test. Its standard error is the standard deviation of its bootstrap coefficients, and its interval collects the values it does not reject at 5% on the same 999 sign vectors. It is not constructed here for Callaway–Sant'Anna or the imputation estimator.

Note 6: The class reads the effective cluster interval in Table 3 against the threshold calibrated for that estimator's point estimate on this research design and dataset (Figure B.1). Class W routes to the cluster-robust Wald test, class F to randomization inference, and class U rejects only when the Wald test, the jackknife and randomization inference all reject. Reject at 5% is that design-adapted test.

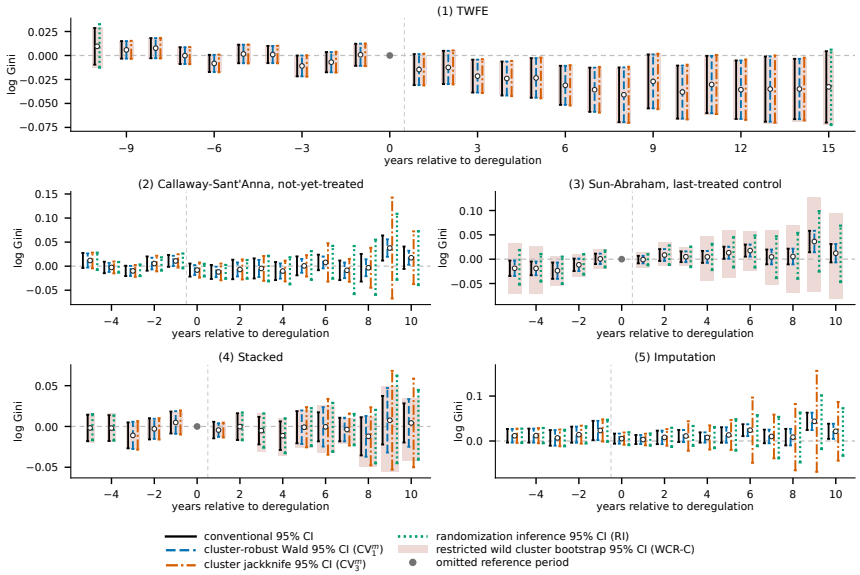
Note 7: The pre-period rows report maximum-statistic calculations for the corresponding panel of Figure 5. Panel C uses the Gaussian multiplier band (G) when every pre-period horizon is class W and the randomization band (R) when every one is class F; otherwise it computes both and reports the larger p value, since a rejection requires both: (1) R 0.382 (G 0.296); (2) R 0.494; (3) R 0.390; (4) R 0.561 (G 0.516); (5) R 0.140. The randomization reference distribution imposes a global sharp null and does not by itself test a pre-trend null that permits unrestricted post-treatment effects. Non-rejection does not establish parallel trends.

the largest state's share of the TWFE coefficient's partial leverage is only 0.037. Because the panel is natively observed at the state-year level, a common state-year shock and an idiosyncratic error are the same thing, the leverage-based and common-shock counts coincide at 46.52, and only serial correlation widens the effective cluster interval, to between 35.01 and 46.52. Both ends lie far above the TWFE threshold of 21.83, and the single-group blind-share diagnostic does not apply to the pooled regression. The classification uses a zero value for this diagnostic by convention, so the TWFE coefficient is class W and we report the state-clustered Wald test used in the article. The robust estimators are different. Restricting the sample to support the stated comparisons leaves 36 states, and their own weights concentrate on a few of them: the largest state carries 0.516 of CS estimator's partial leverage, and the last-adopting state, SA estimator's only comparison, carries 0.731 of SA's. This is the leverage concentration problem of Section 3.2. The effective cluster intervals of the four robust estimators lie between about one and twelve clusters, far below their own thresholds, so all four are class F and are assessed using randomization inference under the stated assignment mechanism. The fixed-weight score diagnostic adds another perspective on these estimates. Ten of the 17 treated cohorts contain a single state, and the last state to deregulate is the only comparison state for SA and for some of the CS comparisons. The variance associated with these single-state groups accounts for 0.365 of the modeled variance for CS and 0.845 for SA. For SA, the value of 0.845 means that 84.5 percent of the modeled variance is associated with groups exposed to the single-cluster score problem. A state can also contribute through other comparisons, so the visibility of its shock in the aggregate score depends on how those comparisons overlap. Table 2 shows the resulting rejection frequencies under a true zero effect in the calibrated simulation. The conventional Wald test rejects 4.0% of the time for TWFE, 29.0% for CS and 35.0% for SA, while the design-adapted calculations range from 3.9% to 5.8% across the five estimators. Randomization inference requires an assumption about the conditional assignment of deregulation dates across states. The evidence in BLL's Table I that pre-existing inequality does not predict deregulation timing is relevant to the design but does not establish that assignment assumption.

5.1.2 Estimates

Table 4 reports the estimates and inference for the bank deregulation design. The TWFE estimate of -0.0220 rejects zero under the reported procedures, with p values between 0.003 and 0.006. In panel (1) of Figure 5, the horizons ten years before and fifteen years after deregulation are class U, while the remaining horizons are class W. The conventional joint pre-period test gives $p = 0.365$, and the reported combined calculation gives 0.382, the larger of the Gaussian value of 0.296 and the randomization value of 0.382. The latter is calibrated

Figure 5: BLL: event studies with conventional and design-adapted 95% intervals.



Note 1: Each panel is one estimator: (1) the two-way regression, (2) Callaway–Sant'Anna, (3) Sun–Abraham, (4) the stacked regression and (5) the imputation estimator. The horizontal axis is years relative to deregulation and the vertical axis is log Gini. Open circles are point estimates. At each horizon the intervals that horizon calls for are drawn a short distance apart, and each interval has its own line style. The black solid bar is the conventional 95% interval, the blue dashed bar is the cluster-robust Wald interval, the orange dash-dot bar is the cluster jackknife, and the green dotted bar is the randomization interval. The pale red box is the restricted wild cluster bootstrap. A gray dot marks the omitted period.

Note 2: The intervals use a nominal 95% confidence level. Randomization intervals invert sharp nulls of constant effects under the specified assignment mechanism. The jackknife leaves out one cluster at a time where reported. Panel (3) omits the reaggregated jackknife because the original target can lose identifying support after state deletion (Appendix A.7). The pale red box is the restricted wild cluster bootstrap of a regression coefficient with 999 draws, using the same procedure as the static intervals in Table 4. It is drawn on every horizon of panels (1), (3) and (4). It is not reported for Callaway–Sant'Anna or the imputation estimator, so panels (2) and (5) have no box.

Note 3: The conventional interval is the one the estimator's own procedure reports, before applying the classification. (1) The two-way regression is cluster-robust by state, 49 clusters, with the small-sample factor and Student's t with 48 degrees of freedom. That is the cluster-robust Wald interval, so the solid bar and the dashed bar have the same ends and are drawn apart. (2) Callaway–Sant'Anna is the multiplier bootstrap, 1,000 draws, using the normal critical value 1.96. It is not the cluster-robust Wald interval; the dashed bar beside it clusters by state and uses Student's t with 35 degrees of freedom. (3) Sun–Abraham is cluster-robust by state, 36 clusters, with the small-sample factor and Student's t with 35 degrees of freedom, the same construction as the cluster-robust Wald interval, so the ends match. (4) The stacked regression is cluster-robust by state × stack, 373 clusters, on Student's t with 372 degrees of freedom, as Baker, Larcker and Wang cluster it. The cluster-robust Wald interval clusters by state, and the ends differ. (5) The imputation estimator's conventional bar has the same ends as the cluster-robust Wald interval by state, on Student's t with 35 degrees of freedom.

Note 4: The classification uses the horizon's class. Class W is read from the Wald interval. Class F is read from the randomization interval; the Wald and jackknife bars beside it do not supply the verdict. Class U rejects only when the Wald test, the jackknife and the randomization test all reject. (1) Horizons −10 and +15 are U, and every other horizon is W. The reported pre-period calculation gives 0.382, the larger of 0.296 and 0.382, and does not reject at 5%; the horizon −3 does ($p = 0.0489$). At −10 and +15 none of the three tests rejects. After deregulation the Wald interval excludes zero from +3 to +8 and from +10 to +14, and not at +1 ($p = 0.0736$). (2) Every horizon is F. The reported pre-period calculation gives 0.494 and does not reject, and no horizon rejects under randomization inference. The Wald interval does reject at −1, +1, +9 and +10. (3) Every horizon is F. The reported pre-period calculation gives 0.390 and does not reject, and no horizon rejects under randomization inference. The Wald interval rejects at −5, −4, −3, +5, +6 and +9. (4) Class U at −5, −4, +2, +3 and +4; class W at −3, −2, −1 and +1; class F from +5 to +10. The reported pre-period calculation gives 0.561, the larger of 0.516 and 0.561, and does not reject. No horizon rejects under its design-adapted test, and at each class-U horizon none of the three tests rejects. (5) Every horizon is F. The reported pre-period calculation gives 0.140 and does not reject, and no horizon rejects under randomization inference. The Wald interval rejects at −1, +6, +9 and +10. Randomization values are calibrated under a global sharp null. The pre-period maximum therefore does not generally test a pre-trend null allowing unrestricted post-treatment effects. Horizon-specific results are pointwise and do not constitute a simultaneous test of the post-treatment path.

Note 5: Panel (1) is the window of Figure 3 in Beck et al. (2010): years −10 to +15 relative to deregulation, omitting the year of deregulation, with the coefficients at −10 and +15 binning every earlier and later year. Panels (2) to (5) are the robust estimators on the 36 states used in the replication of Baker et al. (2022), and they follow that window, from five years before deregulation to ten after.

Note 6: The set of intervals changes from one horizon to the next when the horizon's class changes. Each horizon is classified using its interval and estimator-specific threshold (Figure B.1, Section 3.4), with the single-group diagnostic set to zero by convention for pooled regressions. Besides the conventional interval, a class-W horizon draws the Wald interval and the jackknife, a class-F horizon draws the Wald and randomization intervals and the jackknife when available, and a class-U horizon draws the randomization interval. Panels (2), (3) and (5) are one class on every horizon, so the set does not change along the panel. Panel (1) is class W except at −10 and +15, which are U. Panel (4) is class U at −5, −4, +2, +3 and +4, class W at −3, −2, −1 and +1, and class F from +5 to +10.

under a global sharp null, so neither its non-rejection nor the combined result establishes parallel trends. The coefficient three years before deregulation is individually significant. Pointwise intervals after deregulation exclude zero from the third to the eighth year and from the tenth to the fourteenth year. The recentered coefficient one year after deregulation has $p = 0.074$, while the published interval shifts the bounds for the raw coefficient by the pre-period mean without recomputing the interval for the recentered coefficient. These horizon-specific intervals do not provide a simultaneous test of the entire post-treatment path.

For the original Sun–Abraham aggregate, the leave-one-state calculation loses identifying support after some state deletions, so Table 4 does not report its reaggregated jackknife as an interval for that fixed target. The point estimate and the other reported calculations retain their original specification, and Appendix A.7 describes the support check.

The alternative estimators yield estimates and inferential results that differ from those of the two-way regression. CS's estimate of $+0.0009$ and the stacked estimate of -0.0004 do not reject under the reported procedures. SA's estimate is $+0.0177$ and its conventional test gives $p = 0.0003$, while randomization inference gives $p = 0.363$ for the sharp null. The imputation estimate of $+0.0135$ gives $p = 0.026$ under the conventional Wald test and does not reject under the jackknife or the reported randomization calculation. The simulation for this assignment shows substantial size distortion for the conventional SA test under the specified null disturbances. These comparisons describe sensitivity to the inferential procedure and do not make the sharp randomization null identical to a null about a heterogeneous average effect.

The small-sample procedures provide further evidence on the inference attached to these estimates. WCR-C gives $p = 0.003$ for TWFE and $p = 0.342$ for SA, compared with 0.005 and 0.0003 under the corresponding conventional Wald tests. Thus the original TWFE finding remains significant under the bootstrap, while the conventional rejection for SA is sensitive to the inference method. As a separate comparison, we divide states into six groups by their adoption dates and estimate the TWFE regression within each group. The average of the six estimates is -0.0227, with $p = 0.031$ under the approximate randomization test and $p = 0.033$ under the grouped t-test of Ibragimov and Müller (2016). This group-average estimate gives each group equal weight and need not equal the original pooled TWFE target when effects differ across states or periods (Appendix A.5).

The MDEs in Table 3 place the estimates in the context of the effects that each inference method can detect with 95% power in the calibrated design. The TWFE estimate has an absolute value of 0.0220, compared with a CV_1 MDE of 0.0277. The observed TWFE estimate is significant even though its magnitude is below that MDE, illustrating the distinction between

a result in one sample and power across repeated samples. The SA estimate of 0.0177 is 44.4% of its RI MDE of 0.0399, and its randomization interval is $[-0.0192,\ 0.0550]$. The comparison places the observed estimate alongside an effect that RI would detect in 95 percent of samples generated under the calibrated model. The conventional SA test has an MDE of 0.0257, while its simulated rejection rate under a true zero is 35.0%. The corresponding WCR-C MDE is 0.0744, and WCR-C rejects a true zero in 0.0% of the simulations. The MDE describes rejection across repeated samples under a homogeneous effect, while the randomization interval describes the constant effects not rejected in the observed sample under the stated assignment mechanism. Together these quantities help interpret the precision of the evidence without treating the MDE as a threshold that determines the result of a particular test.

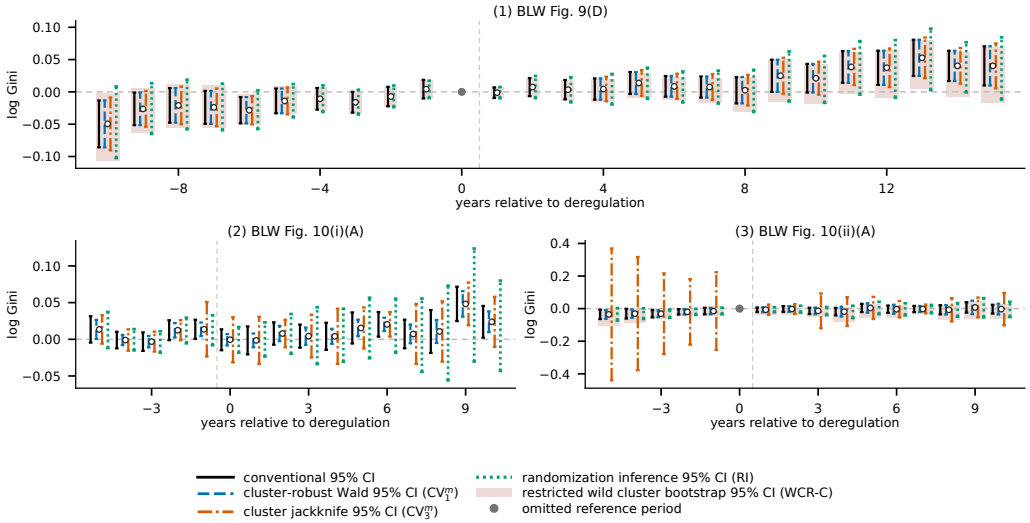
The pre-period calculations also differ across procedures. Conventional joint tests reject in five of the seven alternative event studies in Figures 5 and 6. The reported combined or randomization-calibrated calculations do not reject at the five percent level, although their randomization component is calibrated under a global sharp null. These outcomes do not establish parallel trends or show that the conventional rejections are necessarily spurious. In the two CS event studies, conventional tests reject at several individual horizons, including the ninth and tenth years after deregulation, whereas the randomization calculations do not reject at those horizons.

The comparison complements the reanalysis of Baker et al. (2022) by showing how inference changes when the alternative estimators rely on a smaller and more concentrated set of states. Three of the four alternative estimators reject the sharp null of a constant effect equal to the two-way estimate. This result challenges that constant-effect schedule, while heterogeneity permits the corresponding average effect to equal the two-way estimate. Differences in samples, estimands and identifying assumptions also matter when comparing the estimates.

5.1.3 Robustness

Figure 6 examines the three additional event studies reported in the reanalysis of Baker et al. (2022): the unbinned two-way specification on the restricted 36-state sample, CS using the last-treated state as control, and the stacked regression with controls restricted to states not treated within ten years. In the unbinned specification, removing the bins leaves every horizon's effective cluster interval below the two-way threshold or straddling it, and the design-adapted calculation rejects only at the horizon thirteen years after deregulation. The CS and stacked event studies are class F at every horizon. CS's Wald test rejects at four horizons after deregulation and the stacked regression's at none, but randomization inference rejects at no horizon after deregulation on either, and neither reported randomization-calibrated

Figure 6: BLL: event studies in the replication of Baker et al. (2022).



Note 1: The three panels replicate Fig. 9(D), Fig. 10(i)(A) and Fig. 10(ii)(A) in the replication exercises for the same paper in Baker et al. (2022). The horizontal axis is years relative to deregulation and the vertical axis is log Gini. Symbols, line styles and the omitted-period dot are those of Figure 5.

Note 2: The jackknife and the restricted wild cluster bootstrap are drawn as in Figure 5. The bootstrap is on panels (1) and (3). Panel (2) is Callaway–Sant'Anna, for which it is not defined.

Note 3: The conventional interval is the estimator's own, before applying the classification, as in Figure 5. (1) Fig. 9(D) is cluster-robust by state, 36 clusters, with the small-sample factor and Student's t with 35 degrees of freedom. That is the cluster-robust Wald interval, so the solid bar and the dashed bar have the same ends and are drawn apart. (2) Fig. 10(i)(A) is the multiplier bootstrap, 1,000 draws, using the normal critical value 1.96. It is not the cluster-robust Wald interval; the dashed bar beside it clusters by state and uses Student's t with 35 degrees of freedom. (3) Fig. 10(ii)(A) is cluster-robust by state × stack, as Baker, Larcker and Wang cluster it. The cluster-robust Wald interval clusters by state, 36 clusters, on Student's t with 35 degrees of freedom, and the ends differ.

Note 4: The classification is that of Figure 5. Before routing, the conventional joint pre-trend tests give 0.015, 0.0001 and 0.024 on panels (1) to (3). (1) Class F at -10 to -5 and at $+4$ to $+15$; class U at -4 to -1 and at $+1$ to $+3$. The reported pre-period calculation gives 0.182, the larger of 0.025 and 0.182, and does not reject at 5%. Randomization inference rejects at $+13$ (0.041). No class-U horizon rejects under randomization inference. (2) Every horizon is F. The reported pre-period calculation gives 0.506 and does not reject, and no horizon rejects under randomization inference. The Wald interval rejects at -5, -2, -1, $+5$, $+6$, $+9$ and $+10$. (3) Every horizon is F. The reported pre-period calculation gives 0.080 and does not reject. Randomization inference rejects at -3 and -2 (0.017 and 0.039). The Wald interval rejects at -5, -4, -3 and -2. The jackknife at -5 to -1 is much wider than the other intervals on that panel (at -5 it runs from -0.440 to 0.369), and that width sets the vertical scale. Randomization values are calibrated under a global sharp null. The pre-period maximum therefore does not generally test a pre-trend null allowing unrestricted post-treatment effects. Horizon-specific results are pointwise and do not constitute a simultaneous test of the post-treatment path.

Note 5: Panel (1) is Figure 3 of Beck et al. (2010) unbinned on the 36 states that deregulate after 1976, over the years through 1998: years -10 to $+15$, omitting the year of deregulation. Panel (2) is Callaway–Sant'Anna with the 1999 state recoded as never treated, years -5 to $+10$. Panel (3) is the stacked regression with controls not treated within ten years, years -5 to $+10$, omitting the year of deregulation.

Note 6: Which intervals a class draws is as in Figure 5, and the class is read against the thresholds in Figure B.1. Panels (2) and (3) are class F on every horizon, so the set does not change along the panel. Panel (1) is class F at -10 to -5 and at $+4$ to $+15$, and class U at -4 to -1 and at $+1$ to $+3$.

pre-period calculation rejects under its global sharp null ($p = 0.506$ and 0.080), although randomization inference rejects at two pre-period horizons of the stacked regression.

The robustness exercises illustrate the inferential consequences of changing the comparisons used in the bank deregulation design. The alternative estimators use fewer states and concentrate weight on particular cohorts and comparison groups, which makes their precision sensitive to the procedure used. Their point estimates and reported randomization calculations provide a different perspective on the magnitude of the two-way estimate, while the pre-period results leave the identifying assumptions subject to the usual limitations of pre-trend tests.

The exercise therefore complements the comparison of estimators in Baker et al. (2022) by documenting the information available for inference on each specification.

5.2 Carlin et al. (2023) (CUY)

5.2.1 Research design

Regulators often deputize financial professionals to screen for misconduct without rewarding detection or punishing a failure to detect. Carlin et al. (2023) (CUY) ask whether such permissive laws work, in the setting of elder financial exploitation. In 2016 the North American Securities Administrators Association adopted a Model Act that gives financial professionals two new authorities, to reach out to a client's trusted contact about red flags and to halt disbursements that appear exploitative, without obliging them to use either; by 2020 thirty states had adopted its provisions, at different dates. CUY find that adoption reduced the reports of elder financial exploitation that financial institutions file, and that the effect grows over time. Their headline estimate[7] comes from

$$y_{ct} = \alpha_c + \delta_t + \theta D_{s(c)t} + \gamma' X_{ct} + \varepsilon_{ct}, \tag{17}$$

where y_{ct} is the number of reports of elder financial exploitation that financial institutions in county c file with the U.S. Treasury in month t, winsorized at the 1st and 99th percentiles as in their paper; α_c and δ_t are county and year-month fixed effects; $D_{s(c)t}$ equals one from the month in which the Model Act's provisions take effect in the county's state $s(c)$; and X_{ct} holds ten standardized county controls, nine of them for residents aged 65 and over (credit score, subprime share, low-income share, average age, male share, married share, household income, debt-to-income ratio and log population) and the share holding a bachelor's degree. The specification also controls for county- and state-specific linear trends in the outcome, fitted on the months before July 2016 that precede the state's adoption and extrapolated forward. The panel is 3,139 counties over the 84 months of 2014–2020, 263,676 county-months, and the standard errors are clustered by state, which the article justifies with Bertrand et al. (2004): clustering by state "*works well when the number of groups is large (e.g., 50 states)*". The headline estimate is a drop of about 0.2 in the number of abuse cases per county-month, 15% of the mean, with a standard error of 0.096.

The article traces the dynamics in its Fig. 5(a), which replaces $D_{s(c)t}$ with six-month bins of event time,

$$y_{ct} = \alpha_c + \delta_t + \sum_{k=-8,\, k\neq -1}^{7} \theta_k B_{ct}^k + \gamma' X_{ct} + \varepsilon_{ct}, \tag{18}$$

[7] Column 1 of Table 4 in Carlin et al. (2023).

where B_{ct}^{k} equals one when month t lies $6k$ to $6k+5$ months from the effective month in county c's state, so that bin 0 covers the first six months of treatment and the omitted bin -1 the six months before it. The figure keeps the ten controls but not the trends, uses the county-months inside the window and the never-treated states.[8]

Washington adopted in 2010 and is treated throughout the panel, so our robust estimators drop it. With the ten controls, they are CS with not-yet-treated controls, doubly robust and averaged over months 0 to 47; SA in the six-month bins with never-treated controls; the stacked regression on windows from 48 months before to 47 after each adoption; and the imputation estimator averaged over months 0 to 47, whose event study uses the same six-month bins and also estimates bin -1, which the regressions omit. The article reports no scalar for its own CS estimation, and we average its monthly effects over months 0 to 47. [9]

Table 5 shows why the article's justification for clustering by state does not settle the inference. Treatment is assigned by state, and the panel has 51 clusters, 30 of them treated and 21 never treated. Counted by mass or by leverage, the clusters look plentiful: the mass-based count is 32.60, the leverage-based count of the TWFE coefficient is 27.67, and the largest state holds 0.081 of the observations and 0.102 of the partial leverage. But the panel is not one observation per state-month. Each state-month holds every county of the state, so a shock common to the state-month moves all of them together, and the common-shock channel of Section 3.1 is open: if the whole disturbance were such a shock, the effective number of clusters would be 10.76. Across disturbance mixes and degrees of serial correlation, the effective cluster interval runs from 10.25 to 28.46 and straddles the TWFE threshold of 26.89, so the TWFE coefficient is class U: the design alone cannot say whether the article's Wald test is in the regime where it holds its size. The robust estimators add concentrated weights to the same problem. The other 29 treated states adopt in 21 monthly cohorts, 15 of them a single state, and the fixed-weight score diagnostics associate single-state cohorts with shares of 0.406 and 0.639 of the modeled variance for CS under the idiosyncratic and common-shock endpoints, respectively, and 0.136 and 0.181 for SA. For CS, the increase from 0.406 to 0.639 shows how the modeled contribution associated with single-state groups changes when the disturbance is common to counties in the same state and month. A state can also contribute through other comparisons, so the visibility of its shock in the aggregate score depends on how those comparisons overlap. CS, SA and the article's own CS have

[8] In Carlin et al. (2023), the authors draw 90% intervals and describe the pre-treatment estimates as consistent with parallel trends and the post-treatment declines as increasing over time. Their Internet Appendix Table A9 presents a corresponding tabular specification whose first column includes a county trend. Their Internet Appendix Figure B2 applies the CS estimator to the raw count, with never-treated controls, no covariates and 48 months on either side.

[9] Panel (1) of Figure 7 reproduces Fig. 5(a) of Carlin et al. (2023), and Figure 8 the article's two other event studies.

Table 5: Carlin et al. (2023) (CUY): the design features that govern inference.

	TWFE	CS	SA	Stacked	Imputation
Clusters	51	50	50	50	50
Observations	263,676	260,400	260,400	260,400	260,400
Leverage-based count	27.67	17.88	16.90	30.43	26.24
Common-shock count	10.76	7.48	7.66	12.17	11.48
Effective cluster interval	[10.25, 28.46]	[7.48, 19.11]	[7.66, 21.60]	[11.21, 30.57]	[11.48, 27.46]
Blind share, idiosyncratic	—	0.406	0.136	—	—
Blind share, common shock	—	0.639	0.181	—	—
Design class	U	F	F	U	F
Largest partial-leverage share	0.102	0.106	0.137	0.091	0.093
Common reference effect	0.3757	0.6309	0.8322	0.6780	0.9681
Minimum detectable effect at 95% power					
Design-adapted	0.3926	0.7193	0.8675	0.7137	1.0004
CV_1	0.3636	0.4545	0.5984	0.6498	0.7586
CV_3	0.3829	0.6947†	—	0.6942†	1.0217†
RI	0.3917	0.7193	0.8675	0.7041	1.0004
WCR-C	0.3832	—	0.7009	0.6920	—
WCR-S	0.3848	—	—	0.6937	—
WCR-B	0.3872	—	—	0.6997	—

Note 1: The TWFE column has 30 treated states and 21 never treated; each other column has 29 and 21. The leverage-based count is the effective number of clusters when the disturbance is idiosyncratic, $G^*(0)$, and the common-shock count is the count when the whole disturbance is common to a state-month, $G^*(1)$. The interval, the class and the largest partial-leverage share use that estimator's own weights. The class reads the interval against the threshold calibrated for that estimator's point estimate on this research design and dataset (Figure B.2): 26.89 for TWFE, 56.12 for Callaway–Sant'Anna, 56.42 for Sun–Abraham, 25.41 for stacked and 59.84 for imputation. The article's own Callaway–Sant'Anna specification is not a column of these tables; its interval is [7.42, 23.38], its class is F, and its event study is panel (2) of Figure 8.

Note 2: The mass-based count on the published sample is 32.60. The dominant cluster, the largest state, is treated during the panel and holds 0.081 of the observations.

Note 3: Washington adopted in 2010 and is treated throughout, so it leaves every other column. The other 29 treated states adopt in 21 monthly cohorts, 15 of them a single state.

Note 4: Apart from Washington's, every treated county-month lies where the effect is identified, 0.951 of them.

Note 5: The trend regressors enter the TWFE column only. They are fitted using outcome observations in each state's months up to June 2016 that precede its adoption. Each randomization draw reconstructs the trends using the drawn dates. The other four estimators are fitted without them.

Note 6: The blind-share diagnostic uses fixed weights for CS and SA and reports the fraction of modeled variance associated with single-cluster treatment cohorts or comparison groups under idiosyncratic and common cluster-period disturbances, both without serial correlation. The diagnostic identifies variance exposed to the single-cluster score problem. Contributions can also enter other scores when comparisons overlap, so these entries need not equal the exact variance omitted by the aggregate score. The diagnostic is not defined for the pooled TWFE and stacked coefficients, and its imputation value is not reported for these applications. A dash therefore does not establish that a variance estimator is free of finite-sample bias. Appendix A.4 describes the score construction.

Note 7: The common reference effect is used for every available inference method in the corresponding row of Table 2. The minimum detectable effects (MDEs) use the same calibrated disturbance setting and report the homogeneous effect at which each method reaches 95% rejection at a nominal 5% level. The design-adapted MDE for class U uses joint rejection by all three tests. For each estimator, the disturbance scale matches the expected conventional variance to its estimate in the application. The entries describe repeated-sample rejection, and a smaller MDE can be accompanied by a higher rejection rate under a true zero. The cluster jackknife entries marked † reaggregate the surviving observations after each state deletion. The homogeneous simulated effect remains the same, while aggregation weights may change under heterogeneous effects. The original Sun–Abraham target loses support after some state deletions, so its fixed-target jackknife is not reported. WCR-S and WCR-B hold the stacked regression fixed when constructing the transformed scores. RI tests a sharp null under the stated assignment mechanism. A dash indicates that the corresponding calculation is not reported. Appendix B.4 describes the calculations.

Table 6: Carlin et al. (2023) (CUY): replication, alternative inference, and routing.

	(1) TWFE	(2) CS	(3) SA	(4) Stacked	(5) Imputation
Panel A: Replication with conventional inference					
Model Act	−0.1951	−0.2564	−0.2503	−0.3381	−0.3688
Standard error	0.0960	0.7793	0.1146	0.0529	0.1665
p value	0.0475	0.7421	0.0338	0.0000	0.0314
95% interval	[−0.3880, −0.0023]	[−1.7837, 1.2709]	[−0.4807, −0.0199]	[−0.4418, −0.2344]	[−0.7034, −0.0342]
Joint pre-trend test p	0.3803	0.0169	0.0000	0.2139	0.4326
Pre-trend rejection at 5%	NO	YES	YES	NO	NO
Observations	263,676	260,400	260,400	260,400	260,400
Panel B: Comparison among alternative inference methods					
Cluster-robust Wald (CV_1^m)					
Standard error	0.0960	0.0781	0.1146	0.1680	0.1665
p value	0.0475	0.0019	0.0338	0.0497	0.0314
95% interval	[−0.3880, −0.0023]	[−0.4133, −0.0995]	[−0.4807, −0.0199]	[−0.6757, −0.0005]	[−0.7034, −0.0342]
Cluster jackknife (CV_3^m)					
Standard error	0.1003	0.1475	—	0.1768	0.1741
p value	0.0574	0.0884	—	0.0617	0.0392
95% interval	[−0.3966, 0.0064]	[−0.5528, 0.0400]	—	[−0.6933, 0.0171]	[−0.7186, −0.0190]
Randomization inference (RI)					
Standard error	0.0895	0.1283	0.1807	0.2040	0.2054
p value	0.041 (0.0063)	0.109 (0.0099)	0.080 (0.0086)	0.041 (0.0063)	0.060 (0.0075)
95% interval	[−0.3861, −0.0113]	[−0.4974, 0.0483]	[−0.6044, 0.0437]	[−0.6686, −0.0132]	[−0.8166, 0.0188]
Restricted wild cluster bootstrap (WCR-C)					
Standard error	0.0985	—	0.1243	0.1884	—
p value	0.054	—	0.052	0.057	—
95% interval	[−0.4044, 0.0015]	—	[−0.5030, 0.0028]	[−0.6931, 0.0081]	—
Panel C: Routing by the effective cluster interval					
Design class	U	F	F	U	F
Reject at 5%	NO	NO	NO	NO	NO
Pre-period calculation p	0.741	0.084	0.610	0.800	0.492
Calculation rejects at 5%	NO	NO	NO	NO	NO

Note 1: Column (1) is column (1) of the article's Table 4, which prints −0.196 (0.096) on 263,676 observations.

Note 2: Panel A reports conventional inference: for TWFE the article's cluster-robust test by state, on Student's t with 50 degrees of freedom; for Callaway–Sant'Anna the clustered multiplier bootstrap; and for stacked clustering by county × stack (45,034 clusters), the convention of Baker et al. (2022). The article reports neither Sun–Abraham nor imputation estimates, so these two columns repeat Panel B's cluster-robust Wald test, and the imputation pre-trend test applies it to the eight six-month lead bins of panel (5) of Figure 7. The article's own Callaway–Sant'Anna estimation is not a column: averaged over months 0 to 47 it is −0.6752, with $p = 0.0386$ under the analytic variance and 0.091 under randomization inference.

Note 3: Panel B clusters by state on the estimator's own weights. The cluster-robust Wald test, CV_1^m, uses Student's t with one degree of freedom fewer than the number of clusters and, for Callaway–Sant'Anna, holds the aggregation weights fixed. The reported bootstrap keeps the weights' term and averages the influence function within each state, which weighs the 50 states equally whatever their number of counties. Summed within state, the standard error is 0.0773 without the term and 0.1314 with it; with the term and averaged within state, it is Panel A's 0.7793. The cluster jackknife, CV_3^m, leaves out one state at a time. The Sun–Abraham original aggregate is not identified after some state deletions, so its fixed-target jackknife is not reported (Appendix A.7).

Note 4: RI, WCR-C and the design classes are defined as in Notes 4 to 6 of Table 4, with the effective cluster intervals of Table 5 and the thresholds of Figure B.2. Here RI holds fixed the date of the dominant cluster, the largest state, and the three draws in column (2) for which the statistic is undefined are counted as at least as extreme as the observed statistic. The same convention is used for its event study in Figure 7 (Appendix A.9). The reported randomization standard deviation uses the defined estimates. Randomization tests of constant effects concern sharp nulls. Their inversion gives a confidence set for a common effect under the constant-effect model, and rejection of a constant effect does not by itself reject an average effect of the same magnitude under heterogeneity.

Note 5: The pre-period rows report maximum-statistic calculations for the corresponding event study: panels (2) to (5) of Figure 7 for columns (2) to (5) and, for column (1), the event study on the same trend-adjusted outcome, column 1 of the CUY's Internet Appendix Table A9 (panel (1) of Figure 8). Panel C chooses the band as in Note 7 of Table 4: (1) R 0.741 (G 0.641); (2) R 0.084; (3) R 0.610; (4) G 0.800 (R 0.745); (5) R 0.492. The randomization reference distribution imposes a global sharp null and does not by itself test a pre-trend null that permits unrestricted post-treatment effects. Non-rejection does not establish parallel trends.

intervals of about 7 to 23 effective clusters, against thresholds of 56.12 and 56.42 for the CS and SA specifications reported in the table, and the imputation estimator's interval of 11.48 to 27.46 lies far below its threshold of 59.84; all four are class F. The stacked estimator's interval of 11.21 to 30.57 straddles its threshold of 25.41, so it is class U. Table 2 shows the resulting rejection frequencies under a true zero effect in the calibrated simulation. The conventional Wald test rejects 6.7% of the time for TWFE, 7.3% for the stacked estimator, and 39.8%, 29.6% and 20.8% for CS, SA and imputation, respectively. The design-adapted calculations range from 3.6% to 4.9% across the five estimators. Randomization inference requires a specified conditional distribution of adoption dates across states, an assumption stronger than the timing exogeneity used to motivate the article's identifying comparisons. The trend regressors are fitted from outcomes in months that precede each state's adoption, and each randomization draw reconstructs them using the drawn dates.

5.2.2 Estimates

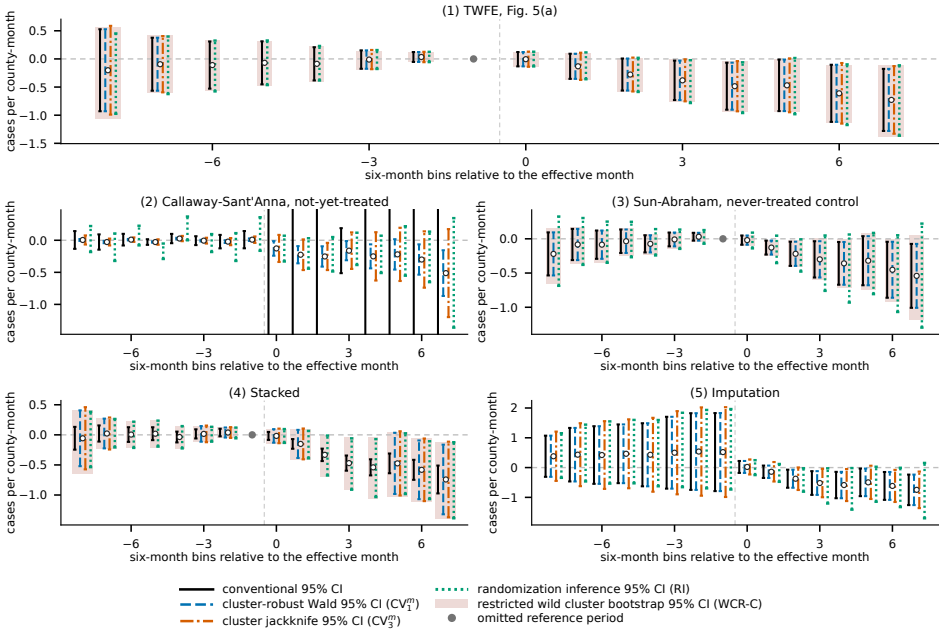
Table 6 reports the estimates for the elder-protection design. The article's two-way Wald test rejects at five percent with $p = 0.0475$, while the jackknife gives $p = 0.057$ and the reported randomization calculation gives $p = 0.041$. The three components therefore do not all reject. In the calibrated simulation, the combined calculation rejects a true zero in 4.0% of replications, compared with 6.7% for the conventional Wald test. The effective-cluster degrees-of-freedom calculation is a sensitivity diagnostic and does not by itself establish the sampling distribution of the statistic.

The alternative estimates are negative, ranging from -0.195 to -0.675 cases per county-month, while their reported inferential results differ. For the stacked estimator, the state-clustered Wald test gives $p = 0.0497$, the jackknife gives 0.062, and the randomization calculation gives 0.041, so the three components do not all reject. The conventional Wald tests for SA and imputation and the analytic standard error for the article's CS specification reject at five percent, whereas their randomization calculations give 0.080, 0.060 and 0.091. These randomization values concern sharp nulls and their non-rejection does not establish that the corresponding average effects are zero.

WCR-C gives $p = 0.054$ for TWFE, 0.052 for SA and 0.057 for the stacked regression, placing each result near the conventional five percent threshold. We also estimate the TWFE specification within six groups of states formed by adoption date. The average of the group estimates is -0.1974, with $p = 0.188$ under the approximate randomization test and $p = 0.130$ under the grouped t-test. These grouped calculations address the group-average effect and complement the inference on the pooled TWFE coefficient.

The MDEs in Table 5 place these estimates in the context of effects that the inference

Figure 7: CUY: event studies with conventional and design-adapted 95% intervals.



Note 1: Each panel is one estimator, in the order of Figure 5: the two-way regression, Callaway–Sant'Anna, Sun–Abraham, the stacked regression and the imputation estimator. The horizontal axis is six-month bins relative to the effective month and the vertical axis is cases per county-month. Symbols, line styles and the omitted-period dot are those of that figure. The published Fig. 5(a) draws 90% intervals; every interval here is 95%.

Note 2: The jackknife and the restricted wild cluster bootstrap are drawn as in Figure 5, and the bootstrap is the one in Table 6. The randomization intervals invert sharp nulls of constant effects under the specified assignment mechanism. Panel (3) omits the reaggregated jackknife because the original target can lose identifying support after state deletion (Appendix A.7). The bootstrap is drawn on panels (1), (3) and (4). It is not reported for Callaway–Sant'Anna or the imputation estimator, so panels (2) and (5) have no box.

Note 3: The conventional interval is the estimator's own, before applying the classification, as in Figure 5. (1) Fig. 5(a) is cluster-robust by state, 51 clusters, with the small-sample factor and Student's t with 50 degrees of freedom. That is the cluster-robust Wald interval, so the solid bar and the dashed bar have the same ends and are drawn apart. (2) Callaway–Sant'Anna is the multiplier bootstrap, 1,000 draws, using the normal critical value 1.96. It is not the cluster-robust Wald interval; the dashed bar beside it clusters by state and uses Student's t with 49 degrees of freedom. The vertical scale of this panel is set by the design-adapted intervals, and the conventional interval runs off it at bins 0, 1, 2 and 4 to 7, where it is about 2 to 8 cases per county-month wide around point estimates between −0.13 and −0.51. The reported bootstrap averages the influence function within each state, which weighs the 50 states equally whatever their number of counties (Table 6, Note 3); summed by state, the same influence function gives standard errors of 0.10 to 0.29 on these bins, the order of the dashed bar. (3) Sun–Abraham is cluster-robust by state, 50 clusters, with the small-sample factor and Student's t with 49 degrees of freedom, the same construction as the cluster-robust Wald interval, so the ends match. (4) The stacked regression is cluster-robust by county × stack, 45,034 clusters, the convention of Baker, Larcker and Wang. The cluster-robust Wald interval clusters by state, and the ends differ. (5) Each horizon is the average of the county-months in that six-month bin. The imputation estimator's conventional bar has the same ends as the cluster-robust Wald interval by state, on Student's t with 49 degrees of freedom.

Note 4: The classification is that of Figure 5. (1) Class U at bins −6, −5 and −4, and class F at the other horizons. The reported pre-period calculation gives 0.900, the larger of 0.797 and 0.900, and does not reject at 5%. Bins +3, +4, +6 and +7 reject under randomization inference (0.035, 0.028, 0.020 and 0.024). Bin +2 does not reject under randomization inference (0.068), and at bin +5 randomization inference gives 0.053. (2) Every horizon is F. The reported pre-period calculation gives 0.084 and does not reject. Randomization inference rejects at bins −5 and −4 (0.008 and 0.050) and at no bin after treatment. (3) Every horizon is F. The reported pre-period calculation gives 0.610 and does not reject. Randomization inference rejects at bin +2 (0.047). (4) Class U at bins −6, −5, −4, +2, +3 and +4, and class F at the other horizons. The reported pre-period calculation gives 0.800, the larger of 0.800 and 0.745, and does not reject. At bins +3 and +4 all three tests reject. Randomization inference also rejects at bins +6 and +7. At bin +2 randomization inference rejects (0.044) and the Wald test and the jackknife do not (0.059 and 0.074), so no rejection is asserted. (5) Every horizon is F. The reported pre-period calculation gives 0.492 and does not reject. Randomization inference rejects at bins +2, +3, +4 and +6 (0.032, 0.036, 0.045 and 0.048), and at no pre-period bin. For panel (2), undefined randomization statistics are counted as at least as extreme as the observed statistic as in Table 6. Randomization values are calibrated under a global sharp null. The pre-period maximum therefore does not generally test a pre-trend null allowing unrestricted post-treatment effects. Horizon-specific results are pointwise and do not constitute a simultaneous test of the post-treatment path.

Note 5: All five panels are six-month bins from 48 months before the effective month to 47 after. Panel (1) is Fig. 5(a) of Carlin et al. (2023), with months −6 to −1 as the omitted bin. Panels (2) to (4) are the robust estimators on the sample that drops Washington, which is treated throughout. Panel (5) is the imputation estimator averaged over the six months in each bin, including bin −1. Panels (1), (3) and (4) omit that bin.

Note 6: Which intervals a class draws is as in Figure 5, and the class is read against the thresholds in Figure B.2. Panels (2), (3) and (5) are class F on every horizon, so the set does not change along the panel. Panel (1) is class U at bins −6, −5 and −4 and class F at the other horizons. Panel (4) is class U at bins −6, −5, −4, +2, +3 and +4 and class F elsewhere.

methods can detect with 95% power in the calibrated design. The TWFE estimate has an absolute value of 0.1951 reports per county-month, compared with a CV_1 MDE of 0.3636. The conventional test rejects zero in the observed sample, while its MDE describes the frequency of rejection across repeated samples under the calibrated model. The estimate is 49.7% of the 0.3926 MDE for the design-adapted calculation that requires all three tests to reject. The absolute CS estimate of 0.2564 is 35.6% of its RI MDE of 0.7193, and its randomization interval is $[-0.4974,\ 0.0483]$ under the constant-effect model used for its inversion. The MDEs describe power across repeated samples under the calibrated disturbances, whereas the intervals summarize uncertainty in the observed sample under their respective assumptions. Their comparison helps assess the precision available for detecting effects of the observed magnitude without treating a non-rejection as evidence that the effect is zero.

For our CS specification, the reported bootstrap standard error is 0.78, compared with 0.078 for the fixed-weight cluster-robust calculation. Aggregating the influence function by state sums while retaining the estimated-weight term gives 0.131. Differences in aggregation and in the treatment of estimated weights contribute to the contrast, and the fixed-weight variance can miss contributions associated with single-state cohorts (Appendix A.4).

The event studies show how the reported evidence varies across horizons and specifications. In panel (1) of Figure 7, every post-treatment bin is class F. The randomization calculation rejects the sharp null at bins +3, +4, +6 and +7, with p values of 0.035, 0.028, 0.020 and 0.024, while the values at bins +2 and +5 are 0.068 and 0.053. In the trend-adjusted study in panel (1) of Figure 8, the randomization calculation gives $p = 0.236$ at bin +1, rejects at bins +3 through +6, and gives $p = 0.067$ at bin +7. The alternative estimators also differ across horizons. These are pointwise results under the stated nulls and do not establish that the effect grows over time or that all later effects are jointly different from zero. As Baker et al. (2026) emphasize, a conclusion about an entire event-study path calls for inference that accounts for the joint distribution across horizons.

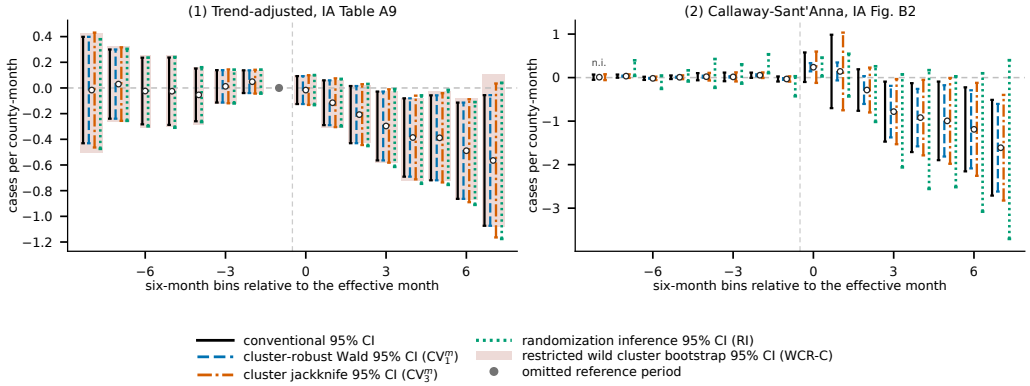
The conventional joint pre-period tests for CS and SA give $p = 0.017$ and a value below 0.0001, respectively, while the randomization-calibrated maximum statistics give 0.084 and 0.610. Randomization inference also rejects at two individual pre-period bins for CS. The randomization reference distribution imposes a global sharp null, and these calculations do not establish parallel trends or provide a general test of pre-period coefficients allowing unrestricted post-treatment effects.

5.2.3 Robustness

Figure 8 presents two additional event studies from the article. Panel (1) adds a county trend and is discussed in Section 5.2.2. Panel (2) uses the article's CS specification, with every

horizon in class F. Its conventional pre-period test gives $p = 0.144$ and the randomization-calibrated maximum statistic gives 0.175 under the global sharp null. Randomization inference rejects at two individual pre-period bins and in the first six months after adoption, where the estimate is positive, and does not reject at the later bins. Non-rejection at those bins does not establish that their average effects are zero.

Figure 8: CUY: event studies in Carlin et al. (2023).



Note 1: The two panels replicate the county-trend specification of Fig. 5(a), Internet Appendix Table A9, column 1, and Internet Appendix Fig. B2 in Carlin et al. (2023). The horizontal axis is six-month bins and the vertical axis is cases per county-month. Symbols, line styles and the omitted-period dot are those of Figure 5. A horizon marked n.i. has a randomization set equal to the whole line.

Note 2: The jackknife and the restricted wild cluster bootstrap are drawn as in Figure 5. The bootstrap is on panel (1), the same procedure as the static intervals in Table 6. Panel (2) is Callaway–Sant'Anna, for which it is not defined.

Note 3: The conventional interval is the estimator's own, before applying the classification, as in Figure 5. (1) Internet Appendix Table A9, column 1, is cluster-robust by state, 51 clusters, with the small-sample factor and Student's t with 50 degrees of freedom. That is the cluster-robust Wald interval, so the solid bar and the dashed bar have the same ends and are drawn apart. (2) Internet Appendix Fig. B2 uses the Callaway–Sant'Anna estimator with the analytic variance clustered by state, with the term for the estimated aggregation weights, using the normal critical value 1.96. It is not the cluster-robust Wald interval; the dashed bar beside it clusters by state and uses Student's t with 49 degrees of freedom.

Note 4: The classification is that of Figure 5. (1) Class U at bins −6, −5 and −4, and class F at the other horizons. The reported pre-period calculation gives 0.741, the larger of 0.641 and 0.741, and does not reject at 5%. Bin +1 does not reject under randomization inference (0.236). Bins +3 to +6 do (0.045, 0.022, 0.044 and 0.023). Bin +7 does not (randomization inference 0.067). (2) Every horizon is F. The reported pre-period calculation gives 0.175, against 0.144 before applying the classification, and does not reject. Randomization inference rejects at bins −7 and −2 (0.041 and 0.017) and at bin 0 (0.038), where the estimate is positive, and at none of the bins after it, where the estimates are negative. Bin −8 puts no weight on its own shift, so its randomization set is the whole line ($p = 0.359$ at zero) and is marked n.i. Randomization values are calibrated under a global sharp null. The pre-period maximum therefore does not generally test a pre-trend null allowing unrestricted post-treatment effects. Horizon-specific results are pointwise and do not constitute a simultaneous test of the post-treatment path.

Note 5: Panel (1) is Fig. 5(a) of Carlin et al. (2023) with the county trend, Internet Appendix Table A9, column 1: the same six-month bins, omitting the bin for months −6 to −1. Panel (2) is Internet Appendix Fig. B2. Its event time is counted from the authors' cohort month, one month before the effective month, and binned in six months, with never-treated controls, no covariates and 48 months on either side.

Note 6: Which intervals a class draws is as in Figure 5, and the class is read against the thresholds in Figure B.2. Panel (1) is class U at bins −6, −5 and −4 and class F at the other horizons, so the set of intervals changes along the panel. Panel (2) is class F on every horizon, so the set does not change, apart from bin −8, whose randomization interval is not drawn.

The elder-protection application shows that estimates and inference can vary across specifications and horizons even when they use the same policy adoption. Several pointwise randomization calculations reject a global sharp null at later horizons, while the reported average-effect results are sensitive to the choice of procedure. The outcome records reports of financial exploitation, so changes in reporting behavior and changes in underlying exploitation can both affect its interpretation. The differences across inference methods therefore matter for the strength of the evidence about the laws and reported exploitation, while non-rejection

alone does not establish that the laws had no effect. The event-study results alone do not establish a progressively growing effect or identify the institutional mechanism behind it.

6 Conclusion

The reliability of inference in staggered difference-in-differences depends on the sample, the estimator and the assumptions used to assess uncertainty. Our census shows that clustered standard errors are widely used in finance research, while information about the clusters supporting inference is reported much less often. As researchers adopt estimators that use different treatment and comparison groups, understanding how those choices affect the distribution of information across clusters becomes important for interpreting empirical findings.

The adoption schedule, cluster sizes, and the comparisons used by an estimator determine which clusters contribute to a treatment effect. Within-cluster dependence also affects those contributions, so a family of disturbances can imply a range of effective cluster counts for the same design. Some score constructions can miss part of the variance associated with a single-cluster cohort or comparison group. These features can differ across estimators and across the horizons of an event study, and the corresponding diagnostics provide complementary information about the reliability of conventional inference.

Exact calculations and simulations show how the behavior of clustered tests varies with these design features. The effective cluster count associated with a close approximation to nominal size differs across estimators in the calibration designs, and the application-based simulations show substantial false rejection for several conventional tests. Alternative procedures have different requirements, including preservation of the original target for a fixed-target jackknife and a stated assignment mechanism for randomization inference. The calibrated thresholds and illustrative classifications show how the diagnostics organize the comparison of inference methods across the designs and disturbances examined.

The two applications show how these considerations inform empirical practice. The original bank deregulation estimate by Beck et al. (2010) remains significant under the reported procedures, while the alternative estimates studied by Baker et al. (2022) and their inference depend on different samples and comparisons. In the elder financial protection study of Carlin et al. (2023), the reported evidence varies across inference methods and event-study horizons. These differences matter for the interpretation of policy effects, while a failure to reject does not establish that an effect is absent. Assessing the sample, the treatment effect of interest and the assumptions of each method helps researchers understand the sensitivity of their conclusions.

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Internet Appendix

How Reliable Is Inference in Staggered Difference-in-Differences?

A Derivations

This section records the procedures of Section 3 and derives the results that section uses. Appendix A.1 sets out the procedures. The derivations then cover the estimators' weights, the score variance under the two-component error and the interval (Appendix A.2), aggregation to cluster-period means (Appendix A.3), groups formed by one cluster and the procedures built on the same scores (Appendix A.4), the coarsened tests on the two designs (Appendix A.5), the exhausted comparison group (Appendix A.6) and the joint pre-trend test (Appendix A.11).

A.1 Setup

There is a finite population $\mathcal{M} = \{1, \ldots, M\}$ of clusters observed over periods $t = 1, \ldots, T$. Cluster m has adoption date $A_m \in \mathcal{G} \cup \{\infty\}$, where ∞ denotes no adoption. Let $Y_{mt}(a)$ be its potential outcome at t under adoption at a, and let $Y_{mt} = Y_{mt}(A_m)$ be the observed outcome. Potential outcomes are cluster-period averages when observations within a cluster are aggregated.

For an adoption cohort a, write $\mathcal{G}_a = \{m : A_m = a\}$. The cohort and period average treatment effect on the treated is

$$\mathrm{ATT}(a,t) = \sum_{m \in \mathcal{G}_a} q_{m|a} \Big[Y_{mt}(a) - Y_{mt}(\infty) \Big], \qquad \theta = \sum_{(a,t) \in \mathcal{I}} w_{at} \mathrm{ATT}(a,t), \tag{A.1}$$

where $t \geq a$, the within-cohort weights $q_{m|a}$ are nonnegative and sum to one, and the aggregation weights w_{at} are nonnegative and sum to one. Equal $q_{m|a}$ give an average over clusters, while weights proportional to the number of units represented by each cluster give the corresponding unit-weighted average. The set of cells and weights define the estimand for the observed design and can differ across estimators and event-study horizons.

Fixing a target at a heterogeneity-robust estimand separates the question of inference from the point-estimator channel studied by Goodman-Bacon (2021) and Baker et al. (2022). With s_g denoting cluster g's share of the observations, the mass-based count $G_{\text{eff}} = 1/\sum_g s_g^2$ describes the panel. The diagnostic in Section 3 uses the weights of the estimator under study.

For the cells on which the target places weight, identification requires the relevant no-anticipation, parallel-trends and overlap conditions, together with the sampling or assignment conditions under which the corresponding observed comparisons identify the stated cohort effects. These conditions are maintained when evaluating inference on an identified estimand.

Tests and sensitivity analysis for parallel trends are studied by Roth and Sant'Anna (2023b) and Rambachan and Roth (2023).

Inference requires a probability model describing the source of uncertainty. Cluster-robust calculations can be motivated by sampling or a model of disturbances, while design-based calculations use assumptions about the assignment of treatment or adoption dates (Aronow and Samii, 2017; Abadie et al., 2020, 2023; Athey and Imbens, 2022). Randomization inference evaluates a sharp null under a specified assignment mechanism (Young, 2019). Under the assumptions of their respective settings, Abadie et al. (2020) and Rambachan and Roth (2026) discuss circumstances in which conventional clustered variance estimators are conservative. These results do not by themselves establish the finite-sample behavior of the tests in the staggered panels studied here. Parallel trends does not generally imply the random assignment mechanism needed for a randomization test. Table A.1 summarizes the hypotheses and assumptions of the procedures considered in the paper, and Appendices A.4 and A.5 discuss the corresponding score and grouped-test calculations.

A.2 The estimators' weights, the two-component error, and the interval

Each estimator brings its own weights. For $\hat{\theta}_m = c_m' y$, cluster g's contribution $c_{m,g}' u_g$ has variance $c_{m,g}' \Omega_g c_{m,g}$, so every display of Section 3 holds with the estimator's own weights, and under independent disturbances $\mathrm{Var}(\hat{\theta}_m) = \sigma^2/\mathcal{A}_m$ with $\mathcal{A}_m = \|c_m\|^{-2}$, the $\mathcal{A}^{-1/2}$ factor of (11). The weights are those of the estimator's own computation: for the two-way and event-study regressions, the stacked regression and the saturated regression of Sun and Abraham (2021), $c_m = Q_m(Q_m'Q_m)^{-1} r_m$ for the regression's design Q_m and the contrast r_m that forms the reported coefficient or aggregate; for Callaway and Sant'Anna (2021), the aggregate of the cell-level difference-in-differences weights, with covariates entering linearly if at all; for the imputation estimator, the weights of Borusyak et al. (2024, p. 3269). Each annihilates unit dummies nested in clusters, through unit dummies in the regressions, stack-by-unit dummies in the stacked regression, differencing within units for Callaway–Sant'Anna and the untreated fit for imputation, so $\sum_{r \in g} c_{m,r} = 0$ and the feasible endpoint is $0/0$ under nesting for every estimator. A panel with one unit per cell has an interval degenerate in λ for every estimator, and the aggregation result of Appendix A.3 holds wherever the weights are cell-constant.

The same substitution produces (8). For a least-squares coefficient the object inside the cluster weight of Carter et al. (2017, eq. (5)) is the row-weight vector $X_g(X'X)^{-1}a$, which for one coefficient with the other regressors partialled out is $\ddot{x}_g/\mathcal{A}$; for $\hat{\theta}_m = c_m' y$ it is $c_{m,g}$. Cluster g's score $S_g = c_{m,g}' u_g$ has variance $c_{m,g}' \Omega_g c_{m,g}$. Under (1), with ν and μ independent and serially uncorrelated, $\Omega_g = \sigma_\mu^2 I + \sigma_\nu^2 Z_g Z_g'$, Z_g the indicator of g's cluster-periods, so $c_{m,g}' \Omega_g c_{m,g} = \sigma_\mu^2 \sum_{r \in g} c_{m,r}^2 + \sigma_\nu^2 \sum_t (\sum_{r \in (g,t)} c_{m,r})^2$, which is $(\sigma_\mu^2 + \sigma_\nu^2)\gamma_g(\lambda)$ of (8);

Table A.1: What each inference procedure tests, assumes, and needs from the design

Procedure (source)	What is tested	Validity rests on	Failure or cost in a staggered design	Checked on the design, before estimation
Panel A. Procedures relying on a central limit theorem across many effective clusters				
Cluster-robust Wald test, t_{G_m-1} critical values (MacKinnon et al., 2023a)	θ_m	conditions for a cluster-level approximation, including sufficient dispersion of variance contributions	size can rise when information is concentrated; some fixed-weight scores miss single-cluster contributions	both ends of the interval, against τ_m; b_m
Wald test with the cluster jackknife variance (MacKinnon et al., 2023b)	θ_m	many effective clusters and a well-defined leave-out target	deletion can change the target or remove identifying support	target and support after each deletion
Unrestricted wild cluster or multiplier bootstrap (Cameron et al., 2008)	θ_m	many conditions for its score approximation	fixed-score versions inherit components absent from the original scores	as the cluster-robust Wald test
Two-way cluster-robust variance (Cameron et al., 2011)	θ_m	many clusters in both dimensions and appropriate dependence assumptions	finite-sample behavior depends on the score construction; the estimated variance can be negative	as the cluster-robust Wald test, and the number of periods
Panel B. Procedures relying on a central limit theorem within each of a fixed number of clusters or groups				
Restricted wild cluster bootstrap, classic version (Cameron et al., 2008; MacKinnon et al., 2023b)	θ_m	few large clusters with homogeneous regressor moments (if studentized, a small size excess), or many clusters	can have substantial size distortion with few informative clusters; the recovery of blind variance depends on the scores and design	the numbers of treated and untreated clusters, and $G^*_m(0)$
Approximate randomization test on q groups of clusters (Canay et al., 2017; Cai et al., 2023)	$\bar{\theta}_j$	a limit of the group estimates invariant to sign changes; the effect identified in every group	with period effects, not identified inside one cluster; $p \geq 2^{1-q}$, so a 5% test needs $q \geq 6$	q fixed beforehand; per group, an adoption in the window, a second status, and its interval
t test on q group estimates (Ibragimov and Müller, 2010, 2016)	$\bar{\theta}_j$	independent, approximately normal group estimates, variances free; the effect identified in every group	with period effects, not identified inside one cluster; valid only at $\alpha \leq 8.3\%$, or 10% if $q \leq 14$	as for the randomization test
Two-sample cluster-level t test (Ibragimov and Müller, 2016)	$\delta_1 - \delta_0$	independent, approximately normal cluster changes over a common window; variances free, means common within type	needs two treated and two never-treated clusters; level bounded as for the t test on groups; the common window costs data; can be conservative	$G_1 \geq 2$, $G_0 \geq 2$, and a non-empty common window
Panel C. Procedures relying on as-good-as-random assignment of adoption dates (design-based; implies parallel trends)				
Randomization inference, studentized, dominant cluster's date held fixed (Hu and Spamann, 2020)	sharp null	the specified conditional assignment distribution and a sharp null; no cluster-level central limit theorem	a weak average-effect null requires additional assumptions beyond studentization	the dominant cluster; the timing assumption is argued, not computed

Notes. Every row maintains identification by (conditional) parallel trends; a row states what a procedure's validity argument assumes beyond it, not whether it holds on a given design. The last column is computed from the dates and the estimator's weights before any outcome is used, or, for timing, argued. What is tested: $\theta_m = c_m' \mathbb{E}y$, the estimand of estimator m, under parallel trends a weighted average of treatment effects with weights the design fixes (Section 3.1); the sharp null, a constant effect in every treated cell, inverted for a confidence set; $\bar{\theta}_j$, the average of the q within-group two-way coefficients, which can differ from the pooled two-way coefficient because the aggregation weights differ; $\delta_1 - \delta_0$, the treated less the never-treated clusters' mean change across a common pre- and post-window. G_1 and G_0 count the treated and the never-treated clusters, and q the groups (Appendix A.5). The interval and $G^*_m(0)$ are defined in Section 3.1; the blind share b_m, and a group of one cluster (a cohort, or a period's comparison group, formed by one cluster), in Section 3.2. τ_m is the estimator's threshold. Figure 4 gives it on the Compustat panel, and Figures B.1 and B.2 give it for each point estimate in calibration designs constructed to match features of the corresponding application. The groups' intervals in Panel B are read against ten, the reference point of Carter et al. (2017, p. 704) (Appendix A.5).

Sources. Panel A: Cameron and Miller (2015); Callaway and Sant'Anna (2021, Remark 10) for the multiplier bootstrap; Appendix A.4. Panel B: Canay et al. (2021, Asm. 2); Djogbenou et al. (2019); MacKinnon and Webb (2017, §§6–7); MacKinnon and Webb (2020, p. 443); Canay et al. (2017, §2.1, p. 1026); Cai et al. (2023, pp. 89, 96); Ibragimov and Müller (2016, pp. 84, 87); Hu and Spamann (2020, p. 9, fn. 12). Panel C: Young (2019, p. 558); Romano (1990); Janssen (1997); Chung and Romano (2013).

G^* is the inverse Herfindahl of the shares $\gamma_g/\sum_h \gamma_h$ and does not change when every γ_g is multiplied by the same positive number. Three boundary facts belong with it. The weights take both signs, so the two channels have no general ordering; by Cauchy–Schwarz $(\sum_{r\in(g,t)} c_{m,r})^2 \le n_{gt} \sum_{r\in(g,t)} c_{m,r}^2$, so the cell term is at most $\max_t n_{gt}$ times the idiosyncratic term, with near-equality when the weights are nearly constant within a cell, the case of a cell-level treatment without controls. With one row per cell each inner sum has one term, the channels coincide and $G^*(\lambda)$ is constant in λ; with one period the cell is the cluster and the cell channel is the cluster-wide term $(\sum_{r\in g} c_{m,r})^2$ of MacKinnon et al. (2023c, eq. (17)). With cluster-by-period fixed effects the cell sums vanish and only the idiosyncratic channel remains. Those effects also absorb a cluster-level treatment.

Nesting of unit effects inside clusters is what makes the feasible count indeterminate. If unit effects are nested in clusters, the cluster indicator is the sum of its units' dummies and lies in the span of the fixed effects, to which $\ddot{x}$ and the weights of every estimator above are orthogonal, so $\sum_{r\in g} c_{m,r} = 0$. In the family $\Omega_g(\rho) = (1-\rho)I + \rho\,\iota\iota'$, $\gamma_g(\rho) = (1-\rho)\gamma_g(0) + \rho(\sum_{r\in g} c_{m,r})^2$ (MacKinnon et al., 2023c, eq. (17)), which is then $(1-\rho)\gamma_g(0)$: $G^*(\rho) = G^*(0)$ for $\rho < 1$, and at $\rho = 1$ every weight is zero. The weights of the feasible measure are proportional to $(\sum_{r\in g} c_{m,r})^2$ (Carter et al., 2017, p. 703), so it is $0/0$. For $\rho < 1$ the resulting count equals $G^*(0)$, including in the limit as ρ approaches one. Every feasible weight vanishes exactly when the weights are orthogonal to every cluster indicator. Nesting is sufficient for that orthogonality. Period effects alone can produce it as well. With period effects only, two clusters of one unit each, two periods, and treatment $(1,0)$ for one unit and $(0,1)$ for the other, the residualized treatment is $(\frac{1}{2}, -\frac{1}{2})$ and $(-\frac{1}{2}, \frac{1}{2})$, so both cluster sums are zero. Period dummies do not span the cluster indicators. Under unit effects that are not nested, $\ddot{x}$ still sums to zero over each unit's rows, so $\sum_{r\in g} \ddot{x}_r$ is the sum, over the units that move, of their rows while in g: units that never leave g contribute nothing, and the feasible weights are the squared partial sums of the movers' rows.

Serial correlation enters the interval through the grid of Section 3.1. Let $x_i \in \mathbb{R}^T$ be unit i's path of weights (zero where it has no row), $b_g = \sum_{i\in g} x_i$ the cluster-period sums, and $R(\rho_e)_{ts} = \rho_e^{|t-s|}$. With both components of (1) first-order autoregressive at ρ_e, cluster g's score variance is proportional to $\gamma_g(\varphi, \rho_e) = \sum_{i\in g} x_i' R(\rho_e) x_i + \varphi\, b_g' R(\rho_e) b_g$, and $G^*(\varphi, \rho_e) = (\sum_g \gamma_g)^2 / \sum_g \gamma_g^2$. At $\rho_e = 0$ the two terms are the channels of (8). The interval is the minimum and maximum over $\varphi \in \{0\} \cup \{10^{-8+0.1j}\}_{j=0}^{160} \cup \{\infty\}$ and the grid of ρ_e of Section 3.1. The family at $\rho_e = 0$ is contained in the family over the grid, and the range of G^* over a larger family contains its range over a smaller one, so relaxing the serial restriction can only widen the interval. The serially correlated within-cluster process of Bertrand et al. (2004), which lies outside (1), cannot narrow it either. A narrow interval computed inside

this family does not clear a design whose dependence lies outside the family.

A.3 Aggregation to cluster-period means

Suppose the treatment, the fixed effects and the controls are constant within cluster-periods. Writing the micro design as $Z = S\bar{Z}$, with S the membership in each cluster and period matrix, gives $Z'Z = \bar{Z}'N\bar{Z}$ and $Z'y = \bar{Z}'N\bar{y}$ with $N = \mathrm{diag}(n_c)$, the normal equations of the n_c-weighted regression on cluster-period means, so $\hat{\beta}$ is the same. The residualized treatment is cell-constant, with value r_c on cell c, so $\|\ddot{x}_g\|^2 = \sum_{c\in g} n_c r_c^2$, which the weighted problem reproduces, as it does the partial leverages, $G^*(0)$, the cluster scores and CV_3; CV_1 differs only through its small-sample factor, whose N and k change. Collapsing and weighting by $\sqrt{n_c}$ leaves the disturbance on cell c equal to $\sqrt{n_c}\,(\nu_c + \bar{\mu}_c)$, with variance $n_c\sigma_\nu^2 + \sigma_\mu^2$, and the weighted regressor $\sqrt{n_c}\,r_c$; the two channels are $\sum_{c\in g} n_c r_c^2 = \gamma_g^{\mathrm{idio}}$ and $\sum_{c\in g} n_c^2 r_c^2 = \gamma_g^{\mathrm{cell}}$, both carried across unchanged, so the collapsed path $G^*(\lambda)$ is the micro path at every λ and the collapsed interval is the micro interval. That premise is sufficient. Writing a control as its between-cell part plus its within-cell part, the micro and collapsed normal equations also coincide when the within-cell regression reproduces the between-cell coefficients, $\tilde{Z}'\tilde{Z}b = \tilde{Z}'\tilde{y}$, which includes the case $\tilde{Z} = 0$. A micro problem that is itself weighted needs weighted cluster-period means and $\sum_{i\in c} w_i$ in place of n_c.

A.4 Groups formed by one cluster, and the procedures on the same scores

Let $\hat{\theta}_m = c_m' y$ be Sun–Abraham's saturated regression, Callaway–Sant'Anna's aggregate of cell effects or the imputation estimator, with aggregation weights held fixed in CV_1^m. In period t the clusters fall into groups: each treated cohort a, from its base period $a-1$ on, and the comparison group, the clusters in their comparison role at t. With w_{gt} cluster g's share of its group's weight at t, $|\sum_{i\in(g,t)} c_{m,i}|$ over the group's sum,

$$f_{gt} \;=\; 1 - 2w_{gt} + \textstyle\sum_h w_{ht}^2. \tag{A.2}$$

Units are nested in clusters throughout.

For Sun–Abraham, without controls, if a treated cohort consists of the units of one cluster g, its relative-period indicators equal g's cluster-period indicators, g's unit indicators complete the reference period, and each period indicator splits into its parts on and off g. The column span is then the direct sum of a subspace supported on g's rows and one supported on the others, so the residual maker is block diagonal between them, its block M_{gg} on g's rows annihilates the first subspace and so is singular, c_m restricted to g lies in that subspace, and $\hat{S}_g = (M_{gg}c_{m,g})'y_g = 0$: CV_1^m depends on the other clusters' outcomes only. With controls, the indicator of every period in which the cohort carries its own relative-period indicator

still lies in the column span, which the residual maker annihilates, so no score depends on g's cluster-by-period shocks there. If the reference cohort is one cluster h, every period indicator minus the treated cohorts' period indicators is h's cluster-period indicator, so no score depends on h's cluster-by-period shocks.

For Callaway–Sant'Anna, with or without covariates, a side's influence term in a cell is $(\Delta Y_i - \bar{m})/n$, or $\{\Delta Y_i - \hat{m}_0(X_i) - \hat{\eta}\}/n$, centred at the side's mean; the estimation-effect terms involve the treated units only through their covariates. With side shares w_h, cluster h's score contains $w_h(\nu_h - \sum_k w_k \nu_k)$ times the cell's weight, so the coefficient of $\mathrm{Var}(\nu_g)$ in $\sum_h \hat{S}_h^2$ is $w_g^2\{(1-w_g)^2 + \sum_{h\neq g} w_h^2\} = w_g^2 f_g$, against w_g^2 in $\mathrm{Var}(\hat{\theta}_m)$. In a cell whose treated units, or whose comparison units, all lie in one cluster, $f = 0$ and that side's terms sum to zero inside the cluster.

For the imputation estimator, $\hat{\tau} - \tilde{\tau}$ sums to zero in each cohort-by-period group, $\tilde{\tau}$ being the group mean and the target's weight constant on the group. At a period at which one cluster h is the only untreated cluster, the period effect is fitted to h's untreated rows alone, so the cell indicator lies in the span of the untreated fit, and the treated-side term moves only through the fitted period effect, common to the treated rows of the period, which the group deviation removes. In both cases no score depends on that cluster's shock. In the form of Borusyak et al. (2024, Appendix A.9), whose independent units (their Assumption 5) are the clusters, $\tilde{\tau}$ is the v^2-weighted mean of the clusters' own weighted means $\hat{T}_{jg}$ in the group, which leaves a group formed by one cluster no deviation either. Their leave-out version, $\tilde{\tau}_{ig}^{\mathrm{LO}} = \sum_{j\neq i} v_{jg}^2 \hat{T}_{jg} / \sum_{j\neq i} v_{jg}^2$, rescales cluster i's residuals by $1/(1-s_i)$, $s_i = v_{ig}^2 / \sum_j v_{jg}^2$: it offsets the treated side's leverage loss, the fraction f being $1 - s_i$ in that form, and is undefined where $s_i = 1$ (their fn. 52). Their bound that also leaves the cluster out of the period effects (Proposition A8) needs $\hat{\delta}_{-h}$ unbiased, which a period whose only untreated cluster is h does not allow.

The same score algebra covers a least-squares contrast. Unless stated otherwise, $\hat{\theta}_m = r_m' \hat{\beta}$ is a least-squares contrast from the regression of y on Q_m, of full column rank and fixed by the design, so that $c_m = Q_m (Q_m' Q_m)^{-1} r_m$ lies in the column span of Q_m; the cluster scores of Callaway and Sant'Anna (2021) are linear in y as well, and the score-bootstrap results apply to them with their own score matrix. Regressors fitted from y are outside this setting. Let M_m be the residual maker of Q_m, $\hat{e}_m = M_m y$, and K_m the $G_m \times n$ matrix with rows $(K_m)_{g\cdot} = \sum_{r\in g} c_{m,r} (M_m)_{r\cdot}$, so that CV_1^m's scores are $\hat{S} = K_m y$ and, for $\mathbb{E}y = \mu$ and $\mathrm{Var} y = \Omega$, $\mathbb{E}\,\mathrm{CV}_1^m = \mathrm{ssc}_m\{\mathrm{tr}(K_m \Omega K_m') + \|K_m \mu\|^2\}$; under the null of a constant effect μ lies in the span and $K_m \mu = 0$. A component H of Ω is blind when $K_m H^{1/2} = 0$. It raises $\mathrm{Var}(\hat{\theta}_m)$ by $c_m' H c_m$ and leaves $\mathbb{E}\,\mathrm{CV}_1^m$ unchanged. Write $L = (L_1^m, \ldots, L_{G_m}^m)'$, so that $\sum_g L_g^m = 1$ and $\sum_g (L_g^m)^2 = 1/G_m^*(0)$, $P_c = c_m c_m' / \|c_m\|^2$, $V_m = c_m' \Omega c_m$, and $M_\perp = M_m + P_c$, the residual

maker of the nuisance terms.

The blind share describes the contribution of shocks associated with single-cluster groups in the specified group-level score construction. With $\Omega = \Omega_0 + \Omega'$ and Ω' supported on cells whose shock directions are annihilated by the full score matrix K_m, $\mathbb{E}\,\mathrm{CV}_1^m = \mathrm{ssc}_m\,\mathrm{tr}(K_m \Omega K_m')$ does not depend on Ω'. A cell that enters other comparisons need not satisfy this condition, even when its contribution to one group-specific score vanishes. $\mathrm{Var}(\hat{\theta}_m)$ rises by $c_m' \Omega' c_m$. Under (1) the components are serially uncorrelated and cell-blocked, so cells contribute additively to $\mathrm{Var}(\hat{\theta}_m)$, and when every such cell enters its own group only, the share lost is $b_m(\lambda)$ of (13). Adding the aggregation-weight term to the cluster sums, as the influence function of Callaway and Sant'Anna (2021, Corollary 2) does, replaces w_{gt} by g's share q_j of all treated units at the horizon: the weight term of a treated cluster j at horizon e is $q_j\{\mathrm{ATT}(a_j, e) - \mathrm{ATT}(e)\}$, its coefficient on the shocks becomes $q_j(\nu_j - \sum_k q_k \nu_k)$, and the algebra above applies with q in place of w. CV_0^m computed with the weight term therefore sees a cluster's own shocks with the leverage it has among all treated clusters at a horizon, and so sees a cohort formed by one cluster unless that cohort is alone at the horizon.

The fraction is exact for every cell that enters one group only, which under never-treated comparisons is every cell; under not-yet-treated comparisons a cluster's pre-adoption cells serve several groups and the fraction is a mixture. Computed for the estimators over six balanced synthetic designs, one of them with a cluster holding half the units, $\mathrm{ssc}_m\{1 - \bar{b}_m(\lambda)\}$ equals CV_1^m's exact expected scale at both ends for Sun–Abraham and for Callaway–Sant'Anna with never-treated comparisons, and is at least as large for not-yet-treated comparisons and for the imputation estimator, whose unit effects, fitted to each unit's own untreated rows, hide a further share; in these computations $\bar{b}_m$ never overstates the loss. A leave-one-cluster-out calculation can reflect the removal of a cohort through a change in the aggregate, but it can also change other estimated coefficients and the aggregation weights. The difference between the full and leave-out estimates therefore cannot generally be attributed to the removed cohort alone. When a positively weighted cell is no longer identified, reaggregation gives a different finite-sample target and does not restore a fixed-target jackknife variance. Appendix A.7 examines this issue in the two applications. A randomization test can accommodate a statistic that is blind in some assignments when the null is sharp, the reference distribution matches the assignment mechanism and the statistic is defined on the full assignment support. Appendix A.9 gives the convention used when the original statistic is undefined.

A score bootstrap draws weights on the same scores. Let a draw be $T^* = \sum_g w_g \hat{S}_g$, the w_g independent of y and of each other, with mean zero and variance one. For a least-squares contrast, T^* is exactly $\hat{\theta}_m^* - \hat{\theta}_m$ under the unrestricted wild cluster bootstrap $y^* = Q_m \hat{\beta} + \mathrm{diag}(w_{g(r)})\hat{e}_m$, since $c_m' Q_m \hat{\beta} = \hat{\theta}_m$ and $c_m' \,\mathrm{diag}(w)\hat{e}_m = \sum_g w_g \hat{S}_g$; with the cluster

means of the influence function in place of $\hat{S}$ and a fixed weight per cluster, it is the clustered multiplier bootstrap. Given y, T^* has mean zero and variance $\|\hat{S}\|^2$, which is CV_0^m. Its conditional law depends on y only through $K_m y$, and a blind component leaves the distribution of $K_m y$, and with it that of the bootstrap law, unchanged: with $u = u_0 + u_H$ independent and $\mathrm{Var} u_H = H$, $\mathrm{Var}(K_m u_H) = K_m H K_m' = 0$. The CV_1^m-studentized unrestricted bootstrap depends on y only through $\hat{e}_m$, because $y^* - Q_m\hat{\beta} = \mathrm{diag}(w)\hat{e}_m$, and is invariant in the same sense to every component whose columns lie in the span of Q_m, among them the cluster-by-period shocks of a group formed by one cluster. A score bootstrap thus replaces the t_{G_m-1} reference by the law of a weighted sum of the same scores. The shape of the reference distribution can change, and the scale remains the scale of those scores.

The restricted wild cluster bootstrap imposes the null before the draw. With $\theta_m = \theta_0$, $\tilde{\beta}$ is the restricted estimate, $\tilde{u} = y - Q_m\tilde{\beta}$, the draws are $y^* = Q_m\tilde{\beta} + \mathrm{diag}(w_{g(r)})\tilde{u}$, and $d = \hat{\theta}_m - \theta_0$. Since $r_m'(Q_m'Q_m)^{-1}r_m = \|c_m\|^2$, $\tilde{u} = \hat{e}_m + c_m d/\|c_m\|^2$, so exactly $\hat{\theta}_m^* - \theta_0 = \sum_g w_g \tilde{S}_g$ with $\tilde{S}_g = \hat{S}_g + L_g^m d$, $\sum_g \tilde{S}_g = d$, and

$$\mathrm{Var}^*(\hat{\theta}_m^*) \;=\; \|\hat{S}\|^2 + 2d\sum_g L_g^m \hat{S}_g + \frac{d^2}{G_m^*(0)};$$

the draw with every $w_g = 1$ returns the data. Where one cluster carries almost all the partial leverage, $G_m^*(0)$ is near one and $\tilde{S}_1 \approx \hat{S}_1 + d$: the draws $\pm\tilde{S}_1$ are as large as the estimate, which is then never extreme among them, the under-rejection with one treated cluster of MacKinnon and Webb (2017, §6, pp. 244–245). Under the null, $\tilde{S} = \tilde{K}_m u$ with $\tilde{K}_m = K_m + L c_m'$, whose row g is $\sum_{r\in g} c_{m,r}(M_\perp)_{r\cdot}$: the restricted scores are cluster scores computed with the nuisance terms' residual maker. A blind component H therefore raises $\mathbb{E}\,\mathrm{Var}^*$ by $c_m' H c_m / G_m^*(0)$, so the restricted draws return the share $1/G_m^*(0)$ of what CV_1^m and every score bootstrap miss. If the projection on the nuisance terms is block diagonal by cluster, as it is for unit effects and for cluster-specific terms when there are no period effects, then $\tilde{S}_g = c_{m,g}' u_g$, cluster g's own contribution to $\hat{\theta}_m - \theta_0$, and $\mathbb{E}\,\mathrm{Var}^* = V_m$ for every Ω block diagonal by cluster. With Rademacher weights $\mathrm{diag}(w)$ then commutes with $M_\perp$, every draw's statistic, studentized or not, is the original statistic evaluated at the disturbance $\mathrm{diag}(w)u$, and if the u_g are independent and each symmetric about zero the bootstrap over all 2^{G_m} sign vectors is a randomization test (Canay et al., 2017) with level at most $\lfloor \alpha 2^{G_m-1} \rfloor / 2^{G_m-1}$: the exact case behind Canay et al. (2021). The period effects of a difference-in-differences regression are not nested in clusters, so the exactness is lost. In toy two-way designs with twelve clusters, eight periods and homoskedastic errors, the CV_1^m-studentized Rademacher WCR over all sign vectors rejects a true null at 5% in 0.000 and 0.008 of 10,000 replications with one and two treated clusters ($G^*(0) = 1.19$ and 2.86), against CV_1^m's 0.544 and 0.193;

with three to six treated clusters it rejects 0.052 to 0.064, against 0.059 to 0.104.

Multiway clustering changes the partition used to form the cluster scores. A shock direction that is annihilated by every partition remains invisible to the resulting variance estimate. For a partition π of the rows let K_m^π have rows $\sum_{r\in h} c_{m,r}(M_m)_{r\cdot}$, $h \in \pi$, and $\hat{V}_\pi = \mathrm{ssc}_\pi \|K_m^\pi y\|^2$. The two-way estimator of Cameron et al. (2011) by assignment cluster S and period T is $\hat{V}_{S,T} = \hat{V}_S + \hat{V}_T - \hat{V}_{S\cap T}$, and its expectation is linear in Ω, as $\mathbb{E}\,\mathrm{CV}_1^m$ is. Since $K_m^\pi v = 0$ for every partition and every v in the span of Q_m, a component whose columns lie in the span reaches no partition's scores: if it loads on c_m, as the cluster-by-period shocks of a group formed by one cluster do, it is blind to all of them, and the component remains invisible under both one-way and multiway clustering. For Sun–Abraham on a balanced panel without controls, every cohort-by-period cell indicator lies in the span and c_m is constant within cells, so $K_m^T = 0$, $\hat{V}_T \equiv 0$ and $\hat{V}_{S,T} = \hat{V}_S - \hat{V}_{S\cap T} \le \hat{V}_S$: two-way clustering can only shrink the one-way variance. If clusters are independent, then under the null $\mathbb{E}(\hat{V}_T - \hat{V}_{S\cap T})$ with unit small-sample factors is $\sum_t \sum_{g\neq h} \mathrm{Cov}(C_{gt}, C_{ht})$, with $C_{gt} = \sum_{r\in(g,t)} c_{m,r}\hat{e}_{m,r}$ the residual cell scores: covariances across clusters that the residual maker alone creates. Within (1) the two-way estimator addresses the same sampling variance as CV_1^m but can have different finite-sample behavior and can be negative. The family (1) makes clusters independent given the period effects. Two-way clustering is built for dependence across clusters within a period that period effects do not absorb, such as unequal exposure of states to an aggregate or industry shock, or spatial correlation. There the interval is not derived, and CV_1^m by state omits the covariances across clusters at any G; a design in which such dependence is substantive is outside what the diagnostic of Section 3 certifies.

A.5 The coarsened tests on the two designs

The approximate randomization test (ART) of Cai et al. (2023) and the Ibragimov–Müller t on groups (IM) of Ibragimov and Müller (2016) reach their p-values through estimates formed inside q groups of clusters. Section 3.3 discusses the information within these groups. We report the grouped calculations as supplementary evidence on the average of the group-specific estimates. No theory selects q (Cai et al., 2023, p. 100), and choosing q or the grouping by results would be data snooping (their p. 96), so the groups below are fixed by the design alone.

The approximate randomization test has a floor on its p-value. The ART p-value is the share of sign vectors $\sigma \in \{\pm 1\}^q$ whose statistic is at least the observed one (Canay et al., 2017, eq. (9), p. 1016), (Cai et al., 2023, Algorithm 2.1). Because $T_{-\sigma} = T_\sigma$ and the identity always counts, $p \ge 2^{1-q}$, so the test can reject at level α only if $2^{1-q} \le \alpha$; at 5% that is $q \ge 6$, the bound at which Cai et al. (2023, p. 89) say the test has non-trivial power. IM refers its

statistic to t_{q-1} and has no discreteness floor; its validity under heterogeneous variances, a result of Bakirov and Székely reported by Canay et al. (2017, p. 1017), holds for $\alpha \leq 0.083$ at every $q \geq 2$ and for $\alpha \leq 10\%$ when $2 \leq q \leq 14$. With period effects and a cluster-level treatment the coefficient is not identified inside a single cluster (Cai et al., 2023, eq. (29), p. 96), and on the two designs that holds in every cluster. Under their remedy A, which refits the fixed effects and controls on each group's own rows, every group must therefore carry a cluster that switches inside the window and a second adoption status.

Coarsening partitions the effective clusters the design already has. For any share vector L with $\sum_g L_g = 1$ and any partition into q groups, with $W_j = \sum_{g \in J_j} L_g$ and $G_j^* = W_j^2 / \sum_{g \in J_j} L_g^2$, $\sum_j W_j^2 / G_j^* = \sum_j \sum_{g \in J_j} L_g^2 = 1/G^*$. Since every G_j^* is at least $\min_k G_k^*$, $1/G^* \leq \sum_j W_j^2 / \min_k G_k^*$, which is (14); $\sum_j W_j^2 \leq \max_j W_j$ because the W_j sum to one. The identity holds at every point of the disturbance family; it is exact for restricted shares and approximate once the fixed effects are refitted inside groups. For every one of $q \geq 6$ balanced groups to reach ten, the reference point of Carter et al. (2017, p. 704), a design therefore needs G^* of about sixty or more, while the upper ends of the two designs' TWFE intervals are 46.52 and 28.46.

On the two designs the groups are fixed before any outcome is seen. Sort the clusters by adoption date, never-treated clusters last and ties by their order in the panel, and deal them round-robin into the six groups that the 5% floor requires. On both designs every group then identifies the effect, and none reaches ten: computed on the group's own residualized treatment and over the design's disturbance grid, the weakest group's lower end $G_{j,\mathrm{lo}}^*$ is 4.88 on Beck et al. (2010) and 2.53 on Carlin et al. (2023), and even the upper end of every group's own interval stays below ten, the largest being 8.69 and 6.80.

For bank deregulation, the equal-weighted average of the six group-specific estimates is -0.0227, and the ART and IM p values are 0.031 and 0.033. For elder protection, the corresponding average is -0.1974, with p values of 0.188 and 0.130. The estimates within each group use that group's observations and fixed effects. Their equal-weighted average differs in general from the pooled TWFE coefficient under heterogeneous effects. The grouped tests depend on their respective distributional and independence conditions, and the effective-cluster counts describe the information available within the groups.

A.6 An exhausted comparison group

Consider a panel in which every cluster has adopted by $\bar{g}$, and let $t_* \geq \bar{g}$ be an observed period. Suppose the aggregate in (A.1) places positive total weight

$$W_* = \sum_{a:(a,t_*) \in \mathcal{I}} w_{at_*} > 0$$

on treatment effects at t_*. The conclusion below applies when untreated potential outcomes are unrestricted in level, no anticipation holds, and parallel trends restricts differences in untreated changes across cohorts without imposing an additional restriction on their common time effect.

To see the identification problem, start from any potential-outcome schedule consistent with the observed data and these assumptions. For any real number η, add η to $Y_{mt_*}(\infty)$ for every cluster, and also add η to $Y_{mt_*}(a)$ for every hypothetical adoption date $a > t_*$. Leave all potential outcomes at other periods and all $Y_{mt_*}(a)$ with $a \leq t_*$ unchanged. No anticipation remains satisfied and common shifts preserve the untreated parallel-trends restrictions. Because every observed cluster is treated by t_*, the transformation leaves the observed outcomes unchanged. It changes each treatment effect entering at t_* by $-\eta$ and leaves all other effects entering the target unchanged. Therefore the transformed target equals

$$\theta_\eta = \theta_0 - \eta W_*.$$

The observed data cannot distinguish these schedules, and the identified set for this aggregate is the real line under the stated unrestricted outcome model. A target that assigns no weight to a post-exhaustion cell is not covered by this argument. Cells for cohorts with no observed untreated period also require separate support conditions, and their identification cannot be inferred from parallel trends alone.

Let C be a confidence set with coverage of at least $1 - \alpha$ for every schedule in this family. The schedules share the same distribution of observed data, so the union bound gives

$$\Pr\{\text{diam}(C) \geq |\eta| W_*\} \geq 1 - 2\alpha$$

for every η. Taking $|\eta|$ to infinity and using continuity of probability gives $\Pr\{\text{diam}(C) = \infty\} \geq 1 - 2\alpha$. Thus a uniformly valid confidence set cannot have finite diameter with probability greater than 2α over this family. Restrictions on untreated outcomes after the last adoption date, such as bounds on their evolution, can narrow the identified set (Rambachan and Roth, 2023).

A.7 Support of the bank-deregulation jackknife

The Sun–Abraham implementation on the restricted bank-deregulation sample includes cohort and relative-period indicators and aggregates the retained post-treatment coefficients using their observation weights. We inspect the regressors retained in the full-sample specification and, for each state, identify indicators whose observations all belong to that state. Ten of the 36 states uniquely support at least one retained coefficient with positive weight in the

original aggregate. When one of these states is deleted, the associated indicator is identically zero in the remaining data, and its coefficient is not identified from that leave-one-out sample. The original fixed-weight aggregate therefore cannot be recovered from that sample without additional restrictions. A jackknife that reaggregates the surviving cells can consequently remain finite while changing the weights or the set of cells in the target. In addition, deleting the sole last-treated comparison state leaves the original weighted aggregate outside the row space of the remaining design matrix. We do not use the reaggregated jackknife as a fixed-target variance estimate for this specification. A separate sensitivity analysis ends the sample in 1993, retains two untreated comparison states and includes treatment cohorts with at least two states. It fixes the weights of the supported aggregate before deletion. The full design matrix and both unrestricted and restricted design matrices remain of full column rank after deleting each state. The resulting estimate and jackknife refer to that restricted sample and aggregate. In the elder-protection Sun–Abraham specification, fifteen states each uniquely support at least one coefficient receiving positive weight in the original aggregate. We likewise omit its reaggregated jackknife from the fixed-target comparison. The elder-protection sensitivity sample retains never-treated states and treatment cohorts with at least two states, fixes the aggregate weights and passes the corresponding full-rank checks for every deletion.

A.8 Sharp and average-effect null hypotheses

Let A denote the vector of cluster adoption dates and let $\mathcal{A}$ be its assignment support. Assume no interference across clusters and no anticipation. For a constant c, consider the sharp null

$$H_0^S(c): \quad Y_{mt}(a) = Y_{mt}(\infty) + c\,\mathbf{1}\{t \geq a\} \quad \text{for every } m, t \text{ and } a \in \mathcal{A}_m, \tag{A.3}$$

where $\mathcal{A}_m$ contains the dates cluster m can receive. This null determines the missing outcomes at every assignment in the support from the observed outcomes. By contrast, the average-effect null $H_0^W(c): \theta = c$ imposes only a restriction on the aggregate in (A.1) and permits treatment effects to differ across clusters, cohorts and periods.

Under (A.3), every cohort effect in (A.1) equals c, so $H_0^S(c)$ implies $H_0^W(c)$. The reverse implication does not hold. For example, equally weighted effects $c - d$ and $c + d$ have average c for any nonzero d, while they violate the constant-effect null. Consequently, rejection of $H_0^S(c)$ is evidence against that constant-effect schedule and does not by itself reject $H_0^W(c)$. A randomization confidence set obtained by inverting (A.3) covers a common constant effect under the corresponding model. Without that model, it is not automatically a confidence set

for the average effect.

Studentizing a randomization statistic does not change which potential outcomes are imputed under the null or make a weak-null test finite-sample exact. Asymptotic validity under heterogeneous effects requires additional conditions appropriate to the assignment, statistic and asymptotic sequence. The randomization calculations reported in this paper therefore test the sharp null unless a separate weak-null justification is supplied. In particular, a comparison with a Wald test of the average effect is a comparison of procedures and hypotheses, and its interpretation as a comparison of tests of the same average-effect null requires additional assumptions. If the model restricts all relevant treatment effects to be a common constant, the sharp and weak nulls coincide within that model.

A numerical example shows why studentization alone cannot establish finite-sample validity for the weak null. Consider eight units with four assigned to treatment uniformly at random. Their untreated outcomes and treatment effects are, respectively,

$$Y(0) = (-0.104, -0.063, 0.084, -0.068, 0.050, -0.149, -0.220, 0.116),$$
$$\tau = (2.831, -1.327, -23.523, 5.424, 5.613, 3.543, 2.148, 5.291).$$

The average effect is zero. For each assignment, compute the two-sided studentized difference in means and its permutation p-value under the sharp zero-effect null, including ties. Enumeration of all 70 assignments gives 15 rejections at the five percent level, or a rejection probability of 0.214. This balanced experiment establishes that a general finite-sample weak-null guarantee does not follow from studentization alone.

A.9 Randomization statistics on all assignments

Let the assignment mechanism and a sharp null satisfy the conditions in Appendix A.8, and suppose the randomization statistic is undefined for some assignments. Define an extended statistic by assigning positive infinity to every such assignment and leaving the statistic unchanged elsewhere. If the observed statistic is undefined, the test does not reject. With the observed assignment and independent draws from the actual conditional assignment distribution, the observed extended statistic and the simulated extended statistics are exchangeable under the sharp null. Their upper-tail rank with ties counted as extreme is therefore super-uniform. The corresponding Monte Carlo p value has a numerator equal to one plus the number of simulated extended statistics at least as large as the observed statistic and a denominator equal to one plus the total number of draws. Its rejection probability is no greater than the nominal level, with discreteness and ties potentially making it conservative. Inversion gives a confidence set for a constant effect under the same assignment

and constant-effect assumptions.

For the elder-protection Callaway–Sant'Anna aggregate, three of the random assignments have undefined statistics. Counting them as extreme yields $p = 0.109$ for the sharp zero-effect null. The same convention is applied to each affected event-study statistic and to its maximum statistic. In the simulation of this estimator, one of the random assignments is undefined. Including it as extreme leaves the reported rejection frequencies unchanged at the displayed precision. These calculations do not establish validity under a weak average-effect null or under an assignment mechanism different from the one used to draw adoption dates.

A.10 Conditional validity of the design-adapted calculations

Let Z collect the fixed design information and the quantities on which assignment is conditioned, including the adoption date of a prespecified dominant cluster when it is held fixed. Let $\pi(a \mid Z)$ be the actual conditional probability of assignment a, and let $S(a, Z)$ be the class assigned by the prespecified design-based rule. For a class s with positive probability, define

$$\mathcal{A}_s(Z) = \{a \in \mathcal{A} : S(a, Z) = s\}, \qquad \pi_s(a \mid Z) = \Pr\{A = a \mid Z, S(A, Z) = s\}.$$

The classification and all variables used to determine it are specified without using outcomes. When the class is invariant over the support conditional on Z, conditioning on the class makes no further restriction. Otherwise, a randomization distribution conditional on the selected class must use π_s and the support $\mathcal{A}_s$.

Under $H_0^S(c)$, impute the potential outcomes for every assignment in $\mathcal{A}_s$ and recompute the prespecified statistic, including any assignment-dependent regressors and weights. Assume the statistic is defined throughout this support. With an absolute studentized statistic $T_c(a)$, let

$$p_{R,s}(A) = \sum_{a \in \mathcal{A}_s(Z)} \pi_s(a \mid Z) \mathbf{1}\{T_c(a) \geq T_c(A)\}.$$

For every $\alpha \in [0, 1]$,

$$\Pr\{p_{R,s}(A) \leq \alpha \mid Z, S = s, H_0^S(c)\} \leq \alpha. \tag{A.4}$$

To prove this, order the distinct values of T_c from largest to smallest. The randomization p-value at a value of the statistic is the cumulative assignment probability at that value and all larger values. The assignments for which this cumulative probability is at most α have combined probability at most α. Including ties in the upper tail makes the test conservative when necessary. Thus the class-F test is conditionally valid for the sharp null under the stated assignment mechanism, for any number of clusters.

The same result holds for the usual Monte Carlo test with the observed assignment and independent draws from π_s, using the observed assignment in the numerator and denominator, because these assignments are exchangeable under the conditional null. Removing draws with undefined statistics does not establish this result unless the actual assignment law is also conditioned on the corresponding event, including the observed assignment. Uniform permutation is justified only when it equals the conditional assignment law.

For class U, let p_1, p_3 and p_R denote the three reported p-values and set $p_U = \max(p_1, p_3, p_R)$. For a common null H_0, suppose at least one component, say p_j, satisfies

$$\Pr\{p_j \le \alpha \mid Z, S = U, H_0\} \le \alpha.$$

Then

$$\Pr\{p_U \le \alpha \mid Z, S = U, H_0\} \le \Pr\{p_j \le \alpha \mid Z, S = U, H_0\} \le \alpha. \tag{A.5}$$

The result requires neither independence among the tests nor validity of every component. Under the assignment conditions above, the randomization component supplies this bound for the sharp null. For the weak average-effect null, the same inequality supplies a validity result only if at least one component is valid for that weak null under the same conditional probability law. Agreement among the three tests does not itself establish that condition. The distinction applies even when the statistics use the same estimated coefficient.

Finally, if the selected test in every class has conditional rejection probability at most α for the same target null under the same probability model, the law of total probability gives an overall rejection probability at most α. This conclusion also requires a valid class-W test and does not follow solely from an effective-cluster threshold calibrated on a finite collection of designs. The reported simulations evaluate the rule on their specified designs and disturbances and do not prove these conditional bounds for all designs.

A.11 The joint pre-trend test

Let $\mathcal{E}_-$ be the reported horizons before adoption, other than the reference period, with estimates $\widehat{\theta}_e$ and standard errors $\widehat{\mathrm{se}}_e$. The reported pre-period maximum statistic can be expressed in terms of the uniform band

$$\mathcal{B}_{1-\alpha} = \left\{ (\vartheta_e)_{e \in \mathcal{E}_-} : \ |\widehat{\theta}_e - \vartheta_e| \le \widehat{c}_{1-\alpha}\, \widehat{\mathrm{se}}_e \ \text{ for all } e \in \mathcal{E}_- \right\}$$

The band excludes the zero path when $\max_{e \in \mathcal{E}_-} |\widehat{\theta}_e| / \widehat{\mathrm{se}}_e$ exceeds $\widehat{c}_{1-\alpha}$. The critical value follows the routing of these horizons. Where every one routes W, $\widehat{c}_{1-\alpha}$ is the $1-\alpha$ quantile of $\max_e |Z_e|$ for $Z \sim N(0, \widehat{R})$, with $\widehat{R}$ the correlation matrix of the joint CV_1^m covariance

$\text{ssc}_m \sum_g \hat{S}_g \hat{S}_g'$ of the horizon scores: the Gaussian multiplier band, the analogue of the multiplier sup-t band of Callaway and Sant'Anna (2021). Where every one routes F, $\widehat{c}_{1-\alpha}$ is the $1-\alpha$ quantile of the studentized maximum $\max_e |\widehat{\theta}_e(a_b)|/\widehat{\text{se}}_e(a_b)$ over the timing draws a_b with the dominant cluster's date held, under the global sharp null of no effect at any event time, and the band uses the observed CV_1^m standard errors; the joint p-value is the share of draws whose maximum reaches the observed one, counting the observed draw. Where any horizon routes U, or the horizons route differently, both reference distributions are computed and the larger p-value is reported. The randomization calculation is exact for the global sharp null only under the assignment and conditioning requirements of Appendix A.10. It does not generally test the isolated null of zero pre-treatment coefficients when post-treatment effects are unrestricted, because outcomes under alternative assignments cannot be imputed from that null alone. The randomization and combined values are checks calibrated under the global sharp null. Failure to reject either calculation does not establish parallel trends.

B The Monte Carlo design

This section describes the designs on which the thresholds τ_m of Section 4 are calibrated (Appendix B.1), the corresponding calibration designs constructed to match features of the two published panels (Appendix B.2), and the simulations on the two published designs (Appendix B.3).

B.1 The Compustat panel and the threshold designs

As in Baker et al. (2022), the designs are built on Compustat data. We begin with all firms in the Compustat annual file over the ten-year period from 1986 to 1995 that are incorporated in the United States, are nonfinancial (SIC codes outside 6000 to 6999), report positive total assets and a nonmissing operating income before depreciation, and have one of the fifty states as their state of incorporation in the Compustat company file. Keeping the firms observed in all ten years leaves a balanced panel of 3,290 firms and 32,900 firm-years in 49 states of incorporation; no firm in the sample is incorporated in North Dakota. Delaware is the state of incorporation of 1,580 firms, 48.0% of the panel, and New York, the next largest, of 178.

Only the size of the panel and, in one design, its distribution across states are taken from the data. Baker et al. (2022) draw firm effects, year effects and residuals from the empirical distribution of return on assets. The outcome here is generated from (1),

$$y_{it} = \alpha_i + \delta_t + \theta D_{g(i)t} + \nu_{g(i)t} + \mu_{it}, \qquad \nu_{gt} \sim N(0,\, s), \qquad \mu_{it} \sim N(0,\, 1-s), \tag{B.1}$$

where $g(i)$ is the cluster of firm i, D_{gt} equals one from the year in which cluster g adopts, ν_{gt} is the shock common to the firms of cluster g in year t, μ_{it} is the firm's own shock, and all of them are independent, so that neither component is serially correlated. The share s of the disturbance variance carried by the common shock is the λ of (8). Every design sets $\theta = 0$. Each estimator below removes α_i and δ_t, and the Wald statistic is unchanged when the disturbance is rescaled, so its null distribution depends only on the design and on s; we therefore set $\alpha_i = \delta_t = 0$. The share takes the values $s \in \{0, 0.25, 0.5, 0.75\}$, except on the 32 designs with 49 clusters, where $s \in \{0, 0.5\}$.

The designs differ only in how the firms are divided into clusters and in which clusters adopt. In every design the treated clusters adopt in four cohorts, in 1989, 1990, 1991 and 1992, so that each has at least three pre-treatment years, and the remaining clusters are never treated. The 66 designs fall into three families. The first divides the firms into G clusters whose sizes differ by at most one firm, for the 28 values $G \in \{25, 26, 27, 28, 30, 32, 35, 40, 49, 50, 60, 62, 65, 68, 70, 72, 75, 80, 85, 88, 90, 92, 95, 100, 120, 150, 200, 250\}$; the first $\lfloor G/2 \rfloor$ clusters are treated, in four cohorts as equal as possible with the earlier cohorts taking the remainder, so that with

$G = 50$ the cohorts hold 7, 6, 6 and 6 clusters. The second keeps $G = 50$ and those cohorts and concentrates the firms in one treated cluster: the first cluster of the 1989 cohort holds a share p of the firms, for the seven values $p \in \{0.05, 0.08, 0.10, 0.15, 0.20, 0.35, 0.50\}$, and the other 49 divide the rest equally. The third uses the 49 states of incorporation, of which the first 24 in alphabetical order are treated, in four cohorts of six, and the other 25 are never treated. One design keeps every state at its empirical size, so that Delaware, in the second cohort, adopts in 1990. In each of the other 30, one state holds a share p of the firms, for the fifteen values $p \in \{0.03, 0.04, 0.05, 0.06, 0.07, 0.08, 0.10, 0.12, 0.15, 0.20, 0.35, 0.50, 0.65, 0.80, 0.90\}$, and the other 48 divide the rest equally; that state is the first of the 1989 cohort in 15 designs and never treated in the other 15. Across the three families the lower end of the interval runs from about one to more than 150 for every estimator.

On each design we compute the five estimators of Section 4: the two-way regression with firm and year effects; the estimator of Callaway and Sant'Anna (2021) with not-yet-treated comparison groups over event times -5 to 5; that of Sun and Abraham (2021) over relative periods up to five, with the never-treated clusters as the comparison cohort; the stacked regression over the event window $[-5, 5]$; and the imputation estimator of Borusyak et al. (2024) over horizons 0 to 5. The target is the two-way coefficient and, for the other four, the overall average of the post-adoption effects each estimates. Each target is a fixed linear combination $\hat{\theta}_m = c_m' y$ (Appendix A.2), and CV_1^m of (6) is built on its scores $\hat{S} = K_m y$ (Appendix A.4).

Because the disturbance u is Gaussian and both $\hat{\theta}_m$ and $\hat{S}$ are linear in it, the null rejection probability of the conventional Wald test is an exact probability. At the 5% level, with G clusters, the test rejects when

$$(c_m' u)^2 \;\geq\; t^2_{G-1,\,0.975}\,\mathrm{ssc}_m\,\|K_m u\|^2. \tag{B.2}$$

The vector $\xi = \left(c_m' u, (K_m u)'\right)'$ is normal with mean zero and a covariance $V_m(s)$ fixed by the design and the share, and (B.2) is the event $\xi' A \xi \geq 0$ with $A = \mathrm{diag}(1,\, -t^2_{G-1,0.975}\,\mathrm{ssc}_m I_G)$. Its probability is therefore that of $\sum_k \omega_k \chi^2_{1,k} \geq 0$, where the ω_k are the eigenvalues of $V_m(s)^{1/2} A V_m(s)^{1/2}$ and the $\chi^2_{1,k}$ are independent, and we compute it by the inversion formula of Imhof (1961). The size of a design for estimator m is the largest of these probabilities over its shares. Its lower end is that of the interval of Section 3.1, computed from c_m over $\lambda \in [0, 1]$ and the serial-correlation grid.

A design fails when its size exceeds 0.06, one percentage point above the nominal level. With L_f the largest lower end among an estimator's failing designs and L_p the smallest lower end above L_f, the threshold is $\tau_m = \lceil 100 L_p \rceil / 100$. For the two-way regression the last

failure is the concentrated design with $p = 0.08$, whose lower end is 10.58, and $L_p = 13.912$ belongs to the incorporation-state design whose treated state holds 7% of the firms, so $\tau_{\text{TWFE}} = 13.92$. For the stacked regression the last failure is the equal design with $G = 27$, at 25.31, and $L_p = 27.021$ belongs to the incorporation-state design whose treated state holds 6%, so $\tau_{\text{Stacked}} = 27.03$. The other three thresholds are set by adjacent equal designs: $G = 60$ and 62 for the imputation estimator (53.16 and 54.87, so 54.87), $G = 88$ and 90 for Callaway–Sant'Anna (56.30 and 58.22, so 58.22), and $G = 85$ and 88 for Sun–Abraham (56.95 and 59.26, so 59.26). Above its threshold, every design's size lies between 0.050 and 0.060 for every estimator.

A threshold carries to another panel when size depends on the design through the lower end of the estimator's interval. For the two-way regression the leading terms depend on the design through the concentration that the interval measures. On a balanced panel of N units with unit effects nested in clusters and period effects, c_m sums to zero over every unit and every period, so the restricted scores of Appendix A.4 are $\tilde{S}_g = S_g - \sum_t C_{gt}\bar{u}_t$, with $C_{gt} = \sum_{r\in(g,t)} c_{m,r}$ and $\bar{u}_t$ the mean disturbance of period t. With i.i.d. errors $\bar{u}_t$ is uncorrelated with $\hat{\theta}_m$, and since $\hat{S}_g = \tilde{S}_g - L_g^m\hat{\theta}_m$ under the null,

$$\mathbb{E}\,\mathrm{CV}_0^m = \mathrm{Var}(\hat{\theta}_m)\Big\{1 - \frac{1}{G_m^*(0)} - \pi_m\Big\}, \qquad \pi_m = \frac{1}{N\|c_m\|^2}\sum_{g,t} C_{gt}^2. \tag{B.3}$$

When c_m is constant within each cluster and period, $\pi_m = \sum_g L_g^m\, n_g/N$, with n_g the units of cluster g, which is $1/G$ when clusters are of equal size. The same concentration bounds the departure of the score sum from normality, its Lyapunov ratio being at most $\kappa\, G_m^*(\lambda)^{-1/2}$ with κ the largest standardized third absolute moment of a cluster score, and it enters the effective degrees of freedom of the variance estimator. These are leading terms only: the common shocks, the controls, the number of clusters, the number of treated clusters and the number of observations inside each cluster and period enter beyond them, so the lower end is not a sufficient statistic for size. On a panel with one observation per cluster and period, 49 clusters and 10 of them treated, the two-way lower end is 13.74 and the size 0.072, while the Compustat design that sets τ_{TWFE} has lower end 13.91 and size 0.059. The expected value of CV_1^m is 0.945 of $\mathrm{Var}(\hat{\theta}_m)$ on the first and 0.936 on the second at its least favorable share, and the difference in size comes with a difference in the effective degrees of freedom of CV_1^m, $\{\mathrm{tr}(K_m\Omega K_m')\}^2/\,\mathrm{tr}\{(K_m\Omega K_m')^2\}$ with Ω the covariance of u, which are 12.7 and 17.4. On such panels $\tau_{\text{TWFE}} = 13.92$ is too low, and Appendix B.2 recalibrates it at 21.83. For the cohort-based estimators the corresponding term is the within-group loss $\bar{b}_m$ of (13), the loss in CV_1^m's expected scale for Sun–Abraham and for Callaway–Sant'Anna with never-treated comparisons and a lower bound on it for the others (Appendix A.4). It moves with the lower

end on this panel only because every design has four cohorts: at the same lower end, a design whose cohorts hold fewer clusters loses more, and for such a design their thresholds would be too low.

B.2 Threshold calibration for the two application designs

The classes in Section 4.2 and Section 5 are read off the thresholds calibrated here, and neither published design enters that calibration. The same failure rule is applied to panels constructed with features of the two application designs. A design fails when its largest exact CV_1^m rejection probability over $s \in \{0, 0.25, 0.5, 0.75\}$ exceeds 0.06, and τ_m is the smallest lower end above every failing design, rounded up to the next hundredth. Only designs with $b_m = 0$ enter. A design with $b_m > 0$ is class F when its interval lies below the threshold and class U when the interval straddles the threshold or clears it, so it is excluded from class W and does not set τ_m. For the pooled two-way and stacked regressions, the single-group diagnostic is not defined and the calibration uses $b_m = 0$ as a classification convention. The lower end is computed as in Appendix B.1. Figures B.1 and B.2 plot these designs. Each marker is one design, and the vertical lines are the thresholds.

In the bank-deregulation calibration designs each cluster contributes one observation per period, and the designs use both published windows, $T = 23$ and $T = 31$. In the equal designs the number of clusters takes every value in the Compustat list together with 8, 10, 12, 14, 16, 18, 20, 22, 36, 42, 44, 46, 48, 51 and 71, and half the clusters are treated, in four cohorts at the nearest integers to $0.4T$, $0.5T$, $0.6T$ and $0.7T$, the relative position of 1989–1992 in the ten-year Compustat window. From $G = 16$ every cohort holds at least two clusters. A panel with one observation per cluster has no firm share to concentrate, and its counterpart of the concentration path varies the number of treated clusters. At the two published counts, 49 clusters over $T = 31$ and 36 over $T = 23$, that number runs from one to half the clusters, the treated clusters take the four dates in turn, and the other clusters are never treated. The four values of s return the same size on every design.

The two-way regression never fails on the equal designs, whose largest size is 0.059 at a lower end of 7.05, and it fails on the path in the number of treated clusters, where with 14 of 49 clusters treated the lower end is 21.53 and the size 0.0605. With 13 of 36 treated the lower end is 21.83 and the size 0.057, so $\tau_{\mathrm{TWFE}} = 21.83$. The same path sets the stacked threshold at 22.45 and the Sun–Abraham threshold at 29.85, with last failures at 19 and 23 of 49 clusters treated. The imputation estimator and Callaway–Sant'Anna are set by equal designs, at 59.60 and 59.52. Above each threshold every design's size lies between 0.050 and 0.060. The path moves one treated cluster at a time, so a last failure that sits at 0.06 matters little, and counting each last failure as a pass lowers the two-way, stacked and Sun–Abraham

thresholds by at most 2.3 and leaves the other two unchanged.

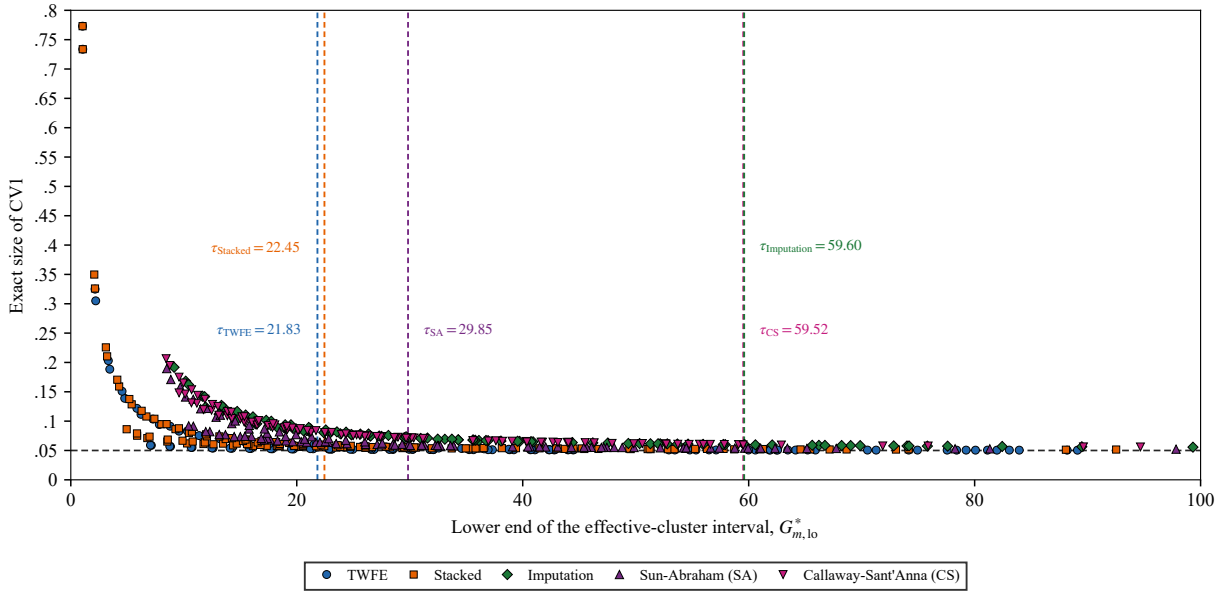
Against the Compustat values of 13.92, 27.03, 59.26, 54.87 and 58.22 for the two-way regression, the stacked regression, Sun–Abraham, the imputation estimator and Callaway–Sant'Anna, the two-way threshold is higher by eight clusters, the stacked threshold is lower by about five, the Sun–Abraham threshold is lower by about thirty, and the other two are within five clusters. Under $\tau_{\text{TWFE}} = 13.92$ the designs with 11 to 14 of 49 clusters treated and those with 10 and 11 of 36 would be class W at sizes from 0.0605 to 0.068, which is the qualification in Appendix B.1. For Sun–Abraham the Compustat threshold lies above the threshold in these calibration designs. Near a lower end of 29 the size is 0.061 here and 0.071 in the Compustat calibration designs.

A further family of bank-deregulation calibration designs uses $T = 23$ and places the treated clusters on dates from period 5 to period 18. The window holds 14 dates, so through $G = 49$ some treated cluster has a date of its own and the cohort-based estimators have $b_m > 0$, and from $G = 60$ every date holds at least two clusters. The two-way regression does not fail on this family, and the stacked threshold is 20.36. Sun–Abraham, the imputation estimator and Callaway–Sant'Anna cross at 130.49, 203.98 and 231.54, beyond the right edge of Figure B.1. Designs in this family with two to six clusters per cohort have sizes up to 0.113 at lower ends past the thresholds drawn in the figure, which is the case of thinner cohorts in Appendix B.1. Sections 4.2 and 5 use the thresholds drawn in Figure B.1, and this family is not part of that calibration. The published bank-deregulation intervals lie on the same side of the thresholds of both families, and on the same side of the Compustat thresholds. The four heterogeneity-robust upper ends run from 1.86 to 11.78, so those rows stay in class F, and the two-way row, with lower end 35.01 and published size 0.050, stays in class W.

In the elder-protection calibration designs the panel has 3,139 units and $T = 84$ periods, split as equally as possible across G clusters for the same list of G, and the four cohort dates are 34, 42, 50 and 59. Two paths at $G = 51$ join the equal designs. On the concentration path one treated cluster holds a share $p \in \{0.05, 0.08, 0.10, 0.15, 0.20, 0.35, 0.50\}$ of the units, and on the other the number of treated clusters runs from one to 25. On the equal designs and the second path the shock share moves the size by at most 0.001. On the concentration path it moves the size by several tenths, and with half the units in one cluster the two-way size is 0.073 at $s = 0$ and 0.480 at $s = 0.75$.

The thresholds are 26.89 for the two-way regression, 25.41 for the stacked regression, 59.84 for the imputation estimator, 56.42 for Sun–Abraham and 56.12 for Callaway–Sant'Anna. The first two are set by the path in the number of treated clusters, with last failures at 16 and 23 treated clusters, and counting each of those as a pass would give 24.39 and 24.33. The other three are set by equal designs. Each threshold other than the two-way one lies

Figure B.1: Exact size of CV_1^m in the bank-deregulation calibration designs



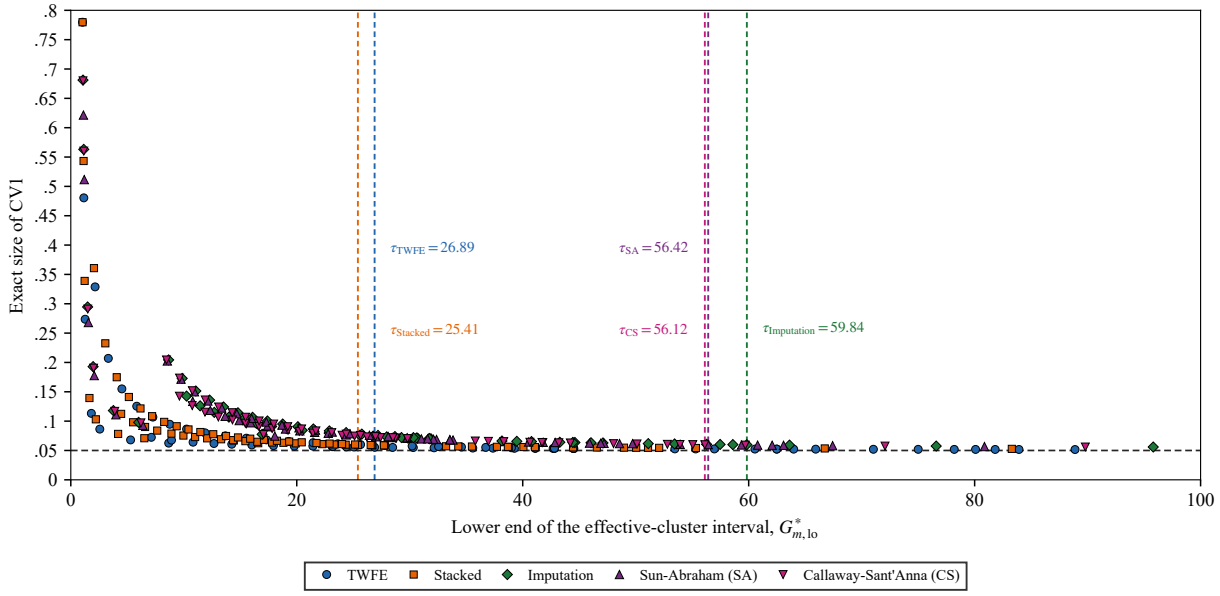
Note 1: Each marker is one design with one observation per cluster and period on the 23-period and 31-period windows. The cohort-based designs have $b_m = 0$, and the pooled regressions use zero as the classification convention for the single-group diagnostic. The horizontal axis is the lower end of the effective-cluster interval and the vertical axis is the largest exact rejection probability of CV_1^m over $s \in \{0, 0.25, 0.5, 0.75\}$. Five series of markers are drawn, one per estimator. The dashed horizontal line is 0.05.

Note 2: The designs are the equal divisions, in which half the clusters are treated in four cohorts, and the designs with 49 and 36 clusters in which the number treated runs from one to half the clusters. A design fails when that probability exceeds 0.06. The vertical lines are the thresholds set by these designs: 21.83 for the two-way regression, 22.45 for the stacked regression, 29.85 for Sun–Abraham, 59.52 for Callaway–Sant'Anna and 59.60 for the imputation estimator.

within five clusters of the corresponding Compustat threshold, and above each threshold every design's size lies between 0.050 and 0.060. The two-way interval from 10.25 to 28.46 straddles 26.89, and the stacked interval from 11.21 to 30.57 straddles 25.41. The other three upper ends, 19.11, 21.60 and 27.46, lie below 56.12, 56.42 and 59.84. All ten published classes are unchanged under these thresholds for either application, and Sections 4.2 and 5 use them.

A further family in the 84-period calibration designs gives every treated cluster its own date through $G = 100$, so every treated cohort is a single cluster, and the cohort-based estimators have $b_m > 0$ on every design of this family and no threshold. The sizes of Callaway–Sant'Anna and Sun–Abraham rise with G, from 0.708 and 0.664 at $G = 30$ to 0.892 and 0.873 at $G = 100$, and that of the imputation estimator stays between 0.645 and 0.662. A cohort of one cluster leaves only the comparison side of each of its groups visible to CV_1^m, and that side's share of the variance falls as G grows. For Sun–Abraham at $s = 0$, $\bar{b}_m$ rises from 0.954 at $G = 30$ to 0.993 at $G = 100$, and on each of these designs the expected value of CV_1^m is $\mathrm{ssc}_m(1 - \bar{b}_m)$ times $\mathrm{Var}(\hat{\theta}_m)$, as Appendix A.4 implies for never-treated comparisons. The two-way threshold on this family is 26.07, and the published interval still straddles it, so that row would stay in class U. The stacked threshold on this family is 37.22. The published stacked upper end is 30.57, so that row would be class F if this family supplied the threshold.

Figure B.2: Exact size of CV_1^m in the elder-protection calibration designs



Note 1: Each marker is one design on an 84-period panel of 3,139 units with four cohorts. The cohort-based designs have $b_m = 0$, and the pooled regressions use zero as the classification convention for the single-group diagnostic. The horizontal axis is the lower end of the effective-cluster interval and the vertical axis is the largest exact rejection probability of CV_1^m over $s \in \{0, 0.25, 0.5, 0.75\}$. Five series of markers are drawn, one per estimator. The dashed horizontal line is 0.05.
Note 2: The designs are equal divisions of the units into G clusters, the concentration path at $G = 51$, and the designs with 51 clusters in which the number treated runs from one to 25. A design fails when that probability exceeds 0.06. The vertical lines are the thresholds set by these designs: 25.41 for the stacked regression, 26.89 for the two-way regression, 56.12 for Callaway–Sant'Anna, 56.42 for Sun–Abraham and 59.84 for the imputation estimator.

Sections 4.2 and 5 use the threshold of 25.41 in Figure B.2, and under that threshold the stacked row stays in class U. No threshold from either family places a published row in class W.

B.3 The two published designs

The second exercise keeps each published design as Section 5 reconstructs it, holding fixed its estimation sample, clusters, adoption dates and each estimator's specification, and draws only the disturbance. On Beck et al. (2010) the two-way regression uses the 49 states over 1976 to 2006, 1,519 state-years; the heterogeneity-robust estimators discard the always-treated states and the years without an untreated state, which leaves 36 states and, for Callaway–Sant'Anna, Sun–Abraham and the imputation estimator, 828 state-years from 1976 to 1998. On Carlin et al. (2023) the two-way regression uses 3,139 counties in 51 states over 84 months, 263,676 county-months, with the article's ten county controls and the county and state linear trends it fits before each state's date; the heterogeneity-robust estimators drop Washington and use 3,100 counties in 50 states, 260,400 county-months, with the ten controls and without the trends, and their largest state-month holds 253 counties. The class of each row of Table 2 is its estimator's interval on the published design read against τ_m.

Each replication draws, for every row j of the estimation sample,

$$u_j(s) = \sqrt{1-s}\, e_j + \sqrt{s}\, \nu_{g(j)t(j)}, \tag{B.4}$$

where $g(j)$ and $t(j)$ are the cluster and period of row j and the e_j and ν_{gt} are independent standard normal draws: the disturbance of (B.1) with unit variance. In the baseline comparison the share is $s = 0.75$ on Carlin et al. (2023). The supplementary exact-size calculations also consider $s \in \{0, 0.25, 0.5, 0.75\}$. On Beck et al. (2010) every state-year holds one row, so the two components coincide and the law of $u(s)$ is the same at every share; one share, $s = 0$, is drawn. Under the null the outcome is $y = a_m u(s)$. Under the alternative it is

$$y = a_m u(s) + \delta_m(s)\, D, \qquad \delta_m(s) = a_m \left\{ t_{G_m - 1,\, 0.975} + t_{G_m - 1,\, 0.95} \right\} \mathrm{sd}_m(s), \tag{B.5}$$

where D is the published treatment indicator, $\mathrm{sd}_m(s)$ is the exact standard deviation of $c_m' u(s)$, and G_m is the number of clusters in estimator m's sample. The effect is homogeneous and every estimator recovers it, $c_m' D = 1$, so that $\hat{\theta}_m - \delta_m(s) = a_m c_m' u(s)$; $\delta_m(s)$ is the approximate reference effect based on (11) at $\alpha = 0.05$ and a target power of $\pi = 0.95$, with G_m in place of G^*_{used} and the exact standard deviation in place of the product of the other factors. Our analysis is based on 1,000 replications at each share, under the null and under the alternative.

The positive scale a_m is selected separately for each application and estimator so that the expected conventional cluster-robust variance under the stated disturbance model equals the variance estimated in the corresponding application. This scaling expresses the reference effect and the method-specific minimum detectable effects in the units of the observed outcome. The same scaled disturbances and homogeneous effect are used for every available inference method within a row of Table 2. The reference effect in these simulations uses the approximation in Section 3.1.2. Under the stated independent-disturbance and weight normalization, the estimator on the used rows has variance $\sigma^2/\mathcal{A}_{\text{used}} = \sigma^2/(\mathcal{A}\rho_{\mathcal{A}})$, and κ represents an adjustment for dependence. The sum of central t quantiles in (11) is a heuristic reference effect and does not generally deliver exact power π. Restricting to the used rows changes the concentration as well: with $\gamma_g = \|\ddot{x}_g\|^2$ and $h' = \sum_g (\gamma_g^{\text{used}})^2 / \sum_g \gamma_g^2$, $G^*_{\text{used}}(0)/G^*(0) = \rho^2_{\mathcal{A}}/h'$, so the restriction leaves the count unchanged only if the deleted rows carry leverage in proportion to each cluster's own. For a unit treated in every period of a balanced panel, $D_{it} \equiv 1$ and the two-way residual is $1 - \bar{D}_i - \bar{D}_t + \bar{D} = \bar{D} - \bar{D}_t$, which does not depend on i; a cluster of n_g such units has $\gamma_g = n_g \sum_t (\bar{D} - \bar{D}_t)^2$.

On every replication the reported procedures are evaluated at the 5% level. The clustered Wald and jackknife calculations concern the specified average-effect target, while

randomization inference concerns a sharp null. Under the simulated homogeneous zero effect these nulls coincide. The conventional Wald test refers $\hat{\theta}_m/(\mathrm{CV}_1^m)^{1/2}$ to t_{G-1}, with G the number of clusters in the estimator's sample. The cluster jackknife replaces CV_1^m with $\mathrm{CV}_3^m = \frac{G-1}{G}\sum_g(\hat{\theta}_{m(g)} - \hat{\theta}_m)^2$, where $\hat{\theta}_{m(g)}$ is the estimate without cluster g, and keeps the reference distribution. The original Sun–Abraham fixed target is not identified after some deletions, so its jackknife result is omitted. Randomization inference holds the adoption date of the dominant cluster, the one with the most units, at its published value and permutes the dates of the others; on Beck et al. (2010), where every state is a single unit, the held state is the first in the panel's order. The $B = 199$ permutations are drawn once and used on every replication. Under each, the weights c_m and the scores are recomputed from the permuted dates. A permutation under which the statistic is not defined is counted as at least as extreme as the observed statistic, as in Appendix A.9. The statistic is $|t| = |\hat{\theta}_m|/(\mathrm{CV}_1^m)^{1/2}$, the p-value is $\{1 + \#(b : |t_b| \geq |t|)\}/(B + 1)$, and the test rejects when it is at most 0.05. On the two-way regression of Carlin et al. (2023) each permutation rebuilds the two trend regressors from the published outcome and the permuted dates, as the application does (Section 5.2). The observed regression keeps the published trend columns, so its interval, its class and its conventional Wald test are those of the published specification. The fifth test, computed for the two-way, Sun–Abraham and stacked regressions, is the restricted wild cluster bootstrap WCR-C. The regression is refit with $\theta_m = 0$ imposed; its residual is $\tilde{u} = \hat{e}_m + c_m\hat{\theta}_m/\|c_m\|^2$ (Appendix A.4), and the bootstrap outcome is the restricted fit plus $\tilde{u}$ times one Rademacher sign per state. The statistic is $|t^*| = |\hat{\theta}_m^*|/(\mathrm{CV}_1^{m*})^{1/2}$ on each of 999 sign vectors, drawn afresh on every replication and shared by the shares and the two effects, and the p-value is $\{1 + \#(b : |t_b^*| \geq |t|)\}/1000$. The stacked regression is bootstrapped on its stacked rows, with the states as clusters. The routed procedure is the conventional Wald test in class W and randomization inference in class F; in class U it rejects only when the conventional Wald test, the jackknife and randomization inference all reject.

For each procedure, Panel A of Table 2 reports the rejection frequency under a true zero effect and Panel B reports the rejection frequency under the common reference effect. Both panels use the same specified disturbance setting for a given application and estimator.

B.4 Power calculations for the applications

The minimum detectable effects in Tables 3 and 5 use the same samples, estimator specifications and calibrated disturbances as Table 2. For each available inference method, we find the magnitude of a homogeneous effect at which its simulated rejection frequency reaches 95 percent at a nominal five percent significance level. The same disturbance draws are used across the effects considered. The common reference effect in Table 2 is calculated

from the approximation in (B.5), whereas the minimum detectable effects are calculated separately for each method from its simulated rejection frequencies. The reference effect need not coincide with any one of the method-specific values. This distinction allows the methods to be compared at a common effect while also describing the effects that each method can detect with the specified power.

The 1,000 simulated samples and the randomization and bootstrap draws are the same as those used to report the rejection frequencies in Table 2. In class U, the minimum detectable effect of the design-adapted calculation is obtained from the frequency with which all three tests reject in the same replication. Randomization inference concerns a sharp null under the stated assignment mechanism. The jackknife calculations for CS, the stacked regression and imputation reaggregate the remaining observations after each state deletion. Under the homogeneous effect used in the simulation, the full and reduced samples target the same common effect, while their weights can differ under heterogeneous effects. The original Sun–Abraham aggregate loses identifying support after some deletions and its fixed-target jackknife is omitted. The stacked bootstrap holds the stacked regression fixed when forming the transformed scores. Because the nominal five percent tests can have different rejection rates under a true zero, the method-specific minimum detectable effects and rejection frequencies under the alternative are interpreted together with the null frequencies.

B.5 Switching and two-stage estimates

The additional calculations use balanced panels with absorbing binary treatment and no covariates. Let the first treatment date of unit i be a_i, and let $\Delta Y_{it} = Y_{it} - Y_{i,t-1}$. For each supported treatment date t, let A_t contain the units first treated at t, let C_t contain units untreated at t, and write their counts as n_{At} and n_{Ct}. We retain a date when the comparison group contains at least two units, and write $N_A = \sum_t n_{At}$. The switching estimate is

$$\widehat{\theta}_D = \frac{1}{N_A} \sum_t n_{At} \left(\overline{\Delta Y}_{At} - \overline{\Delta Y}_{Ct} \right). \tag{B.6}$$

This is the immediate switcher comparison in de Chaisemartin and D'Haultfœuille (2020). Under no anticipation and parallel trends for untreated potential outcomes between switchers and the comparison units, it identifies the unit-weighted mean effect in the first treated period for the retained switchers. The dates, units and weights defining the estimand are held fixed in the score calculation. The restriction on the comparison-group count is a support convention for these calculations and is not a condition that every comparison has at least two independent clusters.

The conditional score of state g for this estimate is

$$S_{Dg} = \frac{1}{N_A} \sum_t \left[\sum_{i \in A_t \cap g} (\Delta Y_{it} - \overline{\Delta Y}_{At}) - \frac{n_{At}}{n_{Ct}} \sum_{i \in C_t \cap g} (\Delta Y_{it} - \overline{\Delta Y}_{Ct}) \right]. \tag{B.7}$$

The resulting variance is the finite-cluster-adjusted sum of squared scores. It is a conditional fixed-weight construction and does not represent every variance or resampling procedure that can be used with this estimator. The centered terms make explicit that a state containing all units in a switching cohort has no switcher deviation for that cohort. The overall score vector can still reflect that state's contribution through other comparisons.

For the two-stage estimator, let X contain unit and time indicators, U be the diagonal matrix selecting untreated observations, and w average the treated observations in the target. One period indicator is omitted so that the untreated first-stage normal equations have full rank. The estimate and its observation weights are

$$\widehat{\gamma} = (X'UX)^{-1}X'UY, \quad \widehat{\theta}_G = w'(Y - X\widehat{\gamma}) = c'Y, \quad c = w - UX(X'UX)^{-1}X'w. \tag{B.8}$$

The target contains either the retained switchers in their first treated period or all treated unit-periods. These are different targets when effects vary over time. The first target uses the same treated observations as the switching estimate but draws on a different set of untreated observations. Identification requires the first-stage untreated outcome model and the corresponding overlap conditions. The first-stage comparison graph is connected in the samples used here.

Let M_g select observations in state g and let $q_g = \mathbf{1}'M_g w$. The complete linear score, including the first-stage estimation contribution, is

$$S_{Gg} = c'M_g(Y - X\widehat{\gamma}) - q_g\widehat{\theta}_G. \tag{B.9}$$

The sums of both sets of scores across states are zero. For each construction the reported standard error is the square root of $G/(G-1)$ times the sum of squared state scores, and the Wald test uses a t reference with $G-1$ degrees of freedom. The state-score multiplier compares the estimate with sums of these scores multiplied by independent random signs. The scores are held fixed within each multiplier calculation. These procedures are evaluated under the specified data-generating processes and are not claimed to be generally valid.

The Compustat exercise uses the same 66 calibration designs and the same disturbance shares as Figure 4. For the two published samples, the disturbance has an independent unit component and a state-period component shared by units within a state. The baseline

uses serially independent Gaussian components. The serial sensitivity retains independent state-period shocks and gives the unit component a stationary autoregressive correlation of 0.8. The state-period shock share ranges from zero to 0.75 in the elder-protection sample. In the bank sample there is one unit per state, and the reported serial calculation uses the unit component. The exact rejection probability follows from the same quadratic-form calculation as Appendix B.1, while the multiplier frequencies use independent Gaussian draws and independent state signs.

For the full-score blind-share calculation in Table B.1, each independent unit-period shock is a separate coordinate. Its contribution to the estimator variance is included in the numerator only if its coefficient in every state score is zero, and the denominator is the variance of the estimate under independent unit-period disturbances. This full-vector definition is specific to the scores above and need not coincide with the group-specific blind-share diagnostic under a different variance construction or covariance model. It is not a sufficient statistic for test size.

In the bank-deregulation sample, the last treated cohort has one remaining untreated comparison state, so the two-unit support convention excludes that cohort from the switching estimate and from the matched first-period two-stage target. Removing individual states eliminates at least one originally supported switching date in eleven of the thirty-six deletions. In elder protection, the corresponding count is fifteen of fifty. Deleting a treated state also removes its treated units from a finite-sample aggregate even where the cohort survives. A jackknife that reweights the remaining units therefore need not preserve the original target without additional assumptions or a separately defined supported target. The original application samples therefore do not provide fixed-target jackknife comparisons for these estimates without additional assumptions or a separately defined supported target.

Table B.2 reports rejection frequencies for a homogeneous nonzero effect under the independent disturbances. Because the corresponding null rejection probabilities in Table B.1 differ across procedures and estimates, these frequencies do not provide a comparison of power at equal size.

B.6 Additional bootstrap and randomization comparisons

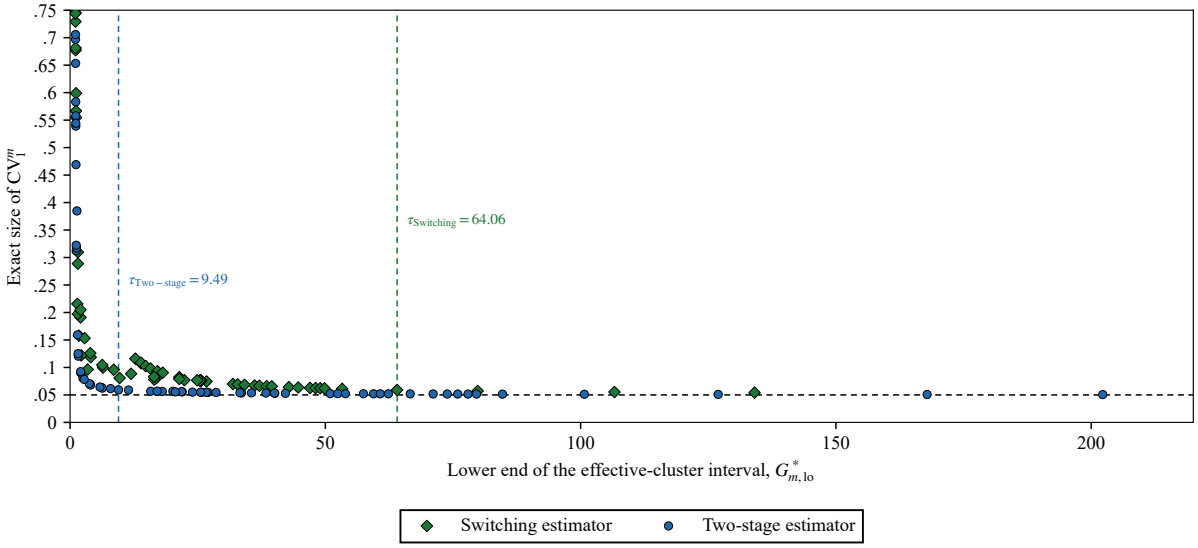
The WCR-S and WCR-B entries in Table 2 use the same published application designs and disturbance draws as the other entries in that table. WCR-S uses the restricted jackknife-transformed cluster scores and conventional studentization, while WCR-B uses those scores and jackknife studentization (MacKinnon et al., 2023b). Both are calculated for the two-way and stacked regressions in the two applications. The original stacked rows are held fixed in the leave-one-state calculations used for the transformed scores. The Sun–Abraham fixed target

Table B.1: Switching and two-stage estimators in the published designs

	Estimate	Cluster s.e.	Effective interval	Wald	Wald, serial	Multiplier	Full-score blind share
BLL, Switching estimator	-0.0055	0.0052	[32.84, 33.48]	0.133	0.128	0.151	0.121
BLL, Two-stage, first period	0.0067	0.0071	[8.26, 33.26]	0.047	0.060	0.060	0.000
BLL, Two-stage, all post periods	0.0195	0.0068	[1.45, 2.20]	0.262	0.353	0.285	0.000
CUY, Switching estimator	-0.0936	0.0417	[6.79, 19.14]	0.491	0.492	0.517	0.461
CUY, Two-stage, first period	0.0602	0.1261	[7.22, 31.21]	0.067	0.067	0.083	0.000
CUY, Two-stage, all post periods	-0.3914	0.1963	[11.52, 24.64]	0.068	0.068	0.086	0.000

Note 1: All specifications use the same balanced sample within each application and exclude covariates. The switching estimator averages effects in the first treated period over switchers with at least two untreated comparison units. The first-period two-stage estimate averages the same switchers, while the post-period estimate averages all treated observations. In bank deregulation the final switching date is excluded from the first-period target because only one untreated comparison state remains. Cluster standard errors use state scores with the stated finite-cluster adjustment. The switching estimator's scores center changes within the switcher and comparison groups, holding membership and aggregation weights fixed. The two-stage scores account for estimation of the first-stage unit and time effects. Wald entries are exact null rejection probabilities. The serial column uses serial correlation of 0.8 in unit disturbances. For elder protection, each entry is the largest over the four common state-month shock shares considered, while bank deregulation has one observation per state and period. The multiplier column uses 10,000 draws and 999 independent state-level signs per draw under serially independent disturbances. The full-score blind share is the fraction of the estimator's variance under independent unit-period disturbances contributed by shock coordinates whose entire state-score vector is zero. The alternatives have different targets and comparison groups, and the reported tests do not establish validity under other disturbance processes. Appendix B.5 provides the construction.

Figure B.3: Exact size of CV_1^m for the switching and two-stage estimators



Note 1: Each marker is one design in the Compustat calibration designs of Section 4.1. The horizontal axis is the lower end of the effective-cluster interval and the vertical axis is that design's largest exact rejection probability of the cluster-robust Wald test across the state-year shock shares computed for it. Two series of markers are drawn, one per estimator. The dashed horizontal line is 0.05.

Note 2: A design fails when that probability exceeds 0.06. The vertical lines are the thresholds τ_m, defined for each estimator as the smallest lower end above the lower end of every failing design, rounded up to the next hundredth. They are 64.06 for the switching estimator and 9.49 for the two-stage estimator. These thresholds summarize the designs and disturbance shares considered.

Note 3: The switching estimate averages first-period effects using fixed-weight centered scores. The two-stage estimate averages all treated periods and accounts for first-stage estimation in its scores. Their targets and comparisons differ. The finite-cluster adjustment can differ from that used for the estimates in Figure 4, so differences between figures reflect the target, score and adjustment as well as the estimator. Appendix B.5 describes the calculations.

Table B.2: Rejection frequencies under a homogeneous alternative

Design and estimate	Wald	Score multiplier
BLL, Switching estimator	0.588	0.619
BLL, Two-stage, first period	0.619	0.658
BLL, Two-stage, all post periods	0.774	0.797
CUY, Switching estimator	0.831	0.841
CUY, Two-stage, first period	0.657	0.700
CUY, Two-stage, all post periods	1.000	1.000

Note 1: Each treated observation has an effect of 0.5 in units of the disturbance standard deviation. Entries are rejection frequencies from 10,000 draws and, for the multiplier, 999 signs per draw. The samples, estimators and scores are those in Table B.1. For elder protection the table reports the smallest frequency over the four common state-month shock shares. The procedures are not adjusted to have equal null rejection probabilities, so differences between entries are not comparisons of power at equal size.

loses identifying support after some state deletions, and the corresponding original-target calculations are not reported. These results describe the procedures and specifications studied in the two designs.

Tables B.3 and B.4 supplement the interpretation of randomization inference. The first holds the bank-deregulation assignment fixed and varies one pattern of heterogeneous effects whose average and expected two-way coefficient are zero. The second compares random assignment with a design in which earlier adoption is associated with untreated trends. The latter departs from both the assumed randomization mechanism and parallel trends, so its rejection frequency does not isolate either departure. Neither experiment establishes the validity of randomization inference for every zero-average null.

Table B.3: Heterogeneous zero-average effects on the bank-deregulation design

Effect heterogeneity	0	0.5	1	2
Clustered Wald	0.040	0.007	0.000	0.000
Randomization calculation	0.038	0.008	0.000	0.000

Note 1: Rejection frequencies at the five percent level use the bank-deregulation two-way specification, 1,000 disturbance draws and 199 randomizations of the adoption dates with one state's date held fixed. The adoption schedule is fixed when generating the outcomes. State-specific effects are constant after adoption and are constructed so that their average over treated state-years and the expected two-way coefficient are both zero. The column heading multiplies a fixed vector of heterogeneous effects whose root mean square is one, with the disturbance standard deviation normalized to one. The zero column satisfies the sharp null. The other columns satisfy the stated average-effect and coefficient nulls while violating the sharp null. These frequencies are conditional on one heterogeneous effect pattern and do not establish weak-null validity in general.

Table B.4: Randomization inference and the assignment of adoption dates

Assignment mechanism	Rejection frequency
Random adoption timing	0.054
Timing related to untreated trends	0.978

Note 1: The experiment uses the state-year structure and observed multiset of adoption dates of the bank-deregulation panel, with one state's date held fixed. Each of 1,000 replications has no treatment effect. Untreated outcomes contain independent disturbances and state-specific linear trends. In the first row the remaining dates are assigned uniformly across states independently of their trends. In the second row earlier adoption is assigned to states with larger untreated trends. Randomization inference uses 199 draws under the first assignment mechanism in both rows. The second row deliberately violates both the randomization assumption and parallel trends, and its rejection frequency measures sensitivity to the joint departure from these assumptions. The Monte Carlo standard error near a rejection frequency of five percent is approximately 0.007.